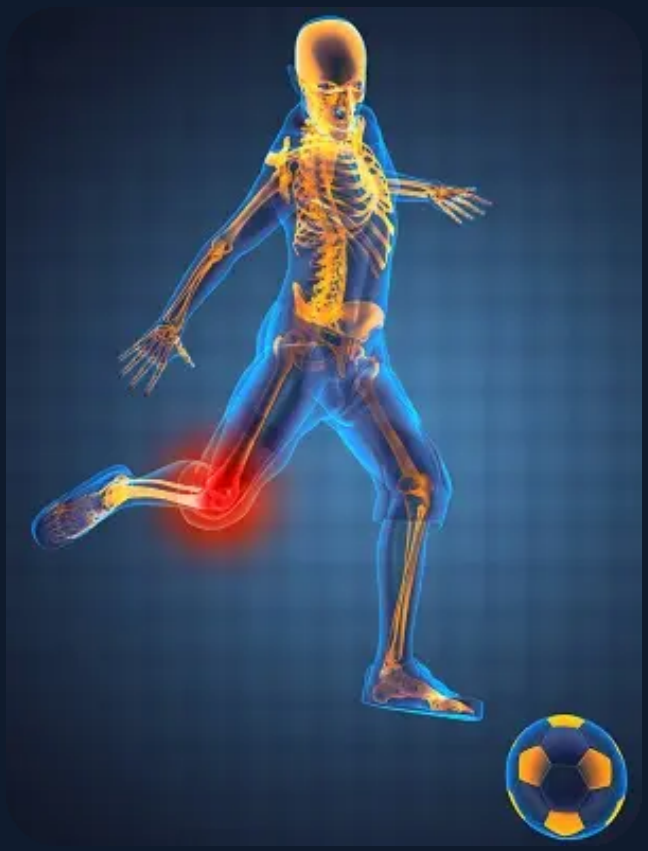
COURSE

Nasr City Ortho tube

Sport Orthopedic

11 lectures



Prof Dr Amr Rashwan

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This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

This transcript is not a replacement for the original lecture or the relevant textbooks and scientific references. The accuracy is generally very good, but AI-assisted transcription can still contain errors. If you are unsure about any point, please check the original lecture or a reliable reference.

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How to use this book

- Click any line in the Contents to jump to that lecture.
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- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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Shoulder Bony Injuries

(Dislocations and Fractures of the Shoulder Girdle)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 10) “2.4 and a half cm from end” → “2.5–4.5 cm from lateral end”

C2 (p. 10) “type IV clavicle moves downward” → “type IV clavicle displaced posteriorly”

C3 (p. 17) “left shoulder 7 to 9 o'clock” → “left shoulder 6 to 9 o'clock”

Queried - kept as spoken, but looks wrong

Q1 (p. 10) As spoken; type V is not the most common Rockwood type - types I-III are far commoner.

Q2 (p. 11) As spoken; conflicts with the 11–13 mm normal coracoclavicular distance given later in the same lecture.

Q3 (p. 13) As spoken; type II AC injuries are normally treated conservatively; arthroscopic fixation targets types III-V.

Q4 (p. 26) As spoken; Hertel's 2004 ischaemia predictors paper appeared in J Shoulder Elbow Surg, not JBJS.

Q5 (p. 28) As spoken; reverse arthroplasty for fracture is generally reserved for elderly patients (about 65-70+), not 40-45.

Classification of Shoulder Injuries Ask AI

- Two groups
- **Bony injuries** → fractures, dislocations
- **Soft tissue injuries** → capsule, glenoid labrum, rotator cuff, biceps tendon

- Most commonly dislocated joint → **glenohumeral joint**
- Discussion starts with **acromioclavicular (AC) dislocation**

Acromioclavicular Dislocation — Anatomy and Diagnosis Ask AI

Anatomy Ask AI

AC joint

- **Acromioclavicular joint** → lateral end of clavicle
  - Articulates with medial end of acromion
- Stabilizers
  - **Acromioclavicular ligaments** → key to joint stability
  - **Coracoclavicular ligaments** → conoid + trapezoid
    - Located 2.5–4.5 cm from lateral end of clavicle *[C1 · transcript: "2.4 and a half cm from end"]*

Pathology Ask AI

AC joint

- Torn → conoid, trapezoid, **AC ligament** → disruption begins
- Usually intact → **coracoacromial ligament**
  - Coracoid → medial end of acromion process

Rockwood Classification Ask AI

- **Rockwood classification** → 6 types

Type	Findings
I	Ligaments stretched only, intact; AC ligaments intact
II	Partial rupture of AC joint ligaments
III	Complete rupture of AC ligaments + coracoclavicular ligament
IV	Superior view → clavicle displaced posteriorly (downward on superior view) *[C2 · transcript: "type IV clavicle moves downward"]*
V	Clavicle displaced upwards; most common; skin tenting; complete separation *[Q1 · As spoken; type V is not the most common Rockwood type - types I-III are far commoner.]*
VI	Clavicle pushed beneath coracoid process; very rare

- Type VI → **very high-energy trauma**
  - Associated injury → brachial plexus or vessels near mid-shaft clavicle

Mechanism Ask AI

AC joint

- Fall on **point of the shoulder**
- **Direct blow** → lateral aspect of joint
- Famous Egyptian case → **Mohamed Salah**, before last World Cup

Imaging Ask AI

AC joint

- **X-ray** → sometimes only minimal displacement
  - ~ Half width of lateral end of clavicle
- Type V → frank appearance; most common
- **CT** → clearer picture
- **Coracoclavicular distance** → average 6–11 mm *[Q2 · As spoken; conflicts with the 11–13 mm normal coracoclavicular distance given later in the same lecture.]*
  - Upper border of coracoid → undersurface of clavicle

Differential Diagnosis Ask AI

AC joint

- Lateral clavicle **tenting** → not always AC dislocation
  - **Coracoid fracture** → especially base of coracoid
    - Radiographic overlap → hidden until further imaging
    - Clue → **coracoclavicular distance** constant, 11–13 mm
  - **Distal (lateral) clavicle fracture** → clavicle rides upwards
    - Ligaments → intact
  - **Physeal separation** → young athletes 14–15 years
    - Lateral physis → not yet ossified
    - Operated case → mimicked AC dislocation

Pitfall

- Coracoid base fracture → mimics AC dislocation; overlap hides it
- Distal clavicle fracture → mimics AC dislocation; ligaments intact
- Lateral physeal separation at 14–15 years → mimics AC dislocation

Acromioclavicular Dislocation — Management Ask AI

Strategy Ask AI

AC joint

- **Ligament reconstruction** → mostly chronic AC dislocation
- **Fixation** → mostly acute
- Reconstruction grafts in chronic cases
  - Semitendinosus
  - Gracilis
  - **Split-thickness tibialis posterior** → widely used abroad
    - More common, more available graft

Modified Weaver–Dunn Ask AI

- **Resect distal (lateral) end of clavicle**
- Intact **coracoacromial ligament** → reattached into distal clavicle from inside
  - → Recreates coracoclavicular ligament

AC joint

Fixation Options Ask AI

- Sutures or **nylon tapes**
- **Hook plate**
- **Coracoclavicular screw**
  - = **Bosworth screw**
- **Cortical flip button** + dog bones
  - Arthroscopic; 2–3 cases, Nasr City hospital
- **K-wires** → least stability for AC joint

Modified Weaver–Dunn — Own Case Ask AI

- **Kocher** under coracoacromial ligament → thick, very clear
- Resect lateral clavicle → **two drill holes** in lateral end
- Ligament held with sutures → pulled into lateral clavicle space
  - → Some stability
- Somewhat **old technique**

Coracoclavicular Reconstruction with Tendon Graft Ask AI

- New technique → **expose coracoid process**
- **Drill holes** → 2.5 cm and 4.5 cm from lateral end of clavicle
  - Graft passed through
- Graft bridged over lateral end of acromion → **sutured onto itself**
- **Very high stability**; not a big procedure
- Drawback → **semitendinosus harvest** from knee
- Stability results → very good

Fixation Methods — Details Ask AI

- **Bosworth screw** → lateral clavicle into coracoid
  - Old; some still use it
  - High **incidence of fracture**
- **Hook plate**
  - Some Egyptian designs → protrude downward → **subacromial impingement**, pain
  - **Remove after 3 months** → coracoacromial ligament healed
- **Arthroscopic technique** → device passed under coracoid and over clavicle → tied
- **K-wires** → least effective
  - Danger → **migration** from inside clavicle
    - → Dangerous structures, or toward lung

Arthroscopic Fixation — Case Ask AI

- Filmed case, Nasr City → **dog bone** + flip button
- Under undersurface of coracoid
- On upper surface of clavicle

Horizontal Instability Ask AI

- Problem → not only **vertical** (superior–inferior translation)
- Also **anteroposterior (horizontal) translation**
- **Imhoff** → figure-of-eight between lateral clavicle and medial acromion
- Arthroscopic dog-bone fixation + figure of eight → **horizontal plane stability**

Take-Home Message Ask AI

AC joint

- **Coracoid fractures** + lateral clavicle fractures → mimic AC dislocation
- Many **modalities** for fixation and reconstruction
  - Use most familiar technique, with fewest complications
- **Arthroscopy** → new trend for fixation (reportedly in type II) *[Q3 · As spoken; type II AC injuries are normally treated conservatively; arthroscopic fixation targets types III-V.]*
  - Long term → unknown; literature support over last 7–8 years
  - **Minimally invasive**; little effort; very high stability

Sternoclavicular Dislocation Ask AI

- Uncommon; overlooked area → ER **reduction** knowledge essential

Anatomy Ask AI

SC joint

- Medial end of clavicle + **manubrium sterni**
- **Four stabilizing ligaments**
  - Anterior sternoclavicular
  - Posterior sternoclavicular
  - Interclavicular
  - Costoclavicular

Danger of Surgery Ask AI

SC joint

- No surgery without **cardiothoracic surgeon**
- Behind plates and screws → **subclavian vessels** (artery, vein), close to aorta
- High-risk structures → **vascular or cardiothoracic surgeon** on standby
  - → Rapid rescue if complication

Stability Ask AI

SC joint

- **Only true articulation** → upper limb ↔ axial skeleton
  - Scapulothoracic + other shoulder articulations → not true articulations
- Secured by **capsule** + four ligaments
  - Anterior and posterior sternoclavicular, interclavicular, costoclavicular

Mechanism Ask AI

SC joint

- **High-energy trauma**
  - Motor vehicle accident
  - **Contact sports** → rugby; sometimes football
- **Direct force** → direct strike to chest (steering wheel)
- Side tackle at SC joint → **posterolateral force** → posterior dislocation
- **Anterior dislocation** → more common
- **Posterior dislocation** → life-threatening
  - Compresses trachea, lung, oesophagus, vessels behind
  - Often asked to **reduce** it

Classification Ask AI

SC joint

- **Subluxation** vs frank dislocation
  - Rheumatoid or **spontaneous osteolysis** of medial clavicle → prominence over SC joint
- Anterior vs posterior
- **Traumatic** vs atraumatic
  - Atraumatic → spontaneous osteolysis of medial clavicle; rheumatoid spontaneous dislocation
- **Chronicity** → acute, chronic, recurrent, congenital

Imaging Ask AI

SC joint

- **CT with 3D reconstruction** → anterior vs posterior
- **Serendipity view** → targeted radiograph, cephalic tilt 40–50°
  - Film behind patient's neck; 40° inclination
- **CT** → investigation of choice
- **MRI** → physeal injury or not

Medial Clavicular Physis Ask AI

- Appears at 16–18 years → **last physis to appear** in body
- **Ossifies** at 25–28 years; recorded up to 30 years
- → Often **physeal injury**, not frank dislocation

High Yield

- Anterior SC dislocation → more common
- Posterior SC dislocation → life-threatening: trachea, lung, oesophagus, vessels
- CT → investigation of choice; MRI → physeal injury
- Medial clavicular physis → last to appear (16–18 years); ossifies 25–28 years

Acute Management Ask AI

SC joint

  - Acute, especially **posterior** emergency → reduction clamp (towel clip)
  - Hand firmly on bone → vessels very close behind
  - Clamp never inserted hazardously
  - Pull **posterior** → **anterior**
- Difficult, **risky** manoeuvre
  - Needed for compression → mediastinum, trachea, vessels, oesophagus

Chronic Management Ask AI

SC joint

- **Avoid plates and screws**
  - Seen → plates out through skin with **infection**; area covered by skin only
  - → Any **hardware** difficult here
- Common → **drilling** + synthetic tape or tendon harvest
  - Passed as **figure of eight** → tightened
- Alternative → **anchor**; sutures through medial clavicle → very good stability

Glenohumeral Joint — Stability and Instability Ask AI

Overview Ask AI

GH joint

- **Glenohumeral joint** → most commonly dislocated joint
- Nearly only joint with full **range of motion**
  - Highest range of motion → most exposed to dislocation
- Directions
  - Anterior
  - Posterior
  - **Inferior** → very rare

Articulations Ask AI

GH joint

- Shoulder → **four articulations**
  - Acromioclavicular
  - Sternoclavicular
  - Glenohumeral
  - Scapulothoracic
- Highest range of motion → at expense of **stability**
  - Large head on small socket
  - **Distensible** glenohumeral capsule

Stabilizers Ask AI

GH joint

- **Dynamic stabilizers**
  - Deltoid
  - Rotator cuff
  - Biceps tendon
- **Static stabilizers**
  - Bony articulation → humeral head on glenoid
  - **Glenoid labrum** → deepens joint
  - Glenohumeral capsule
  - **Glenohumeral ligaments** → most well defined
    - Middle glenohumeral
    - Inferior glenohumeral
- Arthroscopic view → **middle glenohumeral ligament** well developed, sometimes cord-like

Glenoid Labrum Ask AI

- Target of many arthroscopic stabilization operations
- ~ 20% of **glenohumeral stability**
- **Fibrocartilaginous tissue** → deepens glenoid ~ 50%
- Functions
  - Increases **contact surface area**
  - **Buttress**
  - Attachment → glenohumeral ligaments (probable)

Micro-instability vs Impingement Ask AI

- **Shoulder instability** → wide spectrum, starting from micro-instability
- "Shoulder slips out and back in" → **micro-instability**
- Often misdiagnosed as **impingement** → subacromial decompression at 25–30 years
  - Problem → micro-instability, not subacromial impingement
- **Subacromial impingement** → not before 38–40 years
  - Heavy workers
  - Frank **AC joint osteophytes**
  - Narrowed subacromial space
- 25-year-old, slipping or pain → MRI "AC arthritis" / "impingement" → decompression
  - → **Deteriorates** terribly; keeps returning

Types of Glenohumeral Dislocation Ask AI

- **Posterior dislocation**
  - **Most missed** → by us and residents; radiology rarely explained
  - In doubt → **CT**
  - **Electric shock** or convulsions → usually posterior
- **Anterior dislocation** → 97%; more common
  - **Subcoracoid** → beneath coracoid process
- **Inferior dislocation** → very rare
  - One case seen, Kasr Al-Ainy; two cases in literature

Classification of Instability Ask AI

- **Traumatic structural** → articular surface + glenoid labrum
  - → **Bankart lesion**
- **Atraumatic structural** → no trauma; articular surface structure affected
- **Habitual non-structural** → **muscle imbalance**
  - **Scapular dyskinesia** → periscapular imbalance → shoulder moves forwards or backwards

Comparison of Instability Types Ask AI

- Traumatic vs atraumatic vs habitual

Feature	Traumatic	Atraumatic	Habitual
Trauma	Significant	None	None
Articular surface	Bankart lesion	Damaged	No problem
Capsule	—	Stretched, deformed, dysfunctioning	Element of dysfunction
Laterality	Usually unilateral	Not uncommonly bilateral	Usually bilateral
Muscle pattern	Normal	Normal	Abnormal patterning

- Habitual → push against wall → **winging of scapula**
  - Characteristic scapular appearance on back examination

Anterior Glenohumeral Dislocation Ask AI

Frequency Ask AI

Anterior

- **Anterior** → 97%
- Posterior → 3%
- Inferior and superior → very rare
- Image → **inferior dislocation**
- Image → **anterior dislocation**, most common

Mechanism Ask AI

Anterior

- **Abduction + external rotation** → anterior dislocation (sport positions)
- Head leaves glenoid → only labrum in front → **Bankart lesion**
- Repeated dislocation → head engages anterior bony glenoid edge → **Hill–Sachs lesion**
- Engaging Hill–Sachs + glenoid Bankart
  - → **Bipolar lesion**

Labral Lesions Ask AI

- Labral lesion types

Lesion	Features
Bankart	Complete tear and separation of labrum
Osseous (bony) Bankart	Labrum separates with bone piece from anterior glenoid
Reverse Bankart	Posterior dislocation
Perthes	Periosteal sleeve avulsion; delicate; periosteal stripping; labrum stays on glenoid edge
ALPSA	Anterior labroligamentous periosteal sleeve avulsion; labrum displaced onto medial glenoid neck
GLAD	Separation of part of glenoid articular surface + anterior labral detachment

- Image → **anteroinferior separation**

Location of the Bankart Lesion Ask AI

Bankart

- **Glenoid equator** → line joining anterior and posterior
  - Bankart → never above it
- Misconception → any separation = Bankart
- **Bankart lesion** → only below equator
  - Right shoulder → 3 to 6 o'clock
  - Left shoulder → 6 to 9 o'clock *[C3 · transcript: "left shoulder 7 to 9 o'clock"]*
- Above equator → **not Bankart**
- Anteroinferior head dislocation → takes **labrum** along

Reverse Bankart, SLAP and Tackler Injury Ask AI

- **Reverse Bankart** → posterior dislocation; head moves backwards
- Some patients → **SLAP tear** too
- **Tackler injury** → Bankart + SLAP + reverse Bankart
- ~ 270° **tear** arthroscopically; seen once
  - **Air force pilot** → ejected from crashing aircraft in training
  - Parachute caught on aircraft tail → arm pulled
  - Fixed → **SLAP**, posterior labral tear, anteroinferior Bankart

Normal Labral Variants Ask AI

  - Normal variants → **not lesions**
  - Labrum detached 1–3 o'clock → **sublabral foramen** (probable)
  - **Sublabral hole** (probable) ± cord-like middle glenohumeral ligament
  - Absent labrum + cord-like middle glenohumeral ligament → **Buford complex**
- Three variants → very common
  - Often **mistaken for labral detachment** → anchors wrongly placed
- **Never anchor above 3 o'clock** or above equator → unless SLAP lesion

Must-Know

- Bankart → only below equator: 3–6 o'clock right, 6–9 o'clock left
- Sublabral foramen, sublabral hole, Buford complex → normal variants
- Never anchor above 3 o'clock unless SLAP lesion

Reduction Manoeuvres Ask AI

- Standard ER manoeuvres → known by all residents
  - **Hippocratic method** → foot in axilla + pull; when alone
  - **Traction** or traction–counter-traction on arm

Arthroscopic Bankart Repair vs Latarjet Ask AI

- Torn labrum → **arthroscopic Bankart repair**
- 17–18 dislocations → frightening clinic numbers
  - 16–17 dislocations → usually **epileptic** → Bankart repair usually fails
- → **Latarjet** → bony procedure
  - **Coracoid osteotomy**
  - Fixed **below the equator**, anteroinferiorly → two screws
  - → Increases arc, reduces engagement, prevents **re-dislocation**
  - Exception → ongoing **fits** → nothing holds glenohumeral joint

Arthroscopic Case Ask AI

Bankart

- Engaging **Hill–Sachs** on glenoid edge → debrided
  - **Remplissage** → capsulotenodesis of infraspinatus into defect
- Anchor placed → sutures retrieved through cuff → look **anteriorly**
- Torn labrum, completely displaced → **shaver** freshens glenoid surface
- Anchors placed → **bird beak** through labrum → sutures retrieved
- **Knots** tied → final repair

Posterior Glenohumeral Dislocation Ask AI

Mechanism and Features Ask AI

Posterior

- Fall on shoulder in **adduction + internal rotation** → posterior dislocation
- **More commonly missed** → ER, residents
- → **Reverse Bankart tear**
- **Less common** than anterior → ~ 3–5%
- Missed in > 50% on **standard AP view** → without true AP
- History → **electric shock**, seizures, alcoholism, car accident
- Associated lesions
  - **Reverse Bankart lesion**
  - **Reverse Hill–Sachs**
  - **Lesser tuberosity** fracture
  - **Posterior HAGL** lesion
  - **POLPSA** → details skipped
  - Films → all posterior dislocations; not certain on these views

Feature	Anterior	Posterior
Frequency	97%	3–5%
Position	Abduction + external rotation	Adduction + internal rotation
Labral lesion	Bankart	Reverse Bankart
Humeral lesion	Hill–Sachs	Reverse Hill–Sachs

Radiological Signs Ask AI

Posterior

- **Light bulb sign** → head internally rotated; tuberosity not seen
- **Trough line sign** → two parallel overlapping cortical lines
  - Humeral head edge + glenoid edge; head behind
  - → Posterior engagement → **reverse Hill–Sachs**
- Suspicion → **true AP** (head position) or CT

Pitfall

- Posterior dislocation → most missed; > 50% on standard AP
- Electric shock or seizures → suspect posterior
- Light bulb sign; trough line sign
- In doubt → true AP or CT

Neglected Posterior Dislocation Ask AI

- Patients → **very miserable**
  - Health Insurance hospital, Nasr City → **bilateral posterior dislocation** → hemiarthroplasty (believed)
    - Young man, ~ 39 years
  - Often **missed 4–5 months** → defect enlarges → reduction difficult
- Options
  - **Derotation osteotomy**
  - **McLaughlin procedure** → subscapularis into defect
    - = Posterior counterpart of remplissage
- Image → coracoid anterior, acromion posterior; head against **posterior glenoid**
- Suspicion → **CT**

Management of Posterior Instability Ask AI

- **Soft tissue injuries** → arthroscopic; view from anterior, target posterior repair
  - **Capsular shift**
  - **Glenoid labral repair**
- **Bony injuries** → not common
  - **McLaughlin procedure** → subscapularis transfer into reverse Hill–Sachs defect
  - **Glenoid osteotomy** → corrects retroversion
  - **Posterior bone block** (posterior Latarjet) → same steps as anterior

Tackler Injury Ask AI

Posterior

- Rugby; more commonly **American football**
- Arm caught in difficult position → 3 injuries together
  - **SLAP lesion**
  - **Anterior Bankart lesion**
  - **Reverse Bankart lesion**

Multidirectional Instability Ask AI

- Dislocates anteriorly, posteriorly, inferiorly → always big problem
- Best solution → **not to operate**
- If operating → **whole job**
  - **Anterior Bankart repair**
  - Close **rotator interval** → upper subscapularis ↔ supraspinatus
  - Posterior **repair**
  - **Capsular plication**
- Anterior repair only → **posterior dislocation**
- **Lafosse** case → famous French shoulder surgeon
  - Anterior bone block → dislocated posteriorly
  - Posterior bone block → dislocated inferiorly
  - → Same problem easy for any shoulder surgeon
- Australian national team **swimmer** → extreme position; not Photoshop
- Very high **shoulder flexibility** → excel at gymnastics, swimming
- Very **liable** to injury

Superior Shoulder Suspensory Complex (SSSC) Ask AI

Definition Ask AI

SSSC

- Rarely discussed → **MD exam** and oral exam question
- **Soft tissue and bony ring** → between superior and inferior struts
  - **Superior bony strut** → middle of clavicle
  - **Inferior bony strut** → lateral acromion + spine of scapula
- Anterior view → **complete circle**
  - Conoid + trapezoid ligaments
  - Acromioclavicular ligaments
  - Coracoacromial ligaments
  - Acromion + glenoid fossa
- Components
  - Glenoid fossa
  - Coracoid process
  - Coracoacromial ligaments
  - Lateral end of clavicle
  - Acromioclavicular joint
  - Acromion process
- Clearer diagram of ring

Three Units Ask AI

SSSC

- **Three units**
  - Lateral end of clavicle + acromion
  - **C-4 linkage** → clavicle, coracoid, coracoclavicular ligaments
  - **3 scapular processes** → glenoid fossa, coracoid, acromion

Disruption Patterns Ask AI

SSSC

- Single vs double vs triple/quadruple

Disruption	Examples	Stability	Treatment
Single	Clavicle fracture, acromion fracture or AC dislocation alone; coracoclavicular ligaments intact	Usually stable	Conservative
Double	Two components: lateral clavicle, acromion, glenoid neck or coracoid fracture, ± AC dislocation	Unstable	Surgical repair
Triple / quadruple	High-energy trauma	Completely unstable	—

- Triple/quadruple → associated **rib, spine, nerve, head injuries**
- **Comatose** patient + multiple rib fractures + pneumo/haemothorax → examine shoulder on that side
  - Otherwise → diagnosis very delayed after waking

Floating Shoulder Ask AI

- **Glenoid neck fracture** → 2 fragments
  - Distal → glenoid fossa + coracoid + attached ligaments
  - Proximal → acromion, scapular spine, rest of scapular body
- Double disruption → **floating shoulder**
- Conditions → shoulder separated from **axial skeleton**
  - Distal ↔ proximal fragment via **coracoacromial ligament** → must be disrupted
  - Distal fragment ↔ axial skeleton via **coracoclavicular ligaments**
  - Both torn + clavicular fracture + **scapular neck fracture** → held by skin and muscles only
    - → Complete separation from axial skeleton possible
- Cause → **very high-energy trauma**

Shoulder Dropping Deformity Ask AI

- Distal fragment → displaced **anteriorly, medially, inferiorly**
  - Muscle pull + weight of upper extremity
- → **Shoulder dropping deformity**
  - **Brachial plexus** pressure
  - Reduced range of motion
  - Disturbed glenohumeral–acromion balance → subsequent **impingement**
  - Reduced **lever arm** of rotator cuff → weak abduction; subacromial impingement
- Seen in floating shoulder / double disruption

Radiological Views Ask AI

SSSC

- Views for these patients

View	Use
True AP	Reveals posterior dislocation
Scapular Y (lateral scapular)	Head off centre → dislocation; one side anterior, other posterior
Velpeau / axillary	Correctly taken → easy anterior vs posterior dislocation

Ring Concept Ask AI

SSSC

- Like **pelvis** → closed ring
  - High-energy disruption in one place → look for injury elsewhere
- SSSC = **circle**
  - Single disruption with many fragments, comminution → seek second lesion in ring
  - Case → glenoid, scapular spine, acromion fractured + **clavicle** fracture
  - Fixed → clavicle, acromion with body, scapular body

Treatment Algorithm Ask AI

SSSC

- Displacement < 5 mm, no caudal glenoid displacement, angulation < 20° → **conservative**
- Otherwise → **operative**
- Operative → **fix clavicle first**
  - Clavicle reduction + plate and screws → scapular fracture often **reduced**
  - Scapula reduced → accept
  - Scapula unreduced → **scapular fixation**

Exam Pearl

- SSSC → bony–soft tissue ring; superior strut mid-clavicle, inferior strut lateral acromion + scapular spine
- Single disruption → stable → conservative
- Double disruption → unstable → floating shoulder → surgery
- Operative → fix clavicle first; scapula often reduces

Scapular Fractures Ask AI

Scapula

Simplified Classification Ask AI

- Exam → master's, oral or operative committee; often unanswered
  - Simple scheme

Type	Site
A	Body
B or C	Glenoid
D	Scapular neck
E	Acromion
F	Scapular spine
—	Coracoid → very rare, but occurs

- **OTA and AO** → scapula in three parts
  - Body
  - Glenoid
  - Processes → acromion, coracoid

OTA / AO Classification Ask AI

- OTA → Orthopaedic Trauma Association; **AO** → perhaps most famous
- Regions → body, glenoid alone, processes (coracoid, acromion)
  - **OTA classification** → very close to AO

Segment	Code	Pattern
Articular (glenoid)	—	Simple → rim only (anterior or posterior); not into joint
Articular (glenoid)	—	Multifragmentary
Body	B1	Simple → 2 or fewer body fractures
Body	B2	Complex → 3 or 4 body fractures
Process	P1	Coracoid
Process	P2	Acromion
Process	P3	Coracoid + acromion together

Articular Involvement Ask AI

Scapula

- **Type A — extra-articular** → acromion, coracoid or body; glenoid spared
- **Type B — partial articular** → anterior, posterior or inferior rim
  - Anteroinferior part → **bony Bankart** in shoulder dislocation
- **Type C — total articular**
  - Intra-articular with neck
  - Intra-articular with body
  - Extra-articular through **glenoid neck** → whole glenoid fossa drops

Mechanism and Associated Injuries Ask AI

Scapula

- **High-energy trauma**; usually associated injuries
  - **Pulmonary contusions**
  - Haemothorax or pneumothorax
  - Clavicle fracture
  - Shoulder dislocation
  - Multiple rib fractures

Imaging Ask AI

Scapula

- Views
  - **True AP**
  - Scapular lateral
  - True axillary → helpful if obtainable
- Case
  - Large **intra-articular** anteroinferior glenoid fragment
  - Body fragment → lateral scapular view
  - Also **AC dislocation**

Management Ask AI

Scapula

- Majority → **conservative**, unless intra-articular
  - **Broad arm sling** ~ 6 weeks
  - Then pendular exercises + mobilization
- Glenoid involvement or **SSSC injury** → fixation
  - Whole → body, scapular spine, or acromion + clavicle
  - Articular part only → **ORIF** when glenoid fractured with humeral head
- Uncommon; severe accidents → two operated cases
  - Fell from **train** → dragged
  - Car overturned ~ 6–7 times

Arthroscopic Fixation of Glenoid Fractures Ask AI

- **Intra-articular glenoid fracture** → arthroscopic fixation possible
  - Anterior arthroscopic view → inferior + superior fragments
  - **Guide wire** → screw over it → articular surface closes on tightening
- Big risk → **haematoma** limits view
  - Shoulder swells, especially with **fluid**
- Good results → **articular surface** confirmed arthroscopically; satisfactory glenoid position

Proximal Humeral Fractures — Classification and Vascularity Ask AI

Humerus

Introduction Ask AI

- Typical questions → classification, one-/two-part, **Neer classification**, fixation
- Dilemma → hemiarthroplasty, ORIF, K-wires or **PHILOS plate**

Neer Classification Ask AI

- **Four segments**
  - Articular segment → number one
  - **Greater tuberosity** → pulled posteriorly
  - Lesser tuberosity → medial
  - Shaft
- **Displacement** > 1 cm or angulation > 45°
  - → Justifies operating
- Permutations
  - One-part → four
  - Two-part → four
  - Three-part → two
  - Four-part → one
- ± **Fracture–dislocation** → anterior or posterior
- **Head splitting** → articular fragment in 2–3 pieces; much more difficult
- Head superiorly displaced → **impression fracture** (like Hill–Sachs / reverse Hill–Sachs)
  - Images → one-part fracture; Neer classification forms

Hertel's Criteria Ask AI

- Three-/four-part → replacement, hemi, PHILOS/locked plate, or K-wires?
- Key → **vascularity** (ischaemia) of humeral head fragment
- **Hertel** → 2004, JBJS → vascularity depends on three things *[Q4 · As spoken; Hertel's 2004 ischaemia predictors paper appeared in J Shoulder Elbow Surg, not JBJS.]*
- **Metaphyseal extension** of head
  - 8–9 mm or more → viable, vascularized
  - ~ 1 cm extension from metaphysis → good vascularity
- **Medial (calcar) hinge**
  - Intact, undisplaced → supports vascularity → PHILOS successful
  - **Comminuted** → collapse + failure
- **Fracture–dislocation** → poor outcome
- Fourth criterion → **osteoporosis** → more likely AVN
  - → **Failure of fixation** → prefer shoulder hemiarthroplasty

High Yield

- Hertel (2004, JBJS) → head vascularity in 3-/4-part fractures
- Metaphyseal extension 8–9 mm or more → viable head
- Intact medial calcar hinge → PHILOS can succeed
- Comminuted calcar, fracture–dislocation, osteoporosis → collapse / AVN

Blood Supply of the Humeral Head Ask AI

- **Circumflex vessels** → wrap around head
- No benefit from PHILOS or any fixation
  - Fracture–dislocation
  - Comminution without **stable calcar**
  - No metaphyseal fragment

Proximal Humeral Fractures — Fixation and Arthroplasty Ask AI

Calcar Screw Ask AI

- **Calcar screw** → supports head against collapse
- Short or absent → head **collapse**, screw migration
- On plate → fifth screw
- **Humeral head axis** → not 90°; angle adjusted
- Short or absent → collapse → **fixation failure**
- Combine with **Hertel's criteria** → union or not

Rotator Cuff Sutures with Plate Ask AI

- Sutures → subscapularis, supraspinatus, infraspinatus → through **PHILOS plate holes**
  - Restored **soft tissue envelope** → improves vascularity
  - → Construct **stability**
- Cuff not reattached → union without **function**

Intramedullary Nail Ask AI

Humerus

- **Expert nail** → new
  - Screws in multiple directions
  - Includes **calcar screw** → supports head
  - Nice PHILOS fixation without calcar screw → collapse; all ruined
- = **Third-generation intramedullary proximal humeral interlocking nail**
  - Greater tuberosity fragment (displaced backward) → fixed from behind with screw alone
  - Two other directions → stability
  - Calcar screw → some companies; very high cost
- Calcar screw → main **varus** prevention
  - Like **lag screw** of proximal femoral nail → prevents varus angulation + head collapse
  - Position → as **close to articular surface** as possible, e.g. 5 mm (as DHS / PFN lag)

Must-Know

- Calcar screw → supports humeral head; prevents varus + collapse
- Short or absent → collapse, migration, fixation failure
- Place close to articular surface (e.g. 5 mm), like DHS lag screw

Failure and Arthroplasty Options Ask AI

- PHILOS plate failure → revised to **shoulder hemiarthroplasty**
- **Reverse shoulder arthroplasty** in trauma → expanding; very good results
  - Especially young patients needing long-term arm use
  - Price → very big difference

Implant	Cost in Egypt
Reverse	~ 75,000 EGP
Shoulder hemiarthroplasty	~ 20,000

- Over 40–45 years → **reverse** best → much higher function *[Q5 · As spoken; reverse arthroplasty for fracture is generally reserved for elderly patients (about 65-70+), not 40-45.]*
- Hemiarthroplasty + tuberosity sutures → X-ray at 1–2 years
  - **Greater and lesser tuberosities** disappeared
- 45 years, **highly comminuted**, poor vascularity by Hertel → reverse

Metallic Cage Device Ask AI

- New; like **DHS plate** → metallic cage into humeral head
  - For head with expected collapse
  - Screwdriver turned → cage **distends** inside head → prevents collapse
  - Includes **calcar screw**

Calcar Principle Ask AI

- **Hold the calcar** → basis of Expert nail, metal cage plate, PHILOS plate

K-wires Ask AI

Humerus

- **Least efficient** fixation
  - 10–15 minute procedure → seen as easy
  - Usually **fails** → displacement → second procedure

Decision Summary Ask AI

Humerus

- Choice of implant

Situation	Choice
Hertel criteria → good vascularity	Proximal PHILOS plate + calcar screw; good angulation
Criteria not fulfilled	Shoulder hemiarthroplasty
Funds available	Reverse shoulder

Differential Diagnosis of the Painful Shoulder Ask AI

- Exam topic → master's, MD; beyond trauma scope
  - Classified by **level**

Level	Conditions
Bone	Infection (septic shoulder, proximal humerus osteomyelitis); tumour (benign, malignant); avascular necrosis, may be spontaneous, no trauma needed; fractures
Joint — glenohumeral	Osteoarthritis, rheumatoid arthritis, TB
Joint — AC	Osteoarthritis, subacromial impingement
Joint — dislocation	Acromioclavicular, sternoclavicular, glenohumeral
Tendon — rotator cuff	Subacromial impingement, rotator cuff tears
Tendon — biceps	Biceps tendinitis, subluxation, rupture
Capsule	Frozen shoulder (adhesive capsulitis)
Glenohumeral instability	Bankart, Perthes
Nerve entrapment	Spinal accessory, suprascapular
Referred pain	Cervical spine

- Instability → usually **not painful**
  - Exception → dislocation or recurrent episodes

Questions and Answers Ask AI

Micro-instability Misdiagnosed as Impingement Ask AI

- Q → young patients with pain labelled impingement, actually instability?
- Typical patient → **shoulder pain** in clinic
  - Age 25, 27, 30, 35 years
  - No **heavy work**
  - Pain raising hand, or on training and jumping
  - No frank history except mild laxity
  - "Slips out and back", "dislocates for a moment", "clicks back"
- → **Micro-instability** → redundant capsule
  - Or **multi-ligamentous laxity** → head moves very easily
- → **Not impingement**
- MRI templates in radiology centres (esp. residents) → "subacromial impingement"
- **Injection** (shoulder or subacromial space) → no improvement
- **Subacromial decompression** / coracoacromial release → deterioration
  - Coracoacromial ligament → prevents head migration or subluxation
- Careful history in 24-year-old with severe pain
  - **Impingement** → not before 38–40 years
  - Heavy manual workers → sledgehammer, carpenter
  - **Critical shoulder angle** of subacromial space
    - Explained later with subacromial impingement + rotator cuff
    - Measured on X-ray → subacromial decompression not indicated
- Subacromial decompression → done less now; easy arthroscopically; overused for income
- These patients → **deteriorate violently** after first operation
  - Next surgeon → unsure what to do
  - First operation not done conscientiously → easier for next surgeon

Pitfall

- Young (25–35 years) "slipping" shoulder → micro-instability, not impingement
- Subacromial impingement → not before 38–40 years
- Injection fails; decompression or coracoacromial release → deterioration

Treatment of Micro-instability Ask AI

- **Capsular shift** or plication → according to direction of pain
- **Internal impingement** → swimmers, water polo, volleyball
  - Often undiagnosed → arthroscopy without benefit
- First → **physiotherapy** → strengthen anterior + posterior muscles
  - Exclude **scapular dyskinesia**, habitual muscle patterning
- No improvement with skilled physiotherapist (muscle balance) → arthroscopic **capsular plication**

Physiotherapy for Overhead Athletes Ask AI

- Q → tennis, swimming, handball, volleyball, water polo → physiotherapy instructions?
- Internal impingement → **capsular stretches** (next lecture)
  - **Posterior capsular stretch** → hold opposite elbow with hand
  - Swimmers → very good result
  - Frank **scapular dyskinesia** → expert physiotherapist → rebalance around shoulder
- End of session

MRI of the Shoulder

(Principles of MRI and Normal and Pathological Shoulder Anatomy)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 42) “parallel cuts give sagittal / coronal views” → “parallel cuts give coronal views”

Queried - kept as spoken, but looks wrong

Q1 (p. 53) As spoken; type percentages total 104% (50+40+11+3), and Bigliani hooked type is usually about 39%

Q2 (p. 61) As spoken; internal impingement cysts usually sit posterosuperior / posterolateral humeral head (greater tuberosity), not posteromedial

Q3 (p. 61) As spoken; in a Perthes lesion the periosteum is normally intact but stripped medially from glenoid neck

Introduction to MRI Ask AI

Lecture Scope Ask AI

- Topic: **MRI of the shoulder**, graded difficulty
  - What MRI is; meaning of T1 and T2
  - Why tissues appear white or black
- Water: black on T1, white on T2 → low vs **high signal**
- Many labelled MRI images; approximately 120 slides
- Mid-lecture break for questions on first part

Types of MRI Machines Ask AI

- Shoulder → Open or Closed MRI
- **Closed MRI**: old model
  - Whole body enters machine; coffin-like; very loud
  - Problem → **claustrophobia** (fear of closed spaces) in many patients
- **Open MRI**: invented for this reason
  - Vast difference: fear, suffocation, whole-coil entry
- **Extremity MRI**: small machine; patient stays outside
  - Elbow, forearm, wrist, knee, ankle
  - Not for hip or shoulder → joint must lie inside with rest of body
- **Dynamic MRI** = weight-bearing MRI
  - Patient stands or sits
  - Spine, hips, knees, ankle evaluated standing
  - Weight-bearing evaluation needed → especially knee, hip, spine
  - Spine example:
    - Small disc on conventional MRI
    - Weight bearing → disc size increases; **spondylolisthesis** may appear
    - Problem hidden until this MRI done

Advantages of MRI Ask AI

- **Multiplanar** imaging
- No bone artifacts, unlike CT
- **Specific tissue characterization** by signal
- **Superior soft-tissue contrast** → unmatched by other modalities

Safety: Missile Effect Ask AI

- **Missile effect**: very large magnet attracts any metal object
- Patients who cannot enter MRI:
  - **Pacemakers**
  - **Cochlear implants**
  - Foreign body in eye; bullet in spine
- Very high magnet power → harm through metal objects
- Images: patient chair and bed pulled into magnet

MRI Signal: T1 and T2 Ask AI

Physical Basis Ask AI

- MRI = **Magnetic Resonance Imaging**
  - Magnetic field around body + motion of hydrogen ions
  - Body mostly water → 2 hydrogen ions + 1 oxygen ion
  - Images **mobile hydrogen ions**: magnetic field → oscillations → captured in imaging field
- Dots collected → image
- **T1** = quick image
  - Water hydrogen ions move slowly → not captured → water black
  - Anything not captured on T1 → black
- **T2** = prolonged imaging time
  - Water hydrogen ions now captured → water white
- Captured hydrogen ions → white
- Low signal = black; high signal = white

Must-Know

- Water: black on T1, white on T2
- Low signal = black; **high signal** = white

Categories of Hydrogen Ions Ask AI

- **Moderately mobile**: seen on both T1 and T2
- **Slowly mobile** (water): black on T1, white on T2
- **Quickly mobile**: black; not captured on T1 or T2
  - Example: blood flowing inside vessels
  - Solution → **flash technique**: very rapid imaging
    - Captures blood driven at high speed by heart
  - Ordinary MRI: intravascular blood not seen
  - Extravascular blood seen: **hematoma**, intracranial hemorrhage, hemarthrosis, effusions
    - High signal (white)
- **Non-mobile**: no movement in magnetic field; three key sites
  - **Cortical bone**: always black, T1 and T2
  - **Ligaments**: anterior cruciate, posterior cruciate, collateral
    - All black → non-mobile hydrogen ions
  - **Calcification**: physiological or pathological → black
    - Calcific tendinitis of shoulder
    - Chondrosarcoma calcification (pathological)
- **Absent**: no hydrogen ions → black; air
  - Lungs; paranasal sinuses
- Summary:

Hydrogen ions	T1	T2	Example
Moderately mobile	Seen	Seen	—

Hydrogen ions	T1	T2	Example
Slowly mobile	Black	White	Water
Quickly mobile	Black	Black	Blood flow inside normal vessels
Non-mobile	Black	Black	Bone cortices, ligaments, tendons
Absent	Black	Black	Air: paranasal sinuses, lung

- Image requires adequate number of **mobile hydrogen ions**
  - Mobile ions → contrast against black points (fixed / non-mobile ions)

Gray Scale Ask AI

- Adequate mobile ions → **gray-scale** image, black → white, graded

Appearance	Term
Black / dark	Low signal intensity
Gray	Intermediate signal intensity
White / bright	High signal intensity

Signal Combinations Ask AI

- Each structure low or high on T1 and T2 → four combinations

T1	T2	Tissue
Low	High	Water: normal body water, cerebrospinal fluid, effusion
High	Low	Fat: subcutaneous fat, bone marrow
High	High	Hemorrhage: intracranial, hemothorax, neck, hemarthrosis, clotted blood
Low	Low	Non-mobile hydrogen ions: bone cortices, mature fibrous tissue, calcification, intravascular blood, air

- **Water**, all types → slow hydrogen ions → missed on T1, captured on T2
- **Fat**, all types → opposite of water
- **Hemorrhage** → any blood from bleeding or clotted blood
- Low on both:
  - Bone cortices
  - **Mature fibrous tissue**: ACL, PCL, supraspinatus, infraspinatus
  - Calcification: calcific tendinitis, chondrosarcoma calcification
  - Intravascular blood
    - Speed very high → hydrogen ions not imaged
  - Air: very low hydrogen content → lungs, paranasal sinuses
- Table → understand, not memorize; whole lecture builds on it

Example: Knee MRI Ask AI

Knee

- T1 and T2 examples of knee
- Cortices, including tibia → black on both → non-mobile hydrogen ions
- PCL and **ACL** → black
- Fat: white on T1, black on T2
- Effusion: white on T2, black on T1

Shoulder MRI: Indications and Anatomy Ask AI

Indications Ask AI

- **Five main indications** → any joint: shoulder, knee, ankle
  - Pain
  - Swelling: tumor, fibrous dysplasia, chondrosarcoma
  - Trauma: hemarthrosis; rotator cuff tear; greater tuberosity fracture
  - Inflammation: rheumatoid arthritis, septic arthritis, osteomyelitis
  - Tumors

Glenoid and Surrounding Structures Ask AI

- Acromion posterior; coracoid anterior
  - **Coracoacromial ligament** between them
  - Subacromial bursa below
  - Acromioclavicular joint
- Rotator cuff:
  - Supraspinatus → highest
  - Subscapularis
  - Infraspinatus; teres minor
- Biceps tendon → superior labrum → **biceps anchor**
  - Site of SLAP lesion (superior labrum anterior-posterior)
- **Glenoid labrum**: anterior, posterior, inferior
- **Inferior glenohumeral ligament**:
  - Well-developed anterior band; well-developed posterior band
  - Pouch between → axillary pouch
- **Middle glenohumeral ligament** above
- Superior glenohumeral ligament → usually not seen on MRI

Ligaments and Biceps Course Ask AI

- Biceps tendon course:
  - **Supraglenoid tubercle** → oblique through rotator interval → bicipital groove
  - Groove between lesser and greater tuberosities
  - Transverse humeral ligament passes over it
- Subscapularis attachment (cut in picture)
- Middle glenohumeral ligament
- **Coracohumeral ligament**:
  - Main component in frozen shoulder (adhesive capsulitis)
  - Main component of rotator interval
  - Released arthroscopically in frozen shoulder → location must be known

Anterior Aspect Ask AI

- **Subscapularis**: below coracoid → laterally → lesser tuberosity
  - Extension over biceps tendon → anterior edge of greater tuberosity
- **Supraspinatus**: below coracoacromial ligament → top of greater tuberosity
  - Coracoacromial ligament → cut in impingement syndrome
- **Facet arrangement**: supraspinatus → infraspinatus → teres minor
- Subacromial bursa
- Long head of biceps; short head from coracoid
- Above: acromion, clavicle, acromioclavicular joint

Posterior Aspect Ask AI

- Supraspinatus: **supraspinous fossa** → top of greater tuberosity
- Infraspinatus: broad muscle → laterally → inserts behind supraspinatus
- Teres minor below → most inferior facet
- Facet order on greater tuberosity: supraspinatus → infraspinatus → teres minor

Technique, Planes and Reading Template Ask AI

Positioning and Planes Ask AI

- Patient **supine**; arm in slight external rotation
- Surface coil → more accurate images
- Three planes → three films in patient envelope:
  - Coronal
  - Axial
  - Sagittal
- Each plane shows particular structures → know how each is obtained
- Axial plane = **ice cream cone**: glenoid = cone; humeral head = ice cream
- Obtaining planes:
  - Start from **axial view** → basis for whole shoulder
  - Technician / radiologist sets axial view → draws long axis
  - Parallel cuts, anterior ↔ posterior → coronal views *[C1 · transcript: "parallel cuts give sagittal / coronal views"]*
  - Perpendicular cuts → sagittal view

Normal Appearance Ask AI

- Black: cortices, supraspinatus tendon, **glenoid labrum**
- Muscle → different signal
- T2: bone marrow signal changes; tendon and cortices unchanged
- Glenoid labrum: superior and inferior
- Visible pouch → **effusion**
  - Hemorrhage or intra-articular content
  - Or gadolinium injected for MRA → area clearer

Black on T1 and T2 Ask AI

Shoulder

- **Non-mobile protons** (hydrogen ions):
  - Cortical bone
  - Ligaments: superior and middle glenohumeral
  - Tendons: rotator cuff (supraspinatus), biceps
  - Glenoid labrum
  - Calcification
    - Shown better under calcific tendinitis
- T1 vs T2: labrum, cortices, biceps tendon → black on both

Low T1, High T2 Ask AI

Shoulder

- Water-containing: fluid, effusion
- **Articular cartilage** → high water content → low T1, high T2
- High T1, low T2 → fat

Reading Template Ask AI

- Template in mind: what to look for, best cut for each
  - All cuts show most structures
  - Certain cuts show some clearly → tear judgment
- Items:
  - **Ligaments**: superior, middle, inferior glenohumeral; biceps
  - **Bones**: clavicle, acromion, proximal humerus, glenoid
  - **Tendons**: supraspinatus, infraspinatus, teres minor, subscapularis, biceps
  - **Bursae**: subacromial, subcoracoid, subdeltoid (beneath deltoid)
    - Some authors: subacromial + subdeltoid = one bursa
  - **Labrum**

Bursae Ask AI

- Bursae around joint:
  - Subacromial
  - Subcoracoid → below coracoid process
  - Subscapularis bursa
- **Bursitis** or effusion → confusing on MRI
- Early learning → precise identification not required
- Bursae **communicate with joint** capsule
  - Effusion may extend into them
  - MRA gadolinium enters → white image in this area

Coronal View Ask AI

- Reviewed first → important for several items
- Items on coronal:
  - Ligaments: **inferior glenohumeral ligament**
  - Bone: acromioclavicular joint
  - Tendons: supraspinatus (most commonly affected), infraspinatus
  - Bursae: subacromial
- Images reviewed one by one with interpretation

Landmarks Ask AI

Coronal

- Plain X-ray vs coronal MRI → same view
  - X-ray: coracoid process, lateral clavicle, acromion, humeral head
  - Keep X-ray in mind → visualize coronal MRI
- Scroll coronal cuts → two bones articulating → **acromioclavicular joint**
  - Acromion lateral; clavicle medial
- Below → **supraspinatus muscle** → supraspinatus tendon
- First landmark: AC joint → follow muscle laterally → greater tuberosity insertion
- Below → biceps tendon in groove → **supraglenoid tubercle** → superior labrum
- Down along glenoid → inferior labrum → inferior glenohumeral ligament attached
- **Axillary (circumflex) nerve** and vessels directly below
  - Bankart repair → nerve directly beneath
  - Shoulder arthroplasty → no dissection here unless clearly seen
- Suprascapular nerve and vessels may be visible → identification not required
- Sequence: AC joint → supraspinatus → biceps tendon → superior labrum → inferior labrum → inferior glenohumeral ligament
- Normal inferior glenohumeral ligament → **U shape**
  - Remember for HAGL lesion (humeral avulsion of glenohumeral ligament)
  - HAGL → torn, open → U becomes **J shape**

Exam Pearl

- Normal inferior glenohumeral ligament on coronal → **U shape**
- HAGL → J shape

Supraspinatus versus Infraspinatus Ask AI

- Supraspinatus → look for **AC joint**
  - Clear here
- Infraspinatus → **behind** supraspinatus → move posteriorly
  - Know anterior vs posterior in cuts, and start point
- AC joint → anterior cuts; remaining films posterior
- No AC joint; **spine of scapula** appears, well developed
  - Directly below → infraspinatus tendon → infraspinatus muscle
  - Insertion → greater tuberosity
- Inferior glenohumeral ligament: anterior and **posterior band**
  - Posterior band → same level as infraspinatus
- Review:
  - Anterior: AC joint → supraspinatus tendon and insertion → biceps to superior labrum → anterior band
  - Posterior: spine of scapula, AC joint gone → infraspinatus tendon and muscle → posterior band
- Anterior: AC joint, supraspinatus muscle and tendon, long and short heads of biceps
- Posterior: spine of scapula and its broadening → infraspinatus → posterior band
- Anterior → supraspinatus; posterior → infraspinatus

Subscapularis and the Three Landmarks Ask AI

- Anterior cuts → landmark: **coracoid process**
- Two bones one above other → coracoid below, clavicle above
- **Coracoclavicular ligaments** attached to coracoid
  - Ligaments of AC dislocation
- Below coracoid → subscapularis tendon

Landmark	Tendon below
Acromioclavicular joint	Supraspinatus
Spine of scapula	Infraspinatus
Coracoid process	Subscapularis

- Memorize → clue to tendons → MRI reading
- Strands between clavicle and coracoid → **conoid** and trapezoid
  - Four or five strands
  - Not required while learning; missed with quick cuts
- Below coracoid → subscapularis tendon
  - Black line → biceps tendon in groove, under **subscapularis muscle** cover
- Coracoid fades → moving posteriorly → AC joint appears
  - Subscapularis tendon → inserts into lesser tuberosity
- Below AC joint → **supraspinatus** muscle and tendon
  - Superior glenoid labrum with biceps attached

Labrum Ask AI

Coronal

- **Superior labrum** → site of SLAP (superior labrum anterior-posterior avulsion)
  - Black, well-defined triangle → non-mobile hydrogen ions
- Along humeral head curvature → **inferior labrum**, also triangle
  - Attached: inferior glenohumeral ligament = axillary pouch / axillary recess
- Circumflex vessels below; suprascapular vessels above
- Labelled: supraspinatus tendon, superior labrum, inferior labrum, **anterior band**, axillary pouch
- Posterior image: spine of scapula → infraspinatus → greater tuberosity; posterior band
  - Infraspinatus identified → spine of scapula appears

MRA for Biceps Anchor and Superior Labrum Ask AI

- Biceps tendon, superior labral tear, SLAP → need **MRA**
  - MRA = Magnetic Resonance Arthrography → dye inside joint
  - Dye outlines biceps tendon → not this clear on ordinary MRI
  - Superior labrum → long head of biceps
    - Exits in bicipital groove
- Image obtained only if MRA requested
- Suspected SLAP → request MRA
- Explain to patient: dye **inside the shoulder**
  - Some centres misunderstand → inject intravenously as body contrast

Pitfall

- MRA dye → **intra-articular**, not intravenous
- Specify this to radiology centre

Adhesive Capsulitis (Frozen Shoulder) Ask AI

Coronal

- Two points for MRI diagnosis:
  - Look at **inferior glenohumeral ligament**
  - Normal → U shape, thickness 2 to 4 mm
- Abnormal: thickened, **amalgamated** inferior glenohumeral ligament
  - Amalgamated with inferior glenoid labrum → adhesive capsulitis
  - These structures released arthroscopically in frozen shoulder
- Frozen shoulder MRI → coronal: ligament + inferior axillary pouch
  - Normal: thin, 2 to 4 mm
  - Clumped, dark → thickened → **adhesive capsulitis**

Bursae Ask AI

Coronal

- Subacromial and subcoracoid bursae
- White area below AC joint, continuing below deltoid → **subacromial-subdeltoid bursa**
  - Hence not two bursae → one, subacromial → subdeltoid
  - Rotator cuff arthroscopy → all open together
    - One space / one bursa when cleared
- Axial view: anterior to subscapularis → **subscapularis bursa**
- Subacromial bursa above; subcoracoid bursa anteriorly; **effusion** in joint
- Early learning → locating bursae not required
- Effusion (high signal on T2) here → **subacromial bursa**

Summary Ask AI

Coronal

Item	Coronal findings
Ligaments	Inferior glenohumeral, anterior + posterior bands → best here; middle and superior not seen
Bones	Acromion, clavicle, AC joint → supraspinatus below; spine of scapula → infraspinatus below
Tendons	Supraspinatus, infraspinatus → locations and landmarks
Labrum	Superior and inferior seen; anterior and posterior not seen
Bursae	Subdeltoid, subacromial

- Inferior glenohumeral ligament → best seen on **coronal** cuts
- Bone evaluation → acromioclavicular joint
- Anterior / posterior labrum not seen → **Bankart lesion** hard to diagnose on coronal
  - Own views → discussed later
- Review: AC joint, supraspinatus, superior labrum, spine of scapula, infraspinatus, posterior band

Axial View Ask AI

Structures Assessed Ask AI

Axial

- **Ice cream cone**: humeral head = scoop; glenoid = cone
- **Os acromiale**: unfused acromial ossification centre
  - Failed fusion with rest of acromion → detailed under pathology
- **Hill-Sachs lesion**: best on axial, at **level of coracoid**
  - Bony dent in humeral head → patient must be a dislocator
  - Find coracoid level → look for Hill-Sachs there
- **Subscapularis tendon**:
  - View that shows subscapularis tear
  - Biceps tendon in place in bicipital groove?
- **Middle glenohumeral ligament** (labelled superior; actually middle)
- Bankart lesion → this area
- Five items: os acromiale, Hill-Sachs, subscapularis + biceps, middle glenohumeral ligament, anterior + posterior labrum

Orientation Ask AI

Axial

- Black, taut structure → **subscapularis tendon**
- Groove → biceps tendon in bicipital groove
- Both identified → anterior, posterior, lateral, medial
- Subscapularis: maximally taut; anterior to glenoid → **lesser tuberosity**
  - Directly lateral → bicipital groove with biceps
- Posteriorly → infraspinatus
  - Infraspinatus tendon and muscle → judged on **coronal** view
- Axial → recognize infraspinatus, subscapularis, bicipital groove
- Middle glenohumeral ligament; anterior labrum; posterior labrum
  - **Bankart lesion** diagnosed on this view
- T2 → Bankart much clearer than T1
- Posterior humeral head → Hill-Sachs site
- Ball with tendons; cone with labra
  - Detached anterior labrum → **Bankart lesion**
  - Left picture subtle; right picture clear
- Hill-Sachs site → **posterolateral** humeral head
- Diagram of glenohumeral ligaments:
  - Middle → orange
  - Superior → very rarely seen
  - Inferior → green; anterior and posterior bands (seen on coronal)
  - **Hammock** between bands = axillary pouch / axillary recess

Middle Glenohumeral Ligament versus Bankart Ask AI

- Black piece detached from glenoid → may be called **Bankart**
  - Well-known disagreement; two professors, two opinions
- Rule: follow **glenoid cartilage**
  - Black triangle attached anteriorly
  - Black triangle attached posteriorly → anterior + posterior labrum intact
- Structure in front of anterior labrum → **middle glenohumeral ligament**
  - Attaches to labrum → not a Bankart
- Bankart → diagnosed at **level of subscapularis tendon**
- Both labra attached + band / black point in front → **middle glenohumeral ligament**
- Second example: subscapularis tendon; biceps in bicipital groove
  - Glenoid cartilage → posterior labrum, anterior labrum → black point in front = middle glenohumeral ligament
- Labels: subscapularis, infraspinatus, posterior labrum, anterior labrum, middle glenohumeral ligament, biceps
- Inferior glenoid labrum → **3 to 6 o'clock**
  - Middle glenohumeral ligament above this level → unrelated
  - Triangle well attached → structure in front = middle glenohumeral ligament
- **Buford complex**: thickened middle glenohumeral ligament
  - Some surgeons → call it detached labrum → repair
  - Reattach to glenoid; explained later
- MRI (left) vs diagram (right): middle glenohumeral ligament vs **anterior inferior glenoid labrum**
  - Mistaken for Bankart → not a Bankart

> **Pitfall**
> - Black structure in front of intact anterior labrum → **middle glenohumeral ligament**
> - Not a Bankart
> - Bankart → subscapularis tendon level, 3 to 6 o'clock

Hill-Sachs Lesion Ask AI

Axial

- **Indentation** in posterolateral humeral head
  - MRI detail → last part of lecture
- Orientation: biceps groove + subscapularis tendon → **anterior** aspect
- Anterior glenoid labrum vs posterior glenoid labrum
- Anterior triangle torn, displaced medially → **Bankart + Hill-Sachs**
- About 45% of dislocators → Hill-Sachs + **Bankart lesion**
- Size: small dent, one third, half, two thirds of head

Buford Complex Ask AI

- **Buford complex** → mistaken for Bankart; not a Bankart
- **Congenital labral variant**:
  - Anterosuperior labrum absent, 1 to 3 o'clock
  - Compensation → thickened middle glenohumeral ligament
- Diagnosis → **axial cuts**; blue and red arrows
  - Not glenoid labrum → Buford complex
- Torn labrum → stays in place or **displaced medially**
  - Displaced anteriorly → Buford complex, not Bankart
- Distinction:

Feature	Buford complex	Bankart
Location	1 to 3 o'clock	3 to 6 o'clock
Labrum	Anterosuperior absent (congenital)	Anteroinferior torn
Position of structure	Displaced anteriorly	In place or displaced medially
Associated	Thickened middle glenohumeral ligament	—

- Band above on MRA → Buford complex / **middle glenohumeral ligament**
  - Outside 3 to 6 o'clock area (anterior inferior labrum)

Summary Ask AI

Axial

Item	Axial findings
Ligaments	Middle glenohumeral
Bone	Humeral head; os acromiale
Tendons	Subscapularis (most important); infraspinatus, biceps also seen
Labrum	Anterior and posterior
Bursae	Not of interest; not seen here

- Coronal → inferior glenohumeral; axial → **middle glenohumeral ligament**
- **Subscapularis tear** → diagnosed on axial view
- Review: humeral head defect; biceps + subscapularis → orientation; labra; middle glenohumeral ligament in front
- MRA example: bicipital groove clear; labrum torn, displaced medially; **Hill-Sachs**

> **High Yield**
> - Coronal → inferior glenohumeral ligament
> - Axial → **middle glenohumeral ligament**
> - Axial → subscapularis tear, Bankart, Hill-Sachs

Sagittal View Ask AI

Structures Assessed Ask AI

Sagittal

- Last cut reviewed
- Items:
  - Rotator cuff muscles → **atrophy**
    - Before cuff repair → fatty degeneration → Goutallier classification
  - Middle glenohumeral ligament
  - Bankart lesion, 3 to 6 o'clock
  - **Acromion morphology**
  - Impingement by acromioclavicular joint (area above)
  - Rotator interval; coracohumeral ligament
- Acromion → morphology assessed
- Rotator cuff: supraspinatus, infraspinatus, subscapularis, **teres minor**
- **Rotator interval** → between coracoid and undersurface
  - Between subscapularis and supraspinatus tendons
- Two diagrams → basis of sagittal reading

Orientation by the Coracoid Ask AI

Sagittal

- Glenoid → anterior projection = **coracoid**
- Find glenoid → find coracoid → anterior vs posterior
- **Coracoid process** → orientation
- Coracoclavicular ligaments above coracoid → not required while learning

Position	Structures
Centre	Glenoid
Anterior projection	Coracoid process
Above coracoid	Clavicle
Behind clavicle	Acromion
Posterior to glenoid	Supraspinatus → infraspinatus → teres minor
Anterior to glenoid	Subscapularis tendon; subscapularis / subcoracoid bursa; short head of biceps / coracobrachialis
Superficial	Deltoid

- Anterior: subscapularis tendon, subscapularis bursa
  - Short head of biceps; **deltoid** muscle
- Right figure: glenoid → coracoid; three muscles above and behind → posterior to glenoid
  - Anterior to glenoid → subscapularis
  - Subscapularis / **subcoracoid bursa**; coracobrachialis / short head on coracoid
- Sequence: glenoid, **coracoid**, clavicle, acromion, posterior three, anterior subscapularis
- Applied: clavicle above coracoid; acromion behind; three muscles posterior
  - Subscapularis anterior

- Left diagram vs right MRI → same layout
- Real MRI: coracoid, glenoid; clavicle above-right
  - Acromion behind; three **posterior** muscles; subscapularis anterior
- Anteriorly: subscapularis tendon + **middle glenohumeral ligament**
- Structure between coracoid and clavicle → coracoclavicular ligaments
  - Glimpsed on coronal cuts; optional detail
- Lateral cut → **humeral head** appears
  - Clavicle ended; acromion appears; posterior three; subscapularis
- Rotator interval and coracoid seen

Rotator Interval and Coracohumeral Ligament Ask AI

- Black line → humeral head
  - = **coracohumeral ligament**; main constituent of rotator interval
- Frozen shoulder → must be cut arthroscopically
  - Not released → shoulder unchanged
  - "Came out as I went in" → ligament most likely not cut
- Diagram: coracoid, humeral head, **coracohumeral ligament**
  - Some fibres → bicipital groove; adhere to biceps tendon
- Cut without coracoid → interval inferred
- Rotator interval contents:
  - Long head of biceps
  - Capsule part: upper edge of subscapularis ↔ supraspinatus
  - Coracohumeral ligament → **main constituent**
- Bankart surgery → cannula entry through rotator interval
  - **Anterior portal**: spinal needle → cannula → instruments

> **Clinical Pearl**
> - Failed frozen-shoulder release → **coracohumeral ligament** most likely not cut
> - Unknown location → surgeon doesn't know what he releases
> - Rotator interval → anterior portal in Bankart surgery

Rotator Interval Injury Ask AI

- Athletes: e.g. tennis player
  - Serve → hand and racket too far forward → pop
- MRI: labrum normal, all else normal → likely **rotator interval tear**
- Two signs:
  - **Coracohumeral ligament** not seen
  - High fluid signal → white patch: hemorrhage / hematoma, on T1 or T2
- Coracoid
  - Sagittal orientation; humeral head; band between = **coracohumeral ligament**
- Both images → white arrows: coracoid, humeral head, coracohumeral ligament
- Review: coracoid, clavicle, acromion; posterior three; **subscapularis**; glenoid labrum
  - Arrow should point to clavicle (labelling error)
  - Middle glenohumeral ligament

High Yield
- Rotator interval injury, sign 1 → **coracohumeral ligament** absent
- Sign 2 → white patch: hemorrhage, hematoma, effusion

Acromion Morphology Ask AI

- **Four types**, not three Bigliani types
- Types 3 and 4 → cause problems

Type	Shape	Prevalence	Impingement
1	Flat	~50%	No
2	Curved / concave → follows humeral head	~40%	No
3	Hooked → bony spur onto cuff, tearing it	11%	Yes *[Q1 · As spoken; type percentages total 104% (50+40+11+3), and Bigliani hooked type is usually about 39%.]*
4	Convex undersurface	3%	Yes

- Sagittal → acromion shape + **impingement**
- Types 3 + 4 → impingement → only 14% of people
- X-ray: **Y shape** = sagittal appearance → glenoid + rest of scapula
  - Acromion, clavicle; types 1–4 on X-ray
- Type 3 hooked → touching cuff
  - **Tendinosis** → partial bursal-surface tear → full-thickness tear

Exam Pearl
- Acromion types → **four**, not three
- Types 3 and 4 → impingement → 14% of people
- Hooked → tendinosis → bursal partial tear → full-thickness tear

Summary Ask AI

Sagittal

- Last cut → normal MRI reading complete

Item	Sagittal findings
Ligaments	Middle glenohumeral; coracoclavicular (AC joint); coracohumeral
Bone	Acromial shape, four types; AC arthritis and impingement
Tendons	Rotator cuff tendons; atrophy → Goutallier fatty infiltration
Labrum	Anterior and posterior, not detailed → Bankart needs axial
Bursa	Subcoracoid; not important now
Rotator interval	Absent coracohumeral ligament; hemorrhage / hematoma / effusion between coracoid and humeral head

- **Subacromial impingement** → sagittal cuts give acromion type
  - Coronal → shows a little; no type
- Atrophy → guides rotator cuff tear management (Goutallier)
- Rotator interval injury → two signs above

ABER View and Internal Impingement Ask AI

ABER View Ask AI

- **ABER** = abduction + external rotation
- Bankart: ABER → stretches inferior glenohumeral ligament → tension on **anteroinferior labrum**
  - Intra-articular contrast enters tear → shows Bankart otherwise unseen
- Uses: **labral tears**; cuff tendons
- More precisely → internal impingement

Types of Impingement Ask AI

- Subacromial → only one type
- Subcoracoid
- **Internal**

Internal Impingement Ask AI

- **Overhead athletes**: volleyball, handball, baseball, tennis
- Posterior cuff (infraspinatus, teres minor) ↔ posterosuperior glenoid edge
- Hyperlaxity / overstretch of **posterior capsule**
  - Head rotates posteriorly → cuff trapped against posterior glenoid
- ABER diagnoses:
  - Bankart lesion
  - Rotator cuff tears: partial or full thickness
  - Mostly → **internal impingement**

Reading the ABER View Ask AI

- Arm abducted → not typical ice cream cone
  - Humeral head continuous with shaft → **biceps tendon** in groove visible
- Humeral head; glenoid with anterior + posterior labrum
- Supraspinatus → greater tuberosity → **supraspinatus tear** or labral tear diagnosable
- Moving posteriorly: supraspinatus, anterior labrum
- Further posteriorly: **infraspinatus**, anterior labrum
  - Bicipital groove disappeared → posterior
- Groove present → tendon above = supraspinatus; groove gone → infraspinatus

Labral Variants Ask AI

- Variants (shoulder lectures):
  - **Sublabral foramen**: 1 to 3 o'clock
  - Sublabral recess
    - Mimics **SLAP lesion** → exactly at typical SLAP site
- Even MR arthrography → sometimes cannot separate recess from SLAP
  - Radiologist or skilled MRI-reading surgeon → difficulty
- **Arthroscopy** differentiates:

Feature	Sublabral recess	SLAP lesion
Labral edge	Smooth	Frayed; many fibres / fibrils
Triangle	Intact; opening below, deeper than normal	Torn; tear enters triangle

- Exact SLAP diagnosis → not MRI, not MRA → **scope**

Location versus Bankart Ask AI

- Sublabral foramen, 1 to 3 o'clock → never call it **Bankart**
- Bankart → **below equator**, 3 to 6 o'clock
- SLAP → 2 o'clock to 10 o'clock → easily confused with sublabral recess
- Buford complex (absent anterosuperior labrum) and sublabral foramen → not Bankart
- Bankart → 3 to 6 o'clock, below **equator** of glenoid
  - 1 to 3 o'clock → may be sublabral foramen
  - No labrum + thickened adjacent structure → Buford complex
  - Buford = enlarged / hypertrophied **middle glenohumeral ligament**

Entity	Location	Key feature
Sublabral foramen	1 to 3 o'clock	Variant; not Bankart
Buford complex	1 to 3 o'clock	Absent anterosuperior labrum + thick middle glenohumeral ligament
Sublabral recess	Typical SLAP site	Mimics SLAP
SLAP	2 to 10 o'clock	Triangle torn
Bankart	3 to 6 o'clock, below equator	Anteroinferior labrum torn

Sublabral Recess Ask AI

- Confuses even skilled radiologists → **sublabral recess** vs SLAP
- Three types:
  - Type 1: no recess
  - Type 2: small recess
  - Type 3: large recess → very hard to separate from **SLAP tear**
- Workup → **MRA**
  - Skilled radiologist → may differentiate
  - Otherwise → diagnostic scope; especially with **history of trauma**
- Recess: smooth edge; whole **triangle** intact; recess behind it
- SLAP: triangle itself torn
- Recess: opening below triangle, deeper than normal → confusing
  - Good **diagnostic scope** → resolves
- Not entering triangle → **sublabral recess**
  - Smooth area passing around from inside
  - Enters triangle → SLAP
- Sublabral recess example
- Extent also confusing → still not a **Buford complex**
- Dislocation history: torn labrum → medially or attached in place
  - Never displaced anteriorly
  - Anteriorly displaced, well defined → **Buford complex** (MRA sagittal, red arrows)

Pitfall

- Type 3 **sublabral recess** ↔ SLAP
- MRI and MRA may fail → diagnostic arthroscopy decides

Os Acromiale Ask AI

- Failed ossification / fusion of acromion with **spine of scapula**
- Prevalence: 5% of people
- Plain X-ray: two bone edges overlapping in acromion
  - Some call it **fracture** → fairly common
  - Present in a few percent → os acromiale
- Plate fixation → **curette** gap; not a simple fracture reduction
- Worst error → unrecognized + **decompression**
  - Fragment resected; important muscles attached → very severe pain
- Subacromial decompression distal to **synchondrosis**
  - Greater mobility of remaining os acromiale → impingement worsens
  - Patient much worse instead of better

Pitfall

- Unrecognized **os acromiale** + decompression → severe pain, worse impingement

MR Arthrography Technique Ask AI

- Needle into shoulder under imaging
- Arm in **ABER** position
  - Shows rotator cuff tears, labral tears; internal impingement diagnosed
- After dye: **axillary pouch** outlined excellently
- Labrum sometimes invisible on MRI → much clearer on MRA

Rotator Cuff Disorders Ask AI

Normal Cuff Ask AI

- Supraspinatus muscle and **supraspinatus tendon**
- Spine of scapula → infraspinatus tendon
- Posterior and anterior bands of inferior glenohumeral ligament
- Tendon black / dark gray → **non-mobile** hydrogen ions
- Subscapularis → black on axial view

Tendon Degeneration Ask AI

Cuff

- = **tendinitis** / tendinosis; cause → impingement by AC joint
- Normal tendon: 2 to 4 mm thick, black
- Degeneration:
  - Gray / white signal enters tendon
  - Thickness increases → about 1 cm → adherent to **acromion undersurface**
- Diagnosis → **subacromial impingement** + rotator cuff degeneration
- **Intrasubstance tear** beginning:
  - White spot inside tendon
  - Fibres separate; intact line above and below
  - Thickness not 4 mm → e.g. 8–9 mm

Partial Tears Ask AI

- Three types:

Type	Site
Bursal	Acromion undersurface side → subacromial space
Articular	Humeral head side
Intrasubstance	Within tendon

- Articular tear grades:
  - Type 1 → 2 → 3 with increasing depth; all **articular surface**
- Bursal (from above): grade 1, 2, 3
- Arthroscopy:
  - Thinned / hollow cuff; needle passes easily → **partial articular-surface tear**
- Bursal → from above, acromion undersurface (subacromial space)
- Articular → from below, articular margin
  - Of humeral head
- MRI:
  - Yellow arrow; rest of cuff continuous and attached
  - Tear from articular surface; acromial surface intact → **articular-surface tear**
- Left: articular partial tear; right: **bursal partial** tear

Full-Thickness Tears Ask AI

- Whole cuff torn → much easier to diagnose
  - Cuff does not reach **insertion** → red arrow on arthroscopy
  - Example size ~2.5 cm
- Two checks:
  - **Retraction distance**: more medial → harder arthroscopic repair
    - 1 cm ≠ 2 cm ≠ glenoid neck
    - Retracted to glenoid neck → **massive rotator cuff tear**
  - **Muscle atrophy** → repair doomed to failure despite good suturing
- Atrophy → coronal + sagittal; **sagittal** more important
- Gray tendons (left) vs white / white-gray (right) → **fatty infiltration**
  - Left → repair expected to give result
  - Right
    - Much more difficult; high **failure rate**
- Examples: full-thickness tear; **retracted** ~2.5 cm; arthroscopic image

> **High Yield**
> - Before repair → **retraction distance** + muscle atrophy
> - Retraction to glenoid neck → massive tear
> - Atrophy → sagittal view more important than coronal

Goutallier Fatty Degeneration Ask AI

Sagittal

- Sagittal view → line across scapular **Y**
- Supraspinatus; infraspinatus; subscapularis
- Supraspinatus above line by 40% to 50% → **normal muscle**
- Tear untreated for a year → muscle below line → **Goutallier** 3 or 4
- Repair → no improvement; repair **fails**

Calcific Tendinitis Ask AI

- Best diagnosed by **plain X-ray**; MRI sometimes very misleading
- X-ray: white spot above humeral head → **calcific deposits**
- MRI: tendon not 2 to 4 mm; lumpy area breaking smooth contour
- Diagnosis → plain X-ray
- Calcification, physiological or pathological → **black** on MRI → immobile hydrogen ions

Infraspinatus Tear Ask AI

- Coronal: **spine of scapula** → infraspinatus
- Full tear; infraspinatus **retracted**

Subscapularis Tear Ask AI

- Needs **axial** view (ice cream cone)
  - Normal: maximally taut line; biceps in groove
  - Torn: detached from insertion; biceps dislocated medially
- Subscapularis tear
  - May carry **biceps tendon** out of groove medially
- Medially dislocated **long head of biceps** → subscapularis torn, tear seen or not
- Not every tear → biceps may stay in place
  - Example: tear with biceps in groove

Exam Pearl

- Medially dislocated **long head of biceps** → subscapularis tear
- Biceps in groove → tear not excluded

Impingement Ask AI

Subacromial Impingement Ask AI

- **Three bony causes**:
  - Acromion type 3 and type 4
  - Os acromiale (failed ossification of acromion)
  - **Acromioclavicular arthritis**
- Coronal → AC joint with supraspinatus tendon below
  - Degenerated AC joint
    - **Abutment** on supraspinatus tendon
- Image: flat vs type 3 acromion
- Subacromial impingement → look for one of three **osseous factors**

Internal Impingement on MRI Ask AI

- **Athletes**: handball, volleyball, tennis
- Posterosuperior cuff → caused by head rotation
- Cystic changes possible inside **humeral head**, unless frank tear
- Site: posteromedial humeral head *[Q2 · As spoken; internal impingement cysts usually sit posterosuperior / posterolateral humeral head (greater tuberosity), not posteromedial]*
- Early **infraspinatus** tear posteriorly
- Often not diagnosed → **clinical suspicion**:
  - Overhead athlete; pain especially on posterior cuff
- Request **ABER view** from MRI centre
  - Better → **MRA** → infraspinatus tear or not

Glenohumeral Instability Ask AI

Bankart Lesion Ask AI

- Labral tear → **Bankart lesion**
- Torn: anteroinferior glenoid labrum + **periosteum** of glenoid neck → both
  - Distinguishes from Perthes and ALPSA
- Diagnosed at **level of subscapularis tendon**
- Image: subscapularis tendon; bicipital groove; **detached labrum**
- Torn labrum → very close or displaced medially toward **glenoid neck**
  - Buford complex → higher area
- Buford level higher → far from Bankart, different level
- Equator, 3 o'clock → **Bankart** begins
- 1 to 3 o'clock
  - Not at subscapularis level
- Method: locate **subscapularis tendon** first → detached labrum here → Bankart

Bony Bankart Ask AI

- Dislocation may avulse **glenoid** bone with labrum
  - Red arrow: bone fragment + labrum separated from glenoid
- **CT**: bony Bankart 3 to 6 o'clock; assess fragment size

ALPSA Lesion Ask AI

- **ALPSA** = anterior labroligamentous periosteal sleeve avulsion
- Anterior labrum torn; **periosteal sleeve** stretched, not torn
  - Whole complex displaced
- MRA → periosteum not torn

Perthes Lesion Ask AI

- Anteroinferior labrum separated
- **Periosteum** intact: not stripped, no avulsion, no tear *[Q3 · As spoken; in a Perthes lesion the periosteum is normally intact but stripped medially from glenoid neck]*
- Very **deceptive** → missed by radiologist
  - No MRA → most likely not found
  - **ABER view** (abduction external rotation MRI) → shows Perthes

Comparison of Anterior Labral Lesions Ask AI

- Three types:

Lesion	Anteroinferior labrum	Periosteum	Detection
True Bankart	Torn	Torn	Axial, subscapularis level
ALPSA	Torn	Sleeve stretched, intact; complex displaced	MRA
Perthes	Separated	Intact, not stripped	MRA / ABER view

> **High Yield**
> - Bankart → labrum + periosteum torn
> - ALPSA → periosteal sleeve intact, displaced
> - **Perthes** → periosteum intact; needs MRA / ABER view

Hill-Sachs Lesion Ask AI

Instab

- **Indentation** / fragmentation → posterolateral humeral head
- Dislocators; more dislocations → larger dent
- Orientation: bicipital groove + subscapularis tendon → anterior
- Posterior labral tear = typical Bankart, posterior → **reverse Bankart**

Posterior Instability Ask AI

- **Bony posterior Bankart** → fragmented
- Head dislocated posteriorly → indentation
  - Anteromedial humeral head → **reverse Hill-Sachs**

Feature	Typical Hill-Sachs	Reverse Hill-Sachs
Site	Posterolateral humeral head	Anteromedial humeral head
Head dislocation	—	Posterior
Labral partner	Bankart	Posterior (reverse) Bankart

Biceps-Labral Complex and SLAP Lesion Ask AI

Biceps-Labral Complex Ask AI

- **Long head of biceps** → superior glenoid labrum → rotator interval → bicipital groove
- Subscapularis tendon nearby
- Three types:
  - Type 1: no sulcus (no sublabral recess)
  - Type 2: small recess
  - Type 3: large sulcus (large recess)
    - Confused with SLAP; disputed even with MRA → **arthroscopy** settles

SLAP Lesion Ask AI

- **Detachment of biceps anchor** from superior labrum, anterior → posterior
- Triangle not intact → tear continues into triangle itself
- Frank image → SLAP; may still mimic **sublabral recess**
- Minimal triangle entry → may be recess
- SLAP vs sublabral recess → somewhat difficult

HAGL Lesion Ask AI

- Normal inferior glenohumeral ligament → **U shape**
- Torn, completely detached from **humeral head**, not glenoid
  - HAGL = humeral avulsion of glenohumeral ligament
  - Appearance → **inverted J**
- Arthroscopy: capsular opening on humeral-head side
- Arthroscopic repair possible but difficult:
  - **70° scope**
  - Specific equipment, sutures, accessories

Questions and Answers Ask AI

Bursal versus Articular Partial Tears Ask AI

- Superior surface facing acromion → **bursal** tear
- Toward humeral head / glenohumeral joint → **articular** tear

Identifying the Infraspinatus Ask AI

- **Spinoglenoid notch** → possible but very difficult
  - Cuts rarely capture the notch
- Spine of scapula → triangle / "7" shape
  - Tendon below → **infraspinatus**; notch not needed

Acute versus Chronic Supraspinatus Tear Ask AI

- MRI differentiates acute from **chronic tear**

Feature	Acute	Chronic
Signal	White patch: hematoma, effusion, hemorrhage	—
Muscle	—	Atrophy
Retraction	Not marked; tendon still partly attached	Marked

Report Discordant with Clinical Findings Ask AI

- Clinically no complete tear but report says complete → read MRI yourself
- Case: own MRI reading → **complete tear** → repaired → improved
  - Report said no tear → centre re-read, apologized
- Reading MRI yourself → protection from **medicolegal** trouble
- Report conflicting with clinical diagnosis → return for re-reading, second reader, or another centre

Open MRI Reliability Ask AI

- **Open MRI** for shoulder → reliable; no problems

Indications for MRA Ask AI

- **MRA mandatory**:
  - Recurrent dislocation + ordinary MRI showing intact labrum
  - Multiligamentous laxity / **multidirectional instability**
    - Shows redundant capsule, posterior and anterior
  - Internal impingement → only MRA shows it
  - **PASTA lesion** (partial articular-surface tear) → only MRA shows it
  - SLAP lesion
  - Bankart / **Perthes** lesion → Perthes otherwise unseen

Must-Know

- Five main MRA indications: SLAP, PASTA, internal impingement
- Bankart / **Perthes** lesion
- Recurrent dislocation with normal MRI / multidirectional instability

Further Questions Ask AI

- Nerve questions → later nerve-injury lecture; treatment → outside scope
- **Footprint** on MRI:
  - Top of greater tuberosity → end of humeral articular surface
  - Tissue or fluid between supraspinatus and deltoid
  - Deltoid fibres → descend to **deltoid tuberosity** insertion

Shoulder Soft Tissue Injuries

(Instability, Labral Lesions and Bone Defects - Decision Making)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 87) "humeral head defect less than 25%" → "humeral head defect less than 20%"
- **C2** (p. 94) "labrum heals on anterior acromion edge" → "anterior glenoid edge"
- **C3** (p. 95) "coracohumeral ligament (in the question)" → "coracoacromial ligament"

Queried - kept as spoken, but looks wrong

- **Q1** (p. 75) As spoken; standard treatment of HAGL is open (or arthroscopic) HAGL repair, not Latarjet.
- **Q2** (p. 76) As spoken; suprascapular nerve lies roughly 1.5-2 cm medial to the supraglenoid tubercle, not 2 mm.

Overview of Shoulder Soft Tissue Injuries Ask AI

- Scope → **engaging** vs non-engaging Hill-Sachs; significant glenoid defect size; Bankart problems
  - Decision-making and management; many videos
  - Technical content; simplified for those not working on the shoulder
- Previous lecture → theoretical; this lecture → **technical**
- Previous lecture → **bony injuries**: fractures, dislocations
- This lecture → soft tissue injuries:
  - **Glenoid labrum**
  - **Rotator cuff**
  - Biceps tendon, capsule → minor part, not main topics
- Master's / MD exams → most questions on labrum and **rotator cuff**

Glenoid Labral Lesions Ask AI

Anterior Labral Pathology Ask AI

- Glenoid labrum → **deepens the glenoid cavity**
- Shoulder dislocation → anterior labral tear → **anterior labral pathology**
- Classic Bankart → **detachment of the labrum**
  - Figure: normal labrum (left) vs classic Bankart (right)
  - Sagittal cut through humeral head and glenoid
- Double lesions → not detailed
- Bankart variants → **bony Bankart**, Perthes, ALPSA
- Non-Bankart lesions → capsular detachment, **HAGL**, GLAD
- Most common → **classic Bankart lesion**

Lesion Types Ask AI

- Features:

Lesion	Feature
Classic Bankart	Labrum + capsule detached from glenoid neck; most common
Bony Bankart	Glenoid bone fragment detached with labrum
Perthes	Periosteal sleeve stripped, still attached to glenoid, retracted
ALPSA	Labrum detached, medialised onto glenoid neck; long-standing instability (e.g. 4–5 dislocations)
Capsular detachment	Non-Bankart lesion
HAGL	Humeral avulsion of glenohumeral ligaments; capsule detached from humeral head
GLAD	Labrum detached with part of articular cartilage

- **HAGL** → opposite of Bankart
  - Bankart → capsule + labrum off **glenoid neck** during dislocation
  - HAGL → also with dislocation → **humeral side**, not glenoid side
- GLAD detail → beyond Master's / MD requirement
- Frequency → classic Bankart → **bony Bankart** → HAGL
- Tricky exam question → HAGL patient → **arthroscopic Bankart repair**?
  - Detachment on humeral side, not glenoid → no gain
  - Useless operation → costs anchors + surgery
- Lesion treated = **HAGL**, not classic Bankart

Lesion Locations Ask AI

- Classic Bankart → **anteroinferior**, below the equator
  - Right shoulder → 3 to 6 o'clock
  - Left shoulder → 9 to 6 o'clock
- **Reverse Bankart** → posterior dislocation
- **SLAP** lesion
- **270-degree tear** → wraps posteriorly → continues with SLAP from above

Normal Labral Variants Ask AI

- Three normal variants:
  - **Sublabral foramen**
  - **Sublabral recess**
  - **Buford complex**
    - Deficient anterior (anterosuperior) labrum
    - Middle glenohumeral ligament → **cord-like** structure
    - Mistaken for pathology → normal variant → not treated
- MD exam → **normal variants** of glenoid labrum

Exam Pearl

- Normal labral variants → sublabral foramen, sublabral recess, Buford complex
- Buford → deficient anterosuperior labrum + cord-like MGHL
- HAGL → humeral side → arthroscopic Bankart useless

Arthroscopic Bankart Repair: Indications and Contraindications Ask AI

Indications Ask AI

- **First-time dislocators**
  - Not common; current trend → American Academy + European Society of Arthroscopy
  - Best chance of labral repair
  - Not every first-time dislocator → surgery
  - Operate after first dislocation if MRI → **Bankart** + Hill-Sachs
- **Recurrent traumatic instability** ± hyperlaxity
  - Laxity and MDI → discussed later
- **Recurrent anterior subluxation**
  - No further dislocation; head moves forward, on verge of dislocating → **subluxation**

Contraindications Ask AI

- **Severe glenoid bone loss** → e.g. 16 dislocations
  - Labral repair alone → **deficient glenoid** persists
  - Plus large Hill-Sachs → instability
  - → **on-track / off-track** concept; asked in MD, less in Master's
  - Asked by examiners wanting difficulty
  - First contraindication = severe glenoid loss or large **humeral osseous defect**
- **HAGL** → recurrent dislocation
  - **Glenohumeral ligament disruption** → dislocates; Bankart repair inappropriate
  - Very tricky MD question; also MCQ
- **Chronic instability** + abnormal muscle patterning
  - Basic problem → **muscles**, not dislocation → repair → re-dislocation
- **Chronic voluntary dislocators**
  - **Excessive laxity**, as in MDI → dislocates and reduces own shoulder
  - Awake, asymptomatic → arthroscopic Bankart → **fails in 100%**
  - Day 2–3 → resumes dislocating; **psychological** purposes
- **MDI** → no role for arthroscopic Bankart
  - Dislocates anteriorly, posteriorly, laterally, inferiorly
  - → **open surgical procedure** or arthroscopic Latarjet
  - Arthroscopic Latarjet / **bone blocks** → instruments not yet in Egypt; learning curve low
  - → open procedure advised

Exam Pearl

- HAGL → arthroscopic Bankart repair contraindicated
- Voluntary dislocator → arthroscopic Bankart fails in 100%
- MDI → no role for arthroscopic Bankart
- Severe glenoid loss / large Hill-Sachs → on-track / off-track concept

Preoperative Planning in Instability Ask AI

History Ask AI

- **Mechanism of injury**
- Number of dislocations
- **General anaesthesia** needed for reduction?
- Recurrence; **previous surgery**
  - Failed arthroscopic Bankart → repeat fails; same inputs → same results
  - Exception → previous surgeon unskilled (studies, imaging, exam) → easy **redo**
  - Otherwise → think a thousand times before same procedure

Examination Ask AI

- **Range of motion**
- Rotator cuff intact
- Instability tests → apprehension, relocation, **fulcrum test**
- Instability, cuff, hyperlaxity tests → big difference in clinical examination
- Hyperlaxity tests:
  - Anterior drawer
  - Posterior drawer
  - **Sulcus test**
  - Hyperabduction test

Imaging Ask AI

- **X-rays**
- MRI; **MRA** (MR arthrogram) preferred
  - Small tear → may be invisible
  - **HAGL** → definitive diagnosis by MRA, not plain MRI
  - Torn capsule + glenohumeral ligaments stick to **humeral neck** → hidden on MRI
  - Capsular **redundancy**
  - **Rotator cuff tears**
- Gold standard → **MRI**; MRA preferred, esp. HAGL
- **3D CT** reconstruction → glenoid bony defects; very important

MRI Appearances Ask AI

- Classic Bankart → labrum detached from glenoid neck
  - MR arthrogram → **dye** escapes outside
- HAGL → capsule detached from humeral side
- Detached from humeral head / **neck of proximal humerus**; dye tracks far down
- **J sign** → HAGL lesion

> **Clinical Pearl**
> - HAGL → definitive diagnosis by MRA
> - J sign on MRA → HAGL
> - 3D CT → glenoid bony defects
> - Failed arthroscopic Bankart → same repeat procedure fails

Soft Tissue Procedures for Anterior Labral Pathology Ask AI

- **Arthroscopic Bankart repair** → traditional
  - Most shoulder arthroscopists well trained
- **Remplissage** (infraspinatus capsulotenodesis) → fills large Hill-Sachs
  - Defect → isolated; intra-articular → **extra-articular**
- MDI, arthroscopic → add **rotator interval closure**
- MDI → **posterior capsular plication** → capsule closed posteriorly
- **Bony Bankart** → arthroscopic repair
  - Fragment reasonably sized, fixable arthroscopically
- **HAGL**:

  - Arthroscopic repair → possible, extremely difficult; **70-degree scope** preferable
  - Avoid arthroscopic repair → humeral capsular attachment **very weak**
  - Sutures cut through capsule → failure in most cases
  - → **Latarjet** → increases arc of motion
  - External rotation + abduction → head stays within capsule
- MCQ: HAGL → arthroscopic Bankart repair → incorrect
  - Arthroscopic HAGL repair → possible
  - Best option for HAGL → **Latarjet** *[Q1 · As spoken; standard treatment of HAGL is open (or arthroscopic) HAGL repair, not Latarjet.]*

Shoulder Arthroscopic Portals Ask AI

Portal	Site / entry	Main use
Standard posterior	2 cm inferior + 2 cm medial to posterolateral acromial corner	Primary viewing portal
Standard anterior	Lateral to coracoid; upper border of subscapularis	Working portal
Low anterior	Anterior	Bony Bankart fixation
Anterolateral superior	Above glenoid	Glenoid viewed from above; two-cannula repair
Posterolateral	Posterolateral	Posteroinferior capsular plication; rotator cuff
Standard lateral	Lateral	Rotator cuff repair; little role in Bankart
Superior	Near biceps anchor	SLAP lesions only

Overview Ask AI

Portals

- **Standard posterior portal**
- Anterior portal → anterosuperior, anteroinferior, or **low anterior** (bony Bankart fixation)
- **Posterolateral portal** → posteroinferior capsular plication
  - Multidirectional or bidirectional instability
- Posterior portal → **2 cm** medial and inferior to posterolateral acromial corner
- Mark landmarks early in arthroscopic practice
  - Fluid-filled shoulder; heavier patients, esp. women
  - Otherwise landmarks lost → draw **bony landmarks**
- Posterolateral portal → plication; preferred for **rotator cuff**
- Lateral / posterolateral portals → very good in **rotator cuff repair**
- Standard anterior portal → coracoid marked; entry at upper border of **subscapularis**
  - Outside-in or inside-out
- Anterolateral superior portal → glenoid viewed from above
  - Helps when two cannulas used in **arthroscopic Bankart repair**
- **Standard lateral portal** → little role in Bankart; basic role in rotator cuff repair

Posterior Portal Ask AI

- **Primary viewing portal**; first portal made
- First landmark → **biceps tendon** → up/down, right/left orientation
- Instability → capsule **redundant** → blunt trocar may fail
  - → **sharp trocar** justified
  - Avoid entering humeral head → cartilage injury
- Site → 2 cm inferior, 2 cm medial to **posterolateral acromial corner**
- Entry → between infraspinatus and teres minor → **posterior soft spot**
- Directed anteriorly; finger on **coracoid process**
- **Never medial to coracoid**
  - Vessels + **brachial plexus** → medial to coracoid
  - Aim → lateral to coracoid

Anterior (Working) Portal Ask AI

- Coracoid palpated → cannula / **spinal needle** into shoulder
  - Upper border of subscapularis, as high as possible → labrum reached from below
- **Outside-in technique**
- **Inside-out technique**
  - Blunt trocar / **Wissinger rod** from posterior portal → anteriorly, lateral to coracoid
  - Pushed forward → marked → opened over

Other Portals Ask AI

- **Lateral portal** → third portal
- **Superior portal** → mainly SLAP lesions
  - Limited indications; high risk → **suprascapular nerve** injury
  - Suprascapular nerve → ~2 mm behind biceps anchor *[Q2 · As spoken; suprascapular nerve lies roughly 1.5-2 cm medial to the supraglenoid tubercle, not 2 mm.]*
- **Anterosuperior portal** → last portal

Pitfall

- Never go medial to coracoid → vessels + brachial plexus
- Superior portal → suprascapular nerve ~2 mm behind biceps anchor
- Sharp trocar → risk to humeral head cartilage

Arthroscopic Findings in Instability Ask AI

Expected Findings Ask AI

- **Bankart lesion** → most common
- **Osseous (bony) Bankart** → labrum + bone piece
- **Reverse Bankart** → posterior dislocation
- **Perthes lesion** → periosteum detached from glenoid neck, labrum with it
- **ALPSA** → everything shifted medially onto neck
- **GLAD** → part of articular surface detached with labrum

Diagnostic Arthroscopy Ask AI

Instability

- **Glenohumeral ligaments** + capsule → capsular quality
  - Shredded capsule → multiple dislocations
- Probe capsular attachment at labrum and humeral side
  - → **HAGL** present?
- **Drive-through sign** → scope passes between glenoid and humeral head
  - Difficult after 1–2 dislocations
  - High laxity → passes easily → **inferior axillary pouch**, even behind glenoid
- **GLAD lesion**
- **Bony deficiency**
- Humeral head → **engaging Hill-Sachs**

Arthroscopic Demonstration Ask AI

Bankart

- Scope posterior; cannula anterior
- **Probe** via anterior portal → assess Bankart
  - Probe reaches ~6 o'clock or beyond
  - Lesion → 3 to 6 o'clock inferiorly
- Arm positioned → **engaging Hill-Sachs** shown; large, broken in several pieces

Case: Basketball Player Ask AI

- Complaint → something comes between joint and head → then dislocation
- Bankart expected; inferior view → detached **bone fragment** (Bankart)
  - Fragment between glenoid and humeral head → frequent dislocations

Arthroscopic Stabilisation Techniques Ask AI

Classical Bankart Repair Ask AI

- Cannula introduced
  - **Flexible cannula** → very helpful in Bankart
- Shaver; second cannula → **anterosuperior portal**
- Scope shifted → anterosuperior portal → glenoid **face on**
  - Labrum detached from glenoid neck, **displaced medially**
- **Periosteal elevator** (liberator) → labrum mobilised off neck → returned to glenoid neck
- Adequate mobilisation → **anchor positions** marked
- Anchor placed → sutures through labrum → second suture retrieved → tied
- Camera shifted back → sutures through labrum → **knots** tied
- **Accu-Pass** suture passer → used in many cases
- **Stainless steel wire** → shuttles anchor sutures; easier
- Result after repair shown

Bony Bankart by Fragment Size Ask AI

- Management depends on fragment:

Fragment	Management
Small	Classic repair
Somewhat larger	Bony Bankart bridge technique
More than 25%	Arthroscopic bone block, open Latarjet, or arthroscopic cannulated screws

- **Bony Bankart bridge technique**
  - Anchor → single suture through labrum
  - Two limbs tied → bony fragment enclosed between them
- Video: bony fragment area; **Accu-Pass** → classical Bankart repair
  - Suture or stainless steel wire previously passed
- Suture end grasped → out through cannula; assistant applies **traction**
- Anchor placed → **bird beak** → retrieves suture from labrum
  - Exits behind labrum; repeat, or both sutures taken together
- Knot tied → fallen labrum restored onto glenoid
- Remaining anchors same way → **bumper effect**
  - Bumper over anterior glenoid → no dislocation

Remplissage Ask AI

- Fills **engaging Hill-Sachs** defect → infraspinatus + posterior capsule
- Indication → humeral head defect **less than 25%**
- More than 25% → **Latarjet**
- MCQ → what is remplissage; when done
- Defect **abraded** well; anchors introduced (current method differs from older one)
- Second case → **engaging Hill-Sachs**
  - Shaver → defect debrided
  - Anchor placed → bird beak takes sutures
  - Sutures pulled over **infraspinatus** → defect filled
  - One suture at a time; knots tied over infraspinatus in **subacromial space**
- Older method → bird beak through **muscle itself** → sutures fill Hill-Sachs defect

Must-Know

- Remplissage → infraspinatus + posterior capsule fill engaging Hill-Sachs
- Humeral head defect < 25% → remplissage; > 25% → Latarjet
- Bony Bankart fragment > 25% → bone block, open Latarjet or screws

Rotator Interval Closure Ask AI

- Closes space between **subscapularis** and supraspinatus
- **Rotator interval** → between supraspinatus, subscapularis, coracoid process
- Contents:
  - **Superior glenohumeral ligament** (middle GHL often clearer)
  - **Coracohumeral ligament** → coracoid to humeral head
  - **Long head of biceps**
- Function → limits excessive **glenohumeral motion**; stabilises long head in place
- Many techniques
- Indications:
  - **Multidirectional instability**
  - **Revision cases**
  - Anterior stabilisation / posterior instability + **hyperlaxity**
    - Augment anteriorly and posteriorly
- Video:
  - **Prolene suture** passed inferior → superior
  - Retrieved through another interval opening
  - Prolene tied on itself → **Ethibond** shuttled over it → interval closed
  - Esp. MDI / multidirectional cases
- Result → strong **anterior strut** in MDI

Posterior Capsular Plication Ask AI

- Positive **sulcus sign** + positive drive-through sign → close posterior capsular redundancy
- Scope → **anterosuperior portal**
- Working → posterolateral or posteroinferior portal

Complications Ask AI

Bankart

- **Stiff shoulder**
  - Over-tensioning → middle + inferior glenohumeral ligaments
  - Start rehabilitation properly; avoid prolonged **immobilisation**
- **Recurrence**
  - Missed **bony Bankart** on MRI / CT; defect > 25% of glenoid
  - → operation doomed to fail
  - **Technical error** → inexperienced surgeon; misplaced anchors
  - **Improper patient selection** → hyperlaxity, voluntary dislocators → highest recurrence
- **Missed rotator cuff tear**
  - Capsule + ligaments fixed anteriorly → cuff tear from previous dislocation → shoulder does not move
- **Axillary nerve palsy**
  - Bird beak passed slightly deeper near **inferior glenohumeral ligament** → catches nerve
  - Two cases seen; re-arthroscopy → stitch around nerve in one
  - Stitch released → one improved; other did not
- **Anchor arthritis**
  - Anchors protrude, esp. **metal** → between head and glenoid → arthritis
  - Over-tensioning → head pressed onto glenoid → rubbing → **arthrosis**

Pitfall

- Missed bony Bankart > 25% → Bankart repair fails
- Bird beak too deep near IGHL → axillary nerve palsy
- Over-tensioning MGHL + IGHL → stiff shoulder, arthrosis
- Metal anchors protrude → anchor arthritis

Diagnostic Shoulder Arthroscopy: Ten Points Ask AI

- Ten points → every learner of **diagnostic shoulder arthroscopy**
  - Point 1 → posterior portal → **biceps** → pathology?
  - Points 2–4 → **bare area** of head, posterior cuff
  - Point 5 → **undersurface** of anterosuperior cuff → articular side, not bursal
  - Point 6 → **rotator interval** → long head of biceps in groove, superior + middle GHL
  - Point 7 → **subscapularis**, middle GHL, anterior labrum
  - Point 8 → **anteroinferior labrum**, inferior GHL
  - Point 9 → subacromial space → **coracoacromial ligament**, acromion undersurface
  - Point 10 → **rotator cuff** from bursal side
- Video sequence:
  - Posterior entry → **biceps tendon**
  - Subscapularis + middle GHL
  - Labrum → torn (Bankart case)
  - Humeral head
  - **Drive-through sign** attempt → inferior recess
  - Undersurface of rotator cuff

Biomechanics of Shoulder Stability Ask AI

Ball and Socket Ask AI

- Remaining topic → **bipolar osseous defects** in recurrent instability
- **Ball and socket joint**
- Glenoid covers only one fourth of humeral head
- **Static and dynamic stabilisers** → known; two points emphasised

End Range vs Mid Range Ask AI

- End range → abduction, external rotation, elevation to 180°
  - **Capsuloligamentous structures** → main stabilisers
  - Tight → prevent further translation of head
- Mid range → abduction 45 or 75°
  - Before full abduction + external rotation
  - **Negative intra-articular pressure**
  - **Concavity compression** effect
- Large head, small glenoid → covers a quarter → beyond that → **capsule**, tight at end range
- Mid range 45–75° → negative pressure + concavity compression → **main stabilisers**

	End range	Mid range
Position	Abduction + ER, elevation to 180° (above 90°)	Abduction / elevation 45–75°
Capsule	Tight	Lax
Main stabiliser	Capsuloligamentous structures	Negative intra-articular pressure + concavity compression
Related defect	Hill-Sachs (humeral head)	Glenoid defect
Tests	—	Anterior drawer, posterior drawer, sulcus test

Pathological vs Physiological Dislocation Ask AI

- End range → capsule tight → no translation
- Force continues into abduction + ER → tight capsule + **inferior glenohumeral ligament** torn
  - → traumatic dislocation → pathological → **surgical** treatment
- Mid range → capsule lax → head translates easily → measured as **laxity**
  - Anterior drawer, posterior drawer, **sulcus test** → performed in mid range
  - Test capsular laxity itself
- **Physiological laxity** → mid range subluxation / dislocation, no symptoms → no treatment
  - Voluntary dislocators; MDI
- Two types:

	Pathological	Physiological
Range	End range	Mid range
Cause	Trauma → tight capsule torn, avulsion	Atraumatic
Symptoms	Present	None
Treatment	Required	Not required

Negative Intra-articular Pressure Ask AI

- Important in **mid range**, with concavity compression
- Arm by side → shoulder muscles silent → arm weight pulls downward
  - → negative pressure → **suction** of head into glenoid cavity
- Negative pressure = arm weight → **equilibrium** → downward translation stops
- **Sulcus test** → arm by side → examiner pulls downward → sulcus sign
  - Sulcus sign → arm weight vs negative pressure
- Greater laxity → mid range subluxation or dislocation → **physiological laxity**
  - No treatment; surgery usually fails

Concavity Compression Ask AI

- Concave surface compressed → head held centred
- Mechanism → **muscle contraction**
  - Subscapularis, infraspinatus, supraspinatus → head pulled against glenoid socket
  - → **resistance force** against translation
- Concavity → glenoid cavity; compression → **muscle contraction force**
- Depends on:
  - Glenoid cavity **depth**
  - Contraction force **magnitude**

Glenoid Depth and Dislocation Direction Ask AI

- Glenoid **superior–inferior** depth = twice anterior–posterior depth
- → Superior–inferior translation force = double **anterior–posterior** force
- → Anterior / posterior dislocations → **most common**
- → Superior / inferior → less common
- Dislocating superiorly / inferiorly → force must be doubled
- → **incidence** of anterior / posterior dislocation ◎ superior–inferior

High Yield

- Glenoid covers one fourth of humeral head
- End range → capsuloligamentous structures stabilise
- Mid range → negative pressure + concavity compression stabilise
- Superior–inferior depth = 2 × anterior–posterior → AP dislocations commoner
- Pathological → end range, traumatic, treated; physiological → mid range, untreated

Bipolar Lesions and the Hill-Sachs Lesion Ask AI

Bipolar Lesions Ask AI

- **Recurrent glenohumeral instability** → challenge for all shoulder surgeons
  - Humeral side → **Hill-Sachs lesion**
  - Glenoid side lesion
- Humeral head + glenoid bony defects → **bipolar lesion**
  - → major predisposing factor for recurrence
- Most common cause of recurrence after arthroscopic procedure → **bipolar lesions**
- Normal → head **centred** on glenoid
- Dislocation → head anterior → labrum torn → **Bankart lesion**
- Anterior glenoid edge impacts head → **posterolateral impression fracture** = Hill-Sachs
- Three results of dislocation:
  - **Bankart lesion** → soft tissue
  - Glenoid **bone defect**
    - Erosion → friction, anterior glenoid loss
    - Fragment broken off → bony Bankart
  - Humeral head → **posterolateral defect**
    - Compression / impression fracture

Incidence after First Dislocation Ask AI

- Osseous defects after **first-time dislocation**:

Finding	Incidence
Neither Hill-Sachs nor glenoid defect	23% (~quarter)
Hill-Sachs only, intact glenoid	30%
Hill-Sachs + glenoid defect	45–50%
Glenoid defect only	0.8% (~1%)

- Hill-Sachs total → 30 + 45 → ~75% → **most common lesion**
- Isolated **glenoid defect** → only 1%; MCQ in MD exam
- No bony defect, humeral or glenoid → ~24%
- Glenoid defect after first dislocation → uncommon, unless **bony Bankart**
- **Hill-Sachs** frequency → 65–70%; ~93% in all (recurrent) cases
- **Glenoid defect** → 5–50%; very wide range
  - Depends on number of dislocations, defect depth

Hill-Sachs Lesion Ask AI

- **Impression fracture** → head strikes anterior glenoid edge, catches on it
- ~90% of anterior dislocations; 25% of **subluxations**
- Very close to **greater tuberosity** → not engaging → on-track lesion

On-Track Concept: Tyre and Road Ask AI

- Very small, near greater tuberosity → far from anterior glenoid edge → **on track**
- Tyre = humeral head; road = **glenoid**
  - Road wider than tyre → tyre stays on
  - Road narrower than head → tyre **falls off**
  - Two factors → tyre size, **road width**
- Small, near greater tuberosity → no dislocation → **on-track lesion**
- Large, more medial, far from greater tuberosity → **off track**
  - Abduction + ER → defect engages anterior glenoid edge → drops in
  - → **off-track lesion**

Importance of On-Track / Off-Track Ask AI

- Determines procedure:
  - **Arthroscopic Bankart** alone
  - **Latarjet**
  - **Remplissage**
  - Remplissage + Latarjet
  - Remplissage + arthroscopic Bankart
- Modalities → matching procedure to patient

Hill-Sachs and the End Range Ask AI

- Hill-Sachs → related to **end range**, not mid range
- Mid range → defect can hardly reach **anterior glenoid edge**
- End range → defect fully covered by glenoid → **on track**
  - Defect contained; tyre smaller than road → no dislocation
- MCQ → Hill-Sachs → **end-range instability**
  - No relation to mid range → defect not yet on glenoid

Off-Track Lesion Ask AI

- Large Hill-Sachs, **larger than glenoid**
  - End range → passes over anterior glenoid edge → anterior re-dislocation → **off track**
- Small, **contained** by glenoid → ball stays on road
- Larger than glenoid → dislocates anteriorly
- **Glenoid bone loss** or defect larger than glenoid → dislocation at end range
- End range → **above** 90°
- Hill-Sachs → end range; **glenoid defect** → mid range
- **Engaging** vs non-engaging:
  - Less contained by glenoid → dislocates more easily
  - Near greater tuberosity, not medial → contained → no dislocation

Hardy Index Ask AI

- Hill-Sachs size relative to head → **Hardy index**
  - Easiest method; X-ray
  - Best-fit **circle** around humeral head cortex
  - Radius of circle; head centre; defect
  - Defect size ÷ **radius** × 100
- → humeral head defect size

Glenoid Bone Defect and Treatment Algorithm Ask AI

Latarjet Threshold Ask AI

- **Old concept** → until 2014–2015
  - More than one third of glenoid lost → **Latarjet**
  - Latarjet → coracoid osteotomised → screwed onto anteroinferior glenoid
- Current threshold → 33–34% → **25%**
- 25% glenoid loss → **bony procedure**
  - Latarjet or arthroscopic bone block (if instruments available)

Glenoid Defect and the Mid Range Ask AI

- End range → 180°, beyond 90° → capsule **very tight** → prevents translation
- Good repaired capsule + glenoid deficit
  - Head stays **centred** at end range → tight capsule protects
- Mid range → capsule **lax** → glenoid defect → dislocation
- Humeral head defect → **end range**
  - End of motion brings defect onto glenoid, intact or not
- Glenoid, intact or defective:
  - End range → **tight capsule** prevents translation, dislocation
  - Mid range → capsule lax → easily dislocates → off track
- New studies → threshold ~25% → address **glenoid bone defect**

Types and Imaging Ask AI

Glenoid

- **Fragment type** → bony Bankart lesion
- **Erosion type** → glenoid rim erosion
- Bony Bankart → **plain X-ray**, CT, MRI

Sugaya Method Ask AI

- **Sugaya** (Japanese) → glenoid bone loss → en face view on CT
  - Humeral head **subtracted** with software → glenoid left
  - Normal glenoid **diameter** measured
  - Missing area relative to it × 100 → exact loss (10%, 20%, 30%)
  - **Best-fit circle** fitted on glenoid
- Second method → **contralateral side**
  - Intact side → reference → affected side compared
  - Defect ÷ diameter → **percentage**

Treatment Algorithm Ask AI

- **Hardy index** + Sugaya en face → treatment algorithm
- < 20% + < 25% → **arthroscopic Bankart** success up to 93–94%
- Remplissage → fills large humeral head defect → **Hill-Sachs lesion**
- Remplissage added to Latarjet → soft tissue fill → prevents **engagement**
- Hardy index > 40% → arthroplasty or **shoulder hemiarthroplasty**

Humeral head defect (Hardy)	Glenoid bone loss (Sugaya)	Treatment
< 20%	< 25%	Arthroscopic Bankart repair
> 20%	< 25%	Bankart repair + remplissage
< 20%	> 25%	Latarjet *[C1 · transcript: "humeral head defect less than 25%"]*
> 20%	> 25%	Latarjet + remplissage
> 40%	—	Replacement: arthroplasty / hemiarthroplasty

Must-Know

- Hardy < 20% + glenoid < 25% → Bankart; success up to 93–94%
- Hardy > 20% + glenoid < 25% → Bankart + remplissage
- Glenoid > 25% → Latarjet; + remplissage if Hardy > 20%
- Hardy > 40% → replacement

Glenoid Track Ask AI

- **Glenoid track** → humeral head area contacting glenoid at end range
- Humeral head defect **contained** within glenoid → on track; no dislocation
- Glenoid defect + large head defect, smaller glenoid → dislocation → **off track**
- Defect contained on glenoid → never dislocates
- Large defect + **defective glenoid** → over anterior glenoid edge → dislocation

Rotator Cuff Tears: Footprint and Classification Ask AI

Overview Ask AI

Cuff

- Rotator cuff → no instability-type complexity
- Instability part → aim → **concepts**; effect of each factor on the other
- Rotator cuff → most common cause of **shoulder pain and disability** in adults
- MRI → tear; scope → tear viewed from **subacromial space**

Footprint Concept Ask AI

- Old concept → cuff attached at **single point** → wrong
- **Footprint concept** → large surface area
  - Supraspinatus, subscapularis → surface area, not point
- Fixation evolution → **single row** → double row → suture bridge
- Footprints → infraspinatus, teres minor; **subscapularis** anteriorly
  - Exact areas less important
- Footprint = area → treatment evolved from single row → **double row**, suture bridge
- **Infraspinatus** footprint larger than supraspinatus on greater tuberosity
  - Opposite of common belief

Classifications Ask AI

Cuff

- **Habermeyer** → first classification; topographic (geographical), three zones:

Zone	Site	Contents
A	Anterior	Subscapularis; coracohumeral ligament / rotator interval
B	Superior	Mainly supraspinatus + part of infraspinatus
C	Posterior	Infraspinatus + teres minor

- **Ellman and Gartsman** → by tear shape:
  - **U-shaped** / crescent
  - Reverse L
  - **L-shaped**
  - Trapezoidal
  - **Massive**, retracted to glenoid
- U-shaped / **crescent tear** shown
  - Anchor being inserted
- By **tear size**:

Size	Extent	Length
Small	—	< 1 cm
Moderate	Edge of humeral articular surface	1–3 cm
Large	Edge of glenoid	3–5 cm

Rotator Cuff Repair Ask AI

Goals Ask AI

- **Minimise gap formation** → approximate; close gap completely
- **High initial strength** → very strong fixation
- **Mechanical stability** → moved after anchor repair → stable
  - No re-tear; sutures not cutting through cuff
- **Tendon–bone interface healing** → basic goal

Repair Techniques Ask AI

- **Single row**
  - Old method; single-point attachment concept
  - Concept changed → cuff = **surface area**
- **Double row** → two lines
  - Medial row → near **articular surface**
  - Lateral row → near **greater tuberosity**
  - → fixation over large surface area
- **Suture bridge** → current concept
  - Medial anchor sutures **criss-crossed** forward and backward
  - Crossing limbs → **cuff compressed** → better tendon-to-bone healing
- **Mason-Allen** single row
  - Two transverse sutures across cuff + stitch between them
  - Limb passed behind cuff → tied over suture
  - **Rip-stop** → two sutures same level + suture between
  - Third suture behind cuff, held with anchor → prevents **cut-through** on pull

Technique	Concept
Single row	Single-point attachment (old)
Double row	Medial + lateral rows → footprint surface area
Suture bridge	Criss-crossed limbs → compression → better healing
Mason-Allen	Rip-stop → resists cut-through

Positioning and Anaesthesia Ask AI

- **Beach chair** (sitting) → most patients; instability and cuff
- **Hypotensive anaesthesia** → GA or interscalene block
- **Examination under GA**
  - Some → manipulation under GA → break **adhesions**
- America → **lateral decubitus**
  - Patient on side; arm pulled up, suspended (ceiling or beside patient)
- **Bony landmarks** → posterior, lateral, anterior portals drawn
  - Esp. obese → portals otherwise lost

Arthroscopic Steps Ask AI

Cuff

- **Glenohumeral joint** examined in every cuff tear
  - Associated → SLAP, Bankart, instability → inspect inside and outside
- **PASTA** → partial articular-sided tear
  - Tear on glenoid (articular) side
  - Direct to subacromial space → **missed**
  - Seen only from inside glenohumeral joint
  - **Bursal side** tear → easy from subacromial space
  - No glenohumeral look → biceps or articular cuff tears missed
- Subacromial entry → **white cloud** → shaver until cuff clear
- First step → **subacromial bursectomy**
- **Subacromial acromioplasty** → abandoned by many
  - Wide space → no bone removal, unless **osteophyte**
  - Narrow space (shown) → undersurface shaved → space enlarged
- Shaver on **footprint** → bleeding bone → tendon–bone healing
  - Avoid excess removal → cancellous bone weakened → weak **anchor** hold
- Pull cuff → must return easily onto **footprint**
  - Not returning → **release**; more cleaning, adhesion removal
  - Footprint cleaned → bleeding **cancellous** surface
- Crescent / U-shaped tear; anchor site marked via different portal
- Anchor at 45° → **Dead Man's Angle**
  - Not 45° to articular surface → **pulls out** easily
- Inserted to **laser mark** → introducer removed
- Pull → confirm firm hold in bone (likely subchondral)
- Suture management → **parking portal**
  - Or cannula → sutures pulled one by one → repair
- **Scorpion** cuff passer → cuff grasped
  - Suture passed
- Old → needle passes, separate instrument retrieves
- Now → **self-retrieving** cuff repair instruments → much easier
- Repair completed

Video: Complete Tear Repair Ask AI

Cuff

- Tear site → anchor site prepared → **anchor** applied
- Sutures parked → taken one by one
- **Quattro** passer → suture through cuff
  - Undersurface → superior (bursal) surface; each suture in turn
- Needle out → suture retrieved; all sutures passed → cuff pulled
- **Bird beak** from posterior → far posterior cuff suture → added strength
- Sutures gathered → **lateral row**
- **Knotless anchor** → tapped into bone laterally to laser mark

Tips and Tricks Ask AI

Cuff

- **Switch scope** → posterolateral, posterior, lateral portals
- **Cannulas** → sutures not tangled in deltoid, bursa
- Better **bursectomy**, thorough shaving → better visualisation
- Assess cuff from **both surfaces** → articular, bursal
- **Debride footprint** → bleeding surface → tendon-to-bone healing
- More **release** → better mobilisation
- Avoid repair in **abduction** → arm by side → no repair tension

> **Clinical Pearl**
> - Enter glenohumeral joint first → PASTA otherwise missed
> - Footprint → bleeding bone, without weakening cancellous bone
> - Anchor at 45° → Dead Man's Angle → resists pull-out
> - Repair with arm by side, not abduction

Biceps Tendon in Cuff Patients Ask AI

Patient	Management
Above 50	Tenotomy
40, intact tendon	Leave
40 torn, manual worker, needs biceps	Tenodesis: subpectoral or suprapectoral

- Above 50 → **biceps tenotomy** → complete cut
  - Only issue → disfigurement → **Popeye sign** (muscle bunched in lower arm)
- 40 years, tendon intact → leave
- Torn at 40, manual worker, biceps function needed → **tenodesis**
  - Subpectoral or suprapectoral (inside joint)

Questions and Answers Ask AI

Rotator Cuff Classifications Ask AI

Q&A

- Six-type cuff classification? → many classifications
  - **Topography** (geographical)
  - Shape
  - **Retraction** → small, medium, large, massive
- Adopt simple classification guiding preparation and work
- **What's common is common**

Anchor Types Ask AI

Q&A

- Bankart anchors → several stages
- **Metal anchors**
  - Inserted to laser mark, no initial protrusion
  - Exercise, physiotherapy → edge protrudes from glenoid rim
  - → humeral head damage → **humeral arthritis** → era declined
- **PEEK / biodegradable** anchors → no metal; dissolve after 6 months–1 year
  - Labrum unhealed by 6–12 months → anchor dissolves →
    - **Repair failure**
    - **Osteolysis** → phagocytes, giant cells eat resorbing material → glenoid defects
- **All-suture anchors** → next era
  - Not metal → no glenoid / head damage
  - Not biodegradable → no **cyst**, no foreign body reaction
- Evolution → metal → biodegradable → **all-suture**; business and research involved

Anchor	Problem
Metal	Protrusion → humeral head damage → arthritis
PEEK / biodegradable	Dissolves 6–12 months → repair failure, osteolysis, cysts
All-suture	Simply suture → no protrusion, no foreign body reaction

Arthroscopy Pump Ask AI

- **Pump** set to patient blood pressure
  - New sensor pumps → pressure follows patient BP automatically
  - **Constant pressure**; inflow = outflow → no shoulder swelling
- **Manual pump** → pressure unknown
  - High / uncontrolled BP → shoulder fills in 10–15 minutes
  - View lost; shoulder tense
  - Some surgeons → **abort** (compartment fear)
- Pump → constant pressure → procedure completed

Intraoperative Glenoid Measurement Ask AI

- Possible; better → patient studied before entry
- Five or six dislocations → expect **glenoid defect**
- Avoid relying on intraoperative measurement
  - Probe via anterior cannula under scope → loss estimated
  - **CT** → much easier; options prepared beforehand
- Glenoid width → ~24 mm (2.4 cm)
- Original probes → **markings** at 5, 10, 15, 20, 25 mm

Working Portals Ask AI

Cuff

- Most common (senior surgeons) → **posterolateral** + lateral portals
  - Scope switched between the two
- **Anterior portal** → posterior tear
- Massive, posterior, anterior tears → **switch portals** per tear shape

Double Row Knotless Anchor Ask AI

- **Medium-sized tear** → ~3 cm from footprint
- Further from footprint → double row better
  - Tension distributed → **medial row** + lateral row
- Single row → very high tension on lateral anchors → higher **failure**

PASTA: Indications for Surgery Ask AI

- **PASTA** = partial articular surface tear
  - Seen from glenohumeral joint; not exposed on bursal side
- Athletes → **racquet sports**, volleyball, handball
- **Internal impingement** → head pinches cuff against posterior glenoid edge
  - Opposite of cuff pinched against **acromion**
  - Overhead activity; arm raised high
- Throwers, arm brought back → infraspinatus + teres minor pinched
  - **Pincer effect** → tear in this area
- Indication → partial tear > 50% **thickness** (arthroscopy or preop MRI) → repair

Conservative Treatment Ask AI

Cuff

- **Partial tear**
  - 60–70 years, low arm demand at home → may have role
  - Biologically → always progresses to **complete tear**
- **Complete tear**
  - Surgically unfit → conservative
  - Fit, uses arm, esp. **dominant hand** → no conservative
  - Deteriorates → returns in 1–2 years with **fatty infiltration** → surgery useless
  - → **absolute indication** for repair
- Delayed 1–2 years → **fatty atrophy**, weak muscle
  - Every suture cuts through → re-tear → failure
- "Fell two years ago, snap, cannot raise arm" → **MRI**
  - Radiologist reports **Goutallier** grade of fatty infiltration → grade 1–4
- Grade 3–4 → muscle fatty → repair **fails**
- Grade 0–2 → **repair**

High Yield

- Partial cuff tear → biologically progresses to complete
- Fit patient, complete tear, dominant arm → absolute indication
- Goutallier 3–4 → repair fails; 0–2 → repair

Biceps Pathology with Cuff Tear Ask AI

- **Very common** → biceps pathology in > 50%
- Pain → night pain, waking pain → **biceps tendon pathology**
  - Not cuff tear itself
- Cuff tear → **weakness**
  - Plate onto high shelf → fails
  - Reaching behind back for shirt → fails
- Patient pointing to front of shoulder → **biceps pathology**

Order: Remplissage and Bankart Repair Ask AI

- Large Hill-Sachs → **Bankart + remplissage**:
  - **Remplissage anchor** first → sutures through infraspinatus → untied
  - Anterior glenoid anchors → tied
  - Remplissage sutures tied last
- Bankart → two or three **anterior anchors**

Postoperative Protocol Ask AI

- Quick protocol → Bankart vs rotator cuff repair
- **Bankart** (alone or + remplissage)
  - Immobilisation 6 weeks → labral healing on anterior glenoid edge *[C2 · transcript: "labrum heals on anterior acromion edge"]*
  - Then physiotherapy
  - First two months → no **abduction + external rotation** → max anterior tension
- **Rotator cuff** → good cuff, sutures, knots, bone quality:
  - **Passive** ROM → after 4 weeks
  - **Active** motion → week 6
  - Full ROM → 8 weeks
- Poor cuff / bone quality → immobilisation up to two months → then physiotherapy
  - Risk → **stiff / frozen shoulder**
  - → **pendulum exercises** sometimes advised
- Sling:
  - Bankart → arm by side, **arm sling**, elbow 90°
  - Cuff → **abduction pillow** → repair relaxed

	Bankart (± remplissage)	Rotator cuff (good quality)
Immobilisation	6 weeks	Passive ROM after 4 weeks
Restriction	No abduction + ER for 2 months	Active from week 6; full ROM at 8 weeks
Sling	Arm by side, elbow 90°	Shoulder abduction pillow

SCR and Latissimus Dorsi Transfer Ask AI

- **Superior capsular reconstruction** (SCR) → indications:
  - Failed **previous cuff repair**
  - Goutallier 3–4 → repair would fail
- SCR → **patch**
  - Glenoid side → two anchors; humeral side → two anchors
  - Humeral head **depressor** → no upward migration
- **Latissimus dorsi transfer** → massive cuff tears
  - Massive, not reducible → only solution
  - Open or **arthroscopic**

All-Suture Anchor Ask AI

- Third stage of anchor evolution
- Simply a **suture** → loosened or fallen inside → still a suture
- Precise instruments; **sleeve** placed correctly; deployed properly → no loosening
- "Magic solution":
  - Not **metal** → no protrusion, no arthritis
  - Not PEEK → no dissolution, no **foreign body reaction**

Partial-Thickness Tears: Bursal vs Articular Ask AI

- Partial-thickness tear → biologically progresses
  - No tear **heals by itself**
- **Bursal side** < 50% thickness
  - Glenohumeral view → no articular extension → **debridement** only
  - Patient warned → may progress
- **Second school** → complete remaining tear even if < 50% → repair

First-Time Dislocators: Common Defect Ask AI

- **Head defect** commoner
  - Hill-Sachs → 75%
  - Glenoid defect → 5–50%
- Hill-Sachs → first common sign in **first-time dislocators**

Coracoacromial Ligament in Subacromial Decompression Ask AI

- **Coracoacromial ligament** → arises at medial acromial edge *[C3 · transcript: "coracohumeral ligament (in the question)"]*
  - Oblique course → confirms origin from coracoid
- Recognised → not cut
- Cut if → knife-like, **impinging** on cuff
- Or narrow space → widen

Cuff Tear and Instability Ask AI

- Rotator cuff tear → **no instability**, unless associated biceps pathology
  - **Subscapularis** tear → biceps subluxes out of bicipital groove → giving way
- First-time dislocator at fall → Bankart + cuff tear together

Predicting Reparability on MRI Ask AI

- Pre-op MRI → read yourself; don't depend on centre report
- Massive tear, signs of **irreparability**:
  - Goutallier 3–4 → most likely not repairable
    - Every suture cuts through
  - Tendon retracted to / **medial to glenoid edge** → irreparable
- **Atrophy** → also a factor
  - Tendon = elastic band → retracts **medially** when torn
  - More medial → longer duration → more atrophy
  - Lost **elasticity** → does not come with pull

Injection in Cuff Disease Ask AI

- **Steroid** in partial tear → relieves pain, congestion
  - Erodes tendon → **complete tear** within 6 months or after 4–5 injections
- Pure **subacromial impingement** or calcific tendinitis → steroid beneficial
- **Hyaluronic acid** better → literature: no difference vs steroid
- **PRP** used by some
- Hyaluronic acid + PRP → safer than steroid, esp. **without MRI**
- No injection before **MRI** → many cuff tears seen after steroid
  - Partial tear → hyaluronic acid or PRP
  - Intact tendon → **steroid** allowed
- After injection → **physiotherapy**
- Goutallier 3–4 → physiotherapy only
  - Or **reverse shoulder arthroplasty** if affordable
  - Reverse → conditional on intact **deltoid**

> **Pitfall**
> - Steroid into partial tear → complete tear within 6 months / 4–5 injections
> - Inject only after MRI
> - Partial tear → hyaluronic acid or PRP; intact tendon → steroid
> - Reverse shoulder arthroplasty → requires intact deltoid

Rotator Cuff Disorders — Part 1

(Anatomy, Pathogenesis and Diagnosis of Rotator Cuff Tears)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 103) “scapulothoracic: pectoralis major and pectoralis minor” → “scapulothoracic: pectoralis minor only”

C2 (p. 103) “subscapularis arises from suprascapular fossa” → “subscapularis arises from subscapular fossa”

C3 (p. 106) “subscapularis has no equivalent, controls superiorly” → “supraspinatus has no equivalent”

Queried - kept as spoken, but looks wrong

Q1 (p. 111) As spoken; retraction categories skip 3-5 cm (usual grading: <1, 1-3, 3-5, >5 cm).

Rotator Cuff: Overview and Anatomy Ask AI

Lecture Scope Ask AI

- Two sessions → this one covers the basics
- Next session → management in detail
  - **Arthroscopic repair**
  - **Mini-open repair**
  - **Tendon transfers**
  - **Reverse shoulder** salvage procedures
  - **Primary cuff repair**
  - **Failed cuff repair** → what to do when the patient presents with it
  - **Irreparable cuff** diagnosed from the start

Epidemiology Ask AI

- **Cuff tears** → among commonest causes of shoulder pain
- Among commonest causes of **disability** in patients > 60 years

Components Ask AI

Cuff

- Four muscles
  - Anterior → **subscapularis**
  - Superior → **supraspinatus**
  - Posterior → **infraspinatus** and teres minor
- Some classifications → **five units**
  - Subscapularis split → upper subscapularis + lower subscapularis

Relations of the Cuff Ask AI

- **Coracoacromial arch** = coracoid process + coracoacromial ligament + acromion
  - Cuff passes beneath it
- **Subacromial bursa** → separates cuff from coracoacromial arch
- Subscapularis → anterior; borders and passes over **long head of biceps**
- Long head of biceps → **intra-articular**
- **Short head of biceps** → arises from coracoid process
- Long head of biceps relations → basis of **biceps pulley injuries**
  - To subscapularis
  - To supraspinatus above it
  - To coracoacromial arch
- **Coracohumeral ligament** → arises from coracoid; important part of biceps pulley

Muscles Moving the Shoulder Ask AI

- **14 muscles** move the shoulder (also covered in reverse shoulder lecture)
  - 8 → **glenohumeral joint**
  - 6 → **scapulothoracic** articulation
- Scapulothoracic six
  - Serratus anterior, trapezius, pectoralis minor, levator scapulae *[C1 · transcript: "scapulothoracic: pectoralis major and pectoralis minor"]*
  - Rhomboid major, rhomboid minor
- Glenohumeral eight
  - **Deltoid**
  - Four rotator cuff muscles
  - Teres major, pectoralis major, latissimus dorsi → **stabilisers** of glenohumeral joint
- Latissimus dorsi, pectoralis major, teres major → frequent **tendon transfers**
  - Indication → irreparable rotator cuff tears (next session)

Individual Cuff Muscles Ask AI

Muscle	Origin	Insertion	Action	Nerve	Artery
Subscapularis	Subscapular fossa	Lesser tuberosity (four facets)	Internal rotation, adduction, centralises head on glenoid	Upper + lower subscapular	Subscapular *[C2 · transcript: "subscapularis arises from suprascapular fossa"]*
Supraspinatus	Supraspinous fossa	Greater tuberosity — superior facet	Abduction	Suprascapular	Suprascapular
Infraspinatus	Infraspinous fossa	Greater tuberosity — middle facet	External rotation	Suprascapular	Suprascapular + circumflex scapular
Teres minor	Lateral border of scapula — superior part	Greater tuberosity — inferior facet	External rotation (with infraspinatus)	Axillary (C5, C6)	Suprascapular + circumflex scapular

- Lesser tuberosity **four facets** → basis of subscapularis tear classification

Rotator Cuff Footprint Ask AI

Point vs Surface Area Ask AI

- Old view → tendon attaches at a **single point** on greater tuberosity
- Newer studies → **surface area**, mostly ~2 cm × 2 cm
- Repair concept changed
  - From → **single anchor** fixation of the cuff
  - To → entirely different concept
    - **Transosseous-equivalent** repair
    - **Double-row** repair
    - **Transosseous** repair
- Cuff holds over a surface area, not a point
  - **Infraspinatus** → very large surface area
  - Teres minor
  - **Subscapularis** → anterior, on lesser tuberosity
- Repair → torn tendon laid over its **whole insertion area**
  - → **proper healing**; not single-point fixation

Insertion Dimensions Ask AI

- Cuff tendons except subscapularis → roughly ~2.5 × 2 cm

Tendon	Insertion size	Note
Subscapularis	~4 × 2 cm	Largest; lesser tuberosity, adjacent to bicipital groove
Supraspinatus	~20 × 15 mm	Value unclear in recording
Infraspinatus	30 × 20 mm	
Teres minor	~30 × 20 mm	

- Infraspinatus → occupies a large area of the greater tuberosity

High Yield
- Footprint = **surface area**, not a point (~2 cm × 2 cm)
- Subscapularis footprint largest → ~4 × 2 cm
- Footprint concept → transosseous-equivalent, double-row, transosseous repair

Posterosuperior Cuff Ask AI

Infraspinat

- **Posterosuperior cuff** important → infraspinatus, not only torn supraspinatus
  - Infraspinatus fibres **merge with supraspinatus** → large part of greater tuberosity insertion
  - Humeral head completely **exposed** → large part of infraspinatus likely torn too
  - Torn infraspinatus retracts **posteriorly and medially**
  - Retrieve it from there → else repaired cuff comes back retracted

Supraspinatus Extension to Lesser Tuberosity Ask AI

- **Sugaya** and Japanese group (since 2009)
- Supraspinatus → continues onto **lesser tuberosity**
- → surrounds biceps tendon
- Biceps pulley → contains **supraspinatus and subscapularis fibres**
- Supraspinatus → shares in **pulley system** of long head of biceps

Cuff Function and Force Couples Ask AI

Actions Ask AI

Cuff

- Forms **force couples** → two couples in the shoulder
  - With deltoid → **abduction and elevation**
- Counteracts **upward pull of deltoid** on humeral head
  - Deltoid abducts → also exerts superior pull on head
  - Cuff tear / massive tear / cuff arthropathy →
    - Head **translates superiorly** → articulates with acromion undersurface
    - → **acetabularization** of the acromion
- Keeps humeral head **centred on glenoid**
  - Anterior or posterior head migration → erodes glenoid edge → **glenoid defects**
- Assists **rotation of humeral head** → important in elevation

Must-Know

- Forms a mechanical **force couple**
- Antagonises **upward pull of deltoid** on humeral head
- Keeps humeral head **centred** on glenoid
- Assists **rotation** of humeral head in elevation

Vertical and Horizontal Couples Ask AI

Couple	Limb	Limb	Net result
Vertical	Deltoid → pulls upward (attached to acromion)	Supraspinatus, infraspinatus, teres minor, subscapularis → pull medially and downward	Abduction without upward head translation
Horizontal	Subscapularis → pulls medially and anteriorly	Infraspinatus + teres minor → pull medially and posteriorly	Humeral head centred on glenoid

- Deltoid contracts → abduction + **upward pull** on humeral head
- Vertical couple absent → head repeatedly hits acromion → **arthritic shoulder**
  - = what happens in a rotator cuff tear
- Cuff line of force → stabilises head, centres it, pulls it downward
- Subscapularis alone → could dislocate head or make it **lax anteriorly**
  - Infraspinatus (other limb) → balances it
- Couple in **equilibrium** → head stays exactly in place

Couple Imbalance Ask AI

- Both couples, vertical and horizontal → must be **balanced**
- Imbalance → mild or complete
- Complete imbalance = **pseudoparalytic shoulder**
- Tolerable disturbance → correctable by **repair**
- Loss anteriorly only / posteriorly only / superiorly only → **horizontal and vertical imbalance**
  - → pseudoparalytic shoulder → likely **reverse shoulder** or large tendon transfer

Golf-Ball Analogy Ask AI

- Humeral head on glenoid = **golf ball on a tee**
- Completely unstable unless surrounded by
  - Capsule, ligaments, **labrum**
  - Four components of the rotator cuff

Five-Unit Cuff Ask AI

- Subscapularis → lesser tuberosity, very close to **bicipital groove**
  - → related to long head of biceps injuries
- Some classifications → subscapularis split into **upper and lower**
  - Two muscles anterior ↔ infraspinatus + teres minor posterior
  - Reason → one muscle but strong and very broad
- **Supraspinatus** → no equivalent; controls superiorly *[C3 · transcript: "subscapularis has no equivalent, controls superiorly"]*

Impingement Syndromes Ask AI

Rotator Cuff Disorders — Three Groups Ask AI

- **Impingement syndrome**
- **Rotator cuff tears**
- **Calcific tendinitis**
- MD essay on rotator cuff disorders → must cover all three

The Five Syndromes Ask AI

- "Impingement" ≠ subacromial only → **five** syndromes
- **Primary subacromial impingement** → intrinsic and extrinsic
- **Subcoracoid impingement**
- **Secondary impingement** (as opposed to primary)
- **Posterosuperior impingement** = internal glenoid impingement
- **Anterosuperior glenoid impingement**
- Subacromial discussed last → largest, most important, weightiest

Exam Pearl

- MD / viva: "Impingement syndromes you know?" → **five** types
- Answering subacromial only → 20% of the question

Subcoracoid Impingement Ask AI

Subcoracoid

- Causes
  - **Idiopathic** → anatomical abnormality: coracoid projects downward → contacts subscapularis muscle or tendon
  - **Iatrogenic**
    - **Latarjet** → bone block passed through split subscapularis
      - Poorly positioned or placed too far → impinges on subscapularis
    - **Posterior glenoid osteotomy** (glenoid neck, version correction)
      - → anterior glenoid edge closer to subscapularis, or coracoid moved out of position
  - **Fracture / malunion** → coracoid process or lesser tuberosity
  - **Space-occupying lesions** → ganglion, calcification
- Pathology → ↓ **coracohumeral distance**
  - Normal → 8.6 mm
  - 6.7 mm → **stenosis**
- **Subcoracoid space** → four structures
  - Articular cartilage of humeral head
  - Subscapularis muscle
  - Subscapularis bursa
  - Rotator interval structures
- Examination → **tenderness over coracoid** (not AC joint)
- **Lidocaine injection** subcoracoid → complete pain relief
- Treatment → **coracoplasty** or physiotherapy
  - Few surgeons favour coracoplasty
  - **French school** → rarely see or do it; literature reports many coracoplasties

Secondary (Extrinsic) Impingement Ask AI

Secondary

- Distinct from primary impingement (subacromial space)
- **Coracoacromial outlet** / subacromial space → completely normal
- Cause → **glenohumeral instability** + scapulothoracic instability (dyskinesia)
- **Overhead athletes** → throwers, volleyball, handball
- **Repetitive microtrauma** + capsular stretch → cuff fatigue
- → **anterosuperior translation** of humeral head
- → **dynamic narrowing** of coracoacromial outlet
  - Head moves upward; space above not narrowed
- MRI
  - Cuff **undersurface degeneration**
  - Partial articular surface tears → **PASTA lesion**
  - **Labral** abnormalities → some relate to internal impingement

Posterosuperior Glenoid (Internal) Impingement Ask AI

Internal

- **Supraspinatus undersurface** vs posterosuperior glenoid labrum
- Position → abduction–external rotation (**ABER**)
- **Baseball pitcher** → forceful external rotation
  - → cuff **tears** and atrophy
- Associated findings
  - **PASTA lesion**
  - **Posterosuperior labral fraying**
  - **Anterior capsular laxity**
- Partial articular surface tear → consider **internal impingement** as cause
  - Cause not corrected, cuff repaired only →
    - may **tear again**
- ABER → cuff caught between **posterosuperior labrum** and humeral head
- **ABER position** → reproduces pain
  - Patient points behind shoulder → glenoid neck / posterosuperior glenoid
- **Jobe relocation test** → posteriorly directed force on humeral head in ABER
  - → opens space between head and posterior labrum
  - → pinched cuff released → **pain relieved**
- MRI, examined in ABER position
  - **PASTA lesion**
  - Posterosuperior glenoid labral fraying
  - Possible **paralabral cyst**
  - **Cystic changes** in greater tuberosity → repeated impingement

> **Clinical Pearl**
> - Partial articular surface tear → think **internal impingement**
> - Repair cuff without correcting the cause → re-tear
> - Jobe relocation test → posterior force in ABER **relieves** pain

Anterosuperior Glenoid Impingement Ask AI

ASGI

- **Biceps pulley** + subscapularis vs anterosuperior glenoid rim / labrum
- Position → arm **elevated**, horizontally adducted, internally rotated
  - Same position → elicits pain on examination

Feature	Posterosuperior (internal)	Anterosuperior
Impinged structure	Supraspinatus undersurface	Biceps pulley + subscapularis
Impinged against	Posterosuperior glenoid labrum	Anterosuperior glenoid rim / labrum
Arm plane	Abduction	Horizontal adduction + elevation
Rotation	External	Internal

Biceps Pulley and Anterosuperior Impingement Ask AI

ASGI

- **Biceps pulley** (shoulder pulley) → four components
  - **Coracohumeral ligament**
  - **Superior glenohumeral ligament**
  - **Supraspinatus fibres**
  - **Subscapularis fibres**
- Function → stabilises long head of biceps in **intra-articular position**
  - Compression / stabilising structure in **bicipital groove**
  - → prevents **intra-articular dislocation**
- Pulley torn → long head of biceps **anterior** or intra-articular medial subluxation
- Detailed pulley anatomy → see Biceps Pulley section
- Subscapularis + supraspinatus tears (from anterosuperior impingement) + **pulley injury**
  - → ↑ risk of anterosuperior migration and impingement
- Vicious circle
  - **Torn pulley** (traumatic or degenerative) → long head of biceps instability → medial subluxation
  - → long head of biceps no longer a **depressor** of humeral head
  - + **supraspinatus and subscapularis** tear → superior and anterior head support lost
  - → anterosuperior impingement → **vicious circle**

Primary Subacromial Impingement Ask AI

- Four others → subcoracoid, secondary / extrinsic, internal / posterosuperior, anterosuperior
- **Subacromial impingement** → what "impingement syndrome" usually means
- Space beneath **coracoacromial arch**
- **Subacromial space** → bursa, rotator cuff tendons, biceps tendon
  - Trapped between humeral head and coracoacromial arch

Intrinsic Ask AI

Subacromial

- Structures passing under arch → **enlarged or inflamed**
  - Bursa, biceps tendon, rotator cuff → **abut against arch** in limited space
- Causes
  - **Tendinosis** / thickening of rotator cuff
  - **Calcific deposits** → calcific tendinitis
  - **Bursitis** / bursal inflammation

Extrinsic Ask AI

Subacromial

- Problem → **space itself** narrowed, not structures beneath arch
- Causes
  - **Acromial spur** → type 2, type 3 or type 4 acromion
  - **Acromial fractures**
  - **Os acromiale** → ununited apophysis of acromion
  - **AC joint osteophytes** → project into subacromial space
  - **Greater tuberosity** malunion or nonunion
  - **Humeral head exostosis**

Primary impingement	Problem	Causes
Intrinsic	Structures under coracoacromial arch enlarged / inflamed	Tendinosis, calcific deposits, bursitis
Extrinsic	Space available for cuff diminished	Acromial spur, acromial fracture, os acromiale, AC osteophytes, malunion, exostosis

Rotator Cuff Tears: Aetiology and Classification Ask AI

Disease Progression Ask AI

- **Oedema and haemorrhage**
  - → bursitis, tendinosis, impingement, calcific deposits
  - → **partial tears**
  - → **full-thickness tears**
  - → **rotator cuff arthropathy** (if untreated)

Aetiology Ask AI

- **Traumatic** → usually < 40 years
  - **High-velocity trauma** → partial or full-thickness tears
  - **Repetitive microtrauma** → overhead athletes, capsular stretching
  - **Overuse** → carpenters
  - Heavy manual work (occupation unclear in recording)
- **Non-traumatic** → > 40 years
  - **Degenerative**, effort-related → housewives, painters, electricians
  - **Subacromial impingement** → os acromiale, type 2 or 3 acromion
  - **Shoulder instability** → secondary extrinsic and posterosuperior impingement
    - Unstable head → superior translation → dynamic narrowing of coracoacromial arch
    - **Anterior capsular redundancy** → cuff pinched between head and posterosuperior glenoid
  - **Inflammatory** → calcific tendinitis, rheumatoid arthritis, crystal deposition

Classification Systems Ask AI

- **Acute** or chronic
- **Degree** → partial or full thickness
- **Aetiology** → traumatic or degenerative
- **Topography** → anterior, superior or posterior
- **Tendon retraction**

Habermeyer Topographic Classification Ask AI

Habermeyer

Zone (sagittal plane)	Position	Tendon
A (1)	Anterior	Subscapularis
B (2)	Superior	Supraspinatus
C (3)	Posterior	Infraspinatus + teres minor

Partial-Thickness Tears Ask AI

Grade	Articular-sided	Bursal-sided
1	< 25% of tendon	25%
2	~50% or < 50%	< 50%
3	> 50%	> 50%

- Cuff → articular side and **bursal side**; graded by tendon involvement

Full-Thickness Tear Patterns Ask AI

- **Crescent**
- **Reversed L**
- **L-shaped**
- **Trapezoid**
- **Massive** / retracted
- Each pattern → own reduction and repair method
  - **Massive** → may not reduce to footprint → irreparable-tear procedures
  - **Reverse L** → bring cuff forward from posterosuperior → anchor → side-to-side repair
  - **Partial articular-sided** →
    - **Intra-articular** repair from inside the joint
    - Or convert to **full-thickness tear** → then repair
    - Many schools → discussed under tear management

Retraction Ask AI

- < 1 cm
- 1–3 cm
- > 5 cm *[Q1 · As spoken; retraction categories skip 3-5 cm (usual grading: <1, 1-3, 3-5, >5 cm).]*
- Based on how far tendon is displaced / drawn **medially**

Superior Migration, Arthropathy and Massive Tears Ask AI

Acromiohumeral Distance Ask AI

- Cuff tear → **opposing couple** to deltoid lost → head pulled superiorly
- **Acromiohumeral distance** → acromion undersurface to upper edge of humeral head
  - Normal → ~11–16 mm
- < 6 mm → sequence
  - Cuff tear → **proximal (superior) migration**
  - → humeral head **abuts** acromion
  - → **acetabularization** of acromion
  - → rotator cuff arthropathy

High Yield

- Normal **acromiohumeral distance** → 11–16 mm
- < 6 mm → abutment → acetabularization → cuff arthropathy

Hamada Classification Ask AI

Hamada

Grade	Finding
1	Acromiohumeral distance > 6 mm
2	Acromiohumeral distance < 5 mm
3	< 5 mm + acetabularization of acromial undersurface
4A	Glenohumeral arthritis, no acetabularization
4B	Glenohumeral arthritis + acetabularization
5	Complete collapse of humeral head → full cuff arthropathy

- Also attributed to **Fukuda**
- **Radiological** classification

Effects of Cuff Deficiency Ask AI

- Loss of **centralising force** on humeral head
- Deltoid–cuff couple lost → **superior migration** of humeral head
- Upward translation → erodes acromion undersurface → **acetabularization**
- Destroys **glenohumeral cartilage**
- **Superior glenoid** erosion
- **Humeral head** destruction
- **Osteoporosis**
- Coracoacromial ligament → torn or **eroded**
- **AC joint** destruction
- Summary → loss of centring → superior migration → joint destruction → superior glenoid erosion
  - → osteoporosis, head destruction, **cuff arthropathy**, AC joint destruction

Collin Classification Ask AI

Collin

- Basis → subscapularis split into upper + lower → **horizontal balance**
- Five units → superior subscapularis, inferior subscapularis, supraspinatus, infraspinatus, teres minor

Type	Torn	Result
A	Supraspinatus + upper half of subscapularis	No proximal migration → lower subscapularis intact
B	Supraspinatus + both parts of subscapularis	Anterosuperior escape; infraspinatus + teres minor intact → head cannot go back
C	Infraspinatus + supraspinatus + upper half of subscapularis	Superior migration; both horizontal limbs partly intact → no anterosuperior / posterosuperior translation
D	Supraspinatus + infraspinatus	Superior translation
E	Supraspinatus + infraspinatus + teres minor (whole posterior cuff)	Posterosuperior escape → head cannot move anteriorly

- **Type B and type E** → pseudoparalytic shoulder
- Also covered in reverse shoulder lecture; again under irreparable cuff

Subscapularis Tears Ask AI

- **Subscapularis tears** → "dark area" for many arthroscopists
- Scope in without noticing tear / without knowing treatment → patient left with a problem

Clinical Detection Ask AI

- Proper preoperative examination → tear predicted, even if most muscle intact
- Tests → **lift-off**, belly press, bear hug
  - Not done well or not compared with **other side** → tears missed

Key Features Ask AI

Subscap

- Tear usually **articular-sided** at lesser tuberosity
- Always enter **glenohumeral joint** in cuff tears
  - Straight to subacromial space → subscapularis tear missed
  - = **hidden tears**; arthroscopy greatly helps detection
- **Associated with biceps tendon injuries**
  - Superior glenohumeral + coracohumeral ligaments → part of reflection pulley of biceps

Pitfall

- Subscapularis tears → **hidden tears**, articular side of lesser tuberosity
- Skipping glenohumeral joint inspection → tear missed
- Lift-off, belly press, bear hug → compare with **other side**

Classification by Extent Ask AI

Subscap

Type	Extent	Notes
1	Partial tear of upper 1/3	
2	Complete tear of upper 1/3	
3	Upper 2/3 torn	
4	Entire subscapularis torn	Head centred; fatty infiltration grade 1–3
5	Complete tear + subluxation	Anterior limb of horizontal couple lost; fatty infiltration grade 3–4

- Classification attributed to Lafosse (name unclear in recording)
- Fatty infiltration → **Goutallier classification** (MRI lecture)
- Type 5 → anterior strut lost, especially with torn supraspinatus
  - → **anterosuperior** or anterior subluxation of head
- Type 4 → head still **centred**
  - → supraspinatus likely intact; no complete weakness or full fatty infiltration
- Type 5 → head **subluxed anteriorly**
  - Anterosuperior escape, complete tear, severe fatty infiltration
  - → pathway of **pseudoparalytic shoulder**

Biceps Pulley and Comma Sign Ask AI

Pulley Components Ask AI

- **Biceps pulley** = reflection pulley = suspension pulley
  - **Coracohumeral ligament**
  - **Superior glenohumeral ligament**
  - **Supraspinatus**
  - **Subscapularis**
- Aim → stabilise long head of biceps in **bicipital groove** and intra-articular position
  - → prevent **subluxation**
- Drawing
  - Grey → **coracohumeral ligament**, above tendon
  - Green → **superior glenohumeral ligament**, surrounding tendon from below
  - Blue → **long head of biceps**, articular portion → bicipital groove
- **Supraspinatus fibres** cross bicipital groove
  - → insert onto **medial lip** of groove via pulley system
- Opened pulley
  - **Coracohumeral ligament** → from coracoid process
  - Floor → **superior glenohumeral ligament** + subscapularis and supraspinatus contribution
  - Structure under traction → biceps tendon

Bennett Classification Ask AI

Bennett

Type	Torn	Intact	Long head of biceps
1	Medial insertion of subscapularis	Coracohumeral + superior glenohumeral ligaments	Medial subluxation; no dislocation
2	Coracohumeral ligament + superior glenohumeral complex	Subscapularis	Medial subluxation; no dislocation
3	Medial subscapularis + medial coracohumeral ligament + superior glenohumeral complex	—	Medial intra-articular dislocation
4	Superior glenohumeral + coracohumeral ligaments	Subscapularis	Anterior subluxation / dislocation; not intra-articular
5	Whole complex → coracohumeral, subscapularis, superior glenohumeral	—	Medial subluxation (subscapularis torn)

- Type 1 → medial lip weakened by **subscapularis tear**
- Classification of **pulley injuries**

Comma Sign Ask AI

- Tissue shaped like a **comma**
- Described by **Lo and Burkhart** (2003)
- Landmark → **upper edge** of subscapularis
  - Chronic tear → subscapularis displaced medially → leading edge unidentifiable
- = **humeral attachment** of superior glenohumeral + coracohumeral ligaments
  - → part of pulley system of long head of biceps
- → subscapularis tears **very commonly associated** with long head of biceps injuries
- Attached to **upper border** of subscapularis
- Must be **retrieved and reattached**
  - Bridging part between subscapularis and supraspinatus
- Not reduced and repaired to head with **anchors** → rest of cuff cannot be properly reduced

Clinical Pearl

- **Comma sign** = humeral attachment of superior glenohumeral + coracohumeral ligaments
- Landmark for upper edge of retracted subscapularis
- Reattach with anchors → else cuff cannot be reduced

Arthroscopy and Repair of Subscapularis Tears Ask AI

Arthroscopic Findings Ask AI

- Subscapularis tear
  - Biceps tendon **subluxed**
  - **Bare area** of lesser tuberosity
  - Subscapularis completely **atrophic** (Goutallier)
- Torn subscapularis with biceps still in groove
  - Can very easily develop **medial subluxation**
    - or dislocation
- No attachment on lesser tuberosity + torn **coracohumeral and superior glenohumeral ligaments**
  - → way open for medial subluxation of biceps
- Long head of biceps dislocation routes (exam question)
  - **Medial intra-articular**
  - **Anterior extra-articular**

Technical Tips Ask AI

Subscap

- Open **rotator interval** well → remove capsular remnants → visualisation
- **Accessory anterior portal**
  - → grasper holds tendon; **traction sutures** pull medially displaced tendon
- **70° scope** → completely different view → greatly facilitates procedure
- **Anterosuperior portal** → work through it
- **Biceps tenodesis** or tenotomy → helps visualisation and repair
  - Most tears with long head subluxation → tendon **frayed** and torn → tenotomy / tenodesis usual
- **Flexion and internal rotation** of arm → exposes tear
- **Posterior humeral head translation** (surgeon or assistant) → exposes tear

Facet Classification Ask AI

Subscap

- Based on **four facets** of lesser tuberosity → A, B, C, D

Clinical Evaluation Ask AI

History Ask AI

- **Site of pain**
- **Weakness**
- **Time lag** before presentation
- **Occupation**
- **Dominant hand**
- **Deep trigger points** in cuff disease
  - Posterior → supraspinatus
  - Anterior → infraspinatus, subscapularis, teres minor

Look, Feel, Move Ask AI

- **Look** → wasting, winging, pseudoparalytic patient unable to lift arm
- **Feel** → medial to lateral
  - Sternoclavicular joint → coracoid process → clavicle → greater tuberosity → acromion → AC joint
- **Move** → passive and active range; compare both sides

Range lost	Cause
Active only	Rotator cuff tear or nerve injury
Active + passive	Mechanical block → frozen shoulder, severe impingement, labral tear

Specific Cuff Tests Ask AI

Muscle	Tests	Lag test (of the four)
Subscapularis	Gerber lift-off, belly press (Napoleon), bear hug, internal rotation lag sign	Lift-off
Supraspinatus	Jobe / empty can, drop arm	Drop arm
Infraspinatus	External rotation lag sign	External rotation lag
Teres minor	Hornblower sign	Hornblower

- **External rotation lag sign** → patient cannot maintain arm in position → arm drops
- **Hornblower sign** → like blowing a horn; patient cannot maintain arm in position
- **Four lag tests** → easy to use in clinical examination exam

Subacromial and AC Joint Tests Ask AI

- Subacromial impingement
  - **Neer sign**
  - **Hawkins–Kennedy** test
  - **Neer impingement test** (by injection)
  - **Jobe / empty can** test
- AC joint
  - **Tenderness**
  - **Cross-body adduction** → scarf test

Radiology Ask AI

X-Ray and MRI Assessment Ask AI

- **X-rays** and MRI
- MRI → each cuff muscle in **sagittal, coronal and axial** views
- Look for
  - **Traction acromial spur**
  - **Acromial type**
  - **Calcific tendinitis**
  - **Massive tear**
  - **Fukuda / Hamada** grade of cuff arthropathy

Critical Shoulder Angle Ask AI

- Line joining upper and inferior glenoid ↔ line to **lateral edge of acromion**
- Related to **rotator cuff tears**

Critical shoulder angle	Significance
< 30°	Greater liability to glenohumeral arthritis
30–35°	Normal
> 35°	Associated with rotator cuff tears

- Case → 39° before surgery → 33° after surgery
- Greater reduction → lower **re-tear risk**

High Yield
- **Critical shoulder angle** normal → 30–35°
- > 35° → cuff tears; < 30° → glenohumeral arthritis
- Reducing it → lower re-tear risk

MRI Interpretation Ask AI

MRI

- Based on **mobile hydrogen ions**
- Signal → black / dark (none), grey (intermediate), white (bright)
- **T1 and T2** → covered in MRI lecture
- Remain black → **non-mobile protons**
  - Cortical bone
  - Ligaments → superior and middle glenohumeral
  - Tendons → rotator cuff and biceps
  - **Glenoid labrum**
- Assess → ligaments, bones, tendons, labrum, **bursa**

Sagittal landmark	Structure beneath
AC joint	Supraspinatus

Sagittal landmark	Structure beneath
Spine of scapula	Infraspinatus
Coracoid process	Subscapularis (biceps tendon lateral to it)

- **Goutallier** cut → clavicle and acromion above; supraspinatus, infraspinatus, teres minor; coracoid and subscapularis
  - → cut where **fatty infiltration** is documented
- **Partial tears** → bursal or articular side
- **Full-thickness** / retracted tear → e.g. ~3 cm retraction
- Always document
  - **Retraction distance** → reparable vs not reparable
  - **Muscle atrophy** → Goutallier classification
- Horizontal line across fossa
  - Muscle bulk above or below → **wasting** / fatty degeneration
- **Calcific deposits** → black
  - on MRI (calcium)
- Subscapularis tear → **biceps tendon** in place
  - Retracted subscapularis tendon
  - Muscle **atrophied**
  - Bare area of lesser tuberosity

Acetabularization Ask AI

Acromion

- Acromion undersurface → normally **straight** (or slightly sloping downward)
- Curved / semicircular depression → humeral

head converted acromion into an **acetabulum** (like acetabulum with femoral head)

Questions and Answers Ask AI

Is the Long Head of Biceps Part of the Cuff? Ask AI

- One of the **depressors** of humeral head
- Not part of the cuff → but **very closely related**
  - Relation in bicipital groove
  - Relation to subscapularis
  - Potential to dislocate **medially or anteriorly**
- Half of a separate muscle → one of the **rotator interval** and joint contents

Unexplained Shoulder Pain Ask AI

- Many causes
  - **Cervical** causes
  - **Suprascapular nerve** entrapment
  - Rotator cuff
  - **Microinstability**
  - Arthritis
- No such thing as unexplained shoulder pain → proper examination + **MRA** or MRI
- Athlete points to **posterosuperior** glenoid
  - MRI report → subacromial impingement → subacromial decompression performed
  - Real story → **articular-surface tear** + anterior capsular laxity
  - Not all corrected → patient let down
- **Internal impingement** → many improve with proper physiotherapy and muscle recruitment
  - Not all need surgery
  - Do not rely on MRI report alone → many present after such operations still in severe pain

Why Fix the Comma Sign? Ask AI

- Part of **biceps pulley** → biceps preserved → must reduce it
- Attached to **superior edge of subscapularis**
  - Not returned to footprint → subscapularis not returned to footprint
- Biceps lies alongside subscapularis and supraspinatus → cuff pathology reaches it
- Except ~30–40% of cases → **biceps tenotomy** or tenodesis usually needed

Hidden Articular-Sided Tears Ask AI

- **Articular-sided partial tears** → sometimes not visible
- Posterosuperior / internal impingement → **MRA in ABER position**
  - Shows tear at this site + posterosuperior labral fraying

Combined Cervical and Shoulder Pathology Ask AI

- Scenario → herniated disc, **nerve root entrapment**, brachialgia + impingement
- Depends on which symptoms more painful
- No **upper motor neuron** lesion; conservative treatment failed (physiotherapy, injection); surgery indicated
  - First → refer to **spine surgeon**
  - Wait for his decision
- Spine surgery planned → wait (unless complete cuff problem needs urgent repair)
  - **Cervical discectomy** + fixation → ready for cuff repair after ~1 month
- Or repair cuff → warn patient that part of the pain

will remain **persistent** (cervical condition)

- **Suprascapular nerve entrapment** → especially with chronic retracted tear
  - → **suprascapular nerve release** in same arthroscopy
- Spine options → discectomy, fixation, injection, **ablation**

MR Arthrography — Routine? Ask AI

- Not in all cases from the start
- **Internal impingement** / posterosuperior pain → skip plain MRI → MRA
- Normal MRI or suspected **capsular redundancy**
  - Capacious capsule → **microinstability**
  - → head migrates superiorly → needs MRA

When to Suspect a Cervical Lesion Ask AI

- **Numbness** extending to forearm or hand → cervical cause likely
  - Shoulder pain → radiates at most to arm (about elbow)
- Cervical problem mistaken for shoulder → improves with **sympathetic**-acting drugs and similar
- **Trapezius trigger points** → injection relieves pain, or physiotherapy
- Superficial examination → female patient not properly exposed
  - → **scapular dyskinesia** missed
  - → abnormal muscle pattern around scapula missed
  - → severe **trapezius myositis** missed → diagnosis completely different

Partial vs Full-Thickness Tears Ask AI

Feature	Partial-thickness	Full-thickness
Extent	Bursal side not reaching down, or articular side not reaching up	Medial retraction (0.5–1 cm)
MRI	Tendon continuous; thinned, not a sharp black band; eroded from above (bursal) or below (articular)	Discontinuous, retracted
Clinical	Movement possible but very weak → power 3 or 4	Power 1 or 2

Painless Full-Thickness Tears Ask AI

- Main **pain generator** → biceps tendon
  - Very strong pain generator + **tenosynovitis**
  - Biceps tenotomy / tenodesis → most pain goes away
- Remaining pain → **arthritis** and bone friction

Closing Case: Examine Before Operating Ask AI

- Young man, 25 years, referred for "the operation"
- **Car accident** → car overturned
- Cannot move any finger
- Not told → **brachial plexus** torn
- MRI brachial plexus → cords

torn

- Had a cuff tear too → but essentially **no functioning plexus**
- Operating → no motor for these muscles → severe **medicolegal problem**
- Cuff arthropathy → check **deltoid** and axillary nerve sensation

  - Axillary nerve injury missed → surgery useless; surgeon blamed for nerve injury
- Listen to every word
  - **Anterior** vs posterior pain
  - Shoulder **slipping or dislocating**
- Give patient time → not quick history + MRI + "you need surgery"
- Overturned car → shoulder may have dislocated and torn cuff → operate without examination → **lawsuit**
- Proper examination → protects surgeon, protects and benefits patient

Pitfall

- Cuff tear + unexamined **brachial plexus** → no motor; medicolegal disaster
- Check **deltoid** and axillary nerve before arthropathy surgery

Rotator Cuff Disorders — Part 2 (Management)

(Management of Impingement, Rotator Cuff Tears and Calcific Tendinitis)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 137) “pectoralis minor (said twice)” → “pectoralis minor or pectoralis major”

C2 (p. 141) “arthrolysis” → “arthrodesis”

C3 (p. 143) “above 6 years” → “above ~60 years”

Queried - kept as spoken, but looks wrong

Q1 (p. 136) As spoken; massive tear is two or more tendons; the example given has only two.

Q2 (p. 136) As spoken; pseudoparalysis is usually defined as active elevation below 90 degrees with full passive range.

Q3 (p. 139) As spoken; acellular dermal allograft is processed human dermis, not a synthetic graft.

Q4 (p. 140) As spoken; reported healing in patients over 70 is roughly 40-50%, not about 10%.

Q5 (p. 144) As spoken; Burkhart's deadman angle for anchor insertion is 45 degrees, not 60.

Scope of Rotator Cuff Management Ask AI

- Not limited to subacromial impingement → several **impingement syndromes**
- Three topics in management
  - **Impingement**
  - **Rotator cuff tears**
  - **Calcific tendinitis**

Subacromial Impingement Syndrome Ask AI

Conservative Treatment Ask AI

- **Conservative treatment** → initial stage, vast majority of cases
  - Rest
  - Physiotherapy
  - **Steroid injection**; nowadays also PRP and hyaluronic acid
- Caution: steroid in **partial rotator cuff tear**
  - Often widens tear → may progress to full-thickness tear
  - Danger greatest when MRI not done from the start

Injection Routes Ask AI

- Posterior aspect of shoulder
- Lateral aspect → site of acromial **lateral portal**
- **Ultrasound-guided** → if skilled in musculoskeletal ultrasound
  - Subacromial bursa easily identified → needle seen inside bursa
  - 100% sure of intrabursal injection → treats **bursitis** of impingement

Operative Indications Ask AI

- Failed physiotherapy, rest, **activity modification** → operative treatment
- Indications
  - Age older than 40 years
  - Age younger than 40 + **type 2 or type 3 acromion** → will damage cuff
  - Existing **rotator cuff tears**
  - **Biceps tenosynovitis**
- Surgery → open, arthroscopic or **mini-open**

Treat the Cause Ask AI

- Manage the **cause** → not just arthroscopic subacromial decompression

Cause of impingement	Treatment
Space-occupying lesion (calcification)	Remove calcification from rotator cuff
Osteophyte impinging on cuff	Excision of joint osteophyte
Secondary to frozen shoulder	Release of frozen shoulder
Rotator cuff tear	Repair (suture) the tear
Biceps tendon tenosynovitis	Tenotomy or tenodesis

- **Instability** → addressed first
  - Anterosuperior, posterosuperior or internal impingement → problem = shoulder instability
  - Subacromial decompression here → no help; worsens condition
- Diagnosis must distinguish
  - Space-occupying lesion → calcification, osteophytes
  - **Secondary to cuff dysfunction** → frozen shoulder, biceps pathology, cuff tear, instability
- Bursectomy + subacromial decompression → easiest operation, learned first in **shoulder arthroscopy**

Must-Know

- Treat the **cause**, not bursectomy + decompression alone
- Instability-related impingement → decompression worsens condition
- Steroid into **partial tear** → may progress to full-thickness

Arthroscopic Subacromial Decompression Ask AI

Structures Removed Ask AI

- Part of anterior + lateral **acromion** → responsible for type 2 or 3 impingement
- **Acromioclavicular (AC) joint** osteophytes

Steps Ask AI

- **Bursectomy** first
  - Arthroscope in → subacromial bursa removed
  - Appearance → **curtains**, fluffy cotton; must be cleared
  - Essential with cuff tear → good visualisation
- **Coracoacromial ligament** → release is a debatable issue
  - Excision justified
    - X-ray: ligament attachment on acromion or **traction osteophyte**
    - Whole structure feels calcified or bony
    - **Leading edge** impinging on cuff → causing tear
  - Routine release (advocated 7–8 years ago) → no longer practised
  - Role retained even in **massive rotator cuff tears**
  - **Restraint** to humeral head → prevents proximal migration
  - Limits **acetabularization** of acromial undersurface
- Remove **anterolateral osteophytes**; excise anterior acromial osteophytes
- Address **AC joint** → commonest cause of subacromial impingement in Egypt
  - Type 2, type 3 or **hooked acromion** → frankly uncommon in Egypt
  - **Manual workers** → AC joint arthritis or AC joint osteophyte
  - Exam: cross adduction (across chest) = **Scarf sign**
  - Plus tenderness over AC joint
  - AC joint injection → marked relief of impingement symptoms
  - AC osteophyte excision → key step of **acromial decompression**
  - Not only bursa + anterior/anterolateral acromial edge osteophytes

Acromioplasty Techniques Ask AI

- Two techniques → chosen by acromion **thickness** and shape (type 1, 2, 3)

Technique	Method
Limited anterior resection	Removes anterior piece only
Cutting block technique	Undersurface of posterior acromion = cutting block; resect level with it, removing everything anterior

- **Thin acromion** + cutting block → easy acromion fracture
  - Especially with sharp **acromionizer** or new instruments
- Mark **depth** with burr on acromial undersurface
  - ~5–6 mm = diameter of acromionizer
  - Widen to this depth → posterior to anterior
- Cutting block technique → arthroscopic image
- View from **lateral portal**; acromionizer enters posteriorly → cuts undersurface
- Mark with acromionizer → everything anterior removed
- **Dotted line** = cutting block plane

Evidence on Acromioplasty Ask AI

- Literature on **acromioplasty** as additional step in cuff tears
- 2018 **systematic review**, ~3 papers pooled
  - No difference in shoulder function or pain scores, with or without acromioplasty
- Acromioplasty in every case → no longer a rule (view ~ten years ago)
- Assess on entry: instruments moving freely? Something pressing → **remove it**
- Repair instruments angled **45 degrees** forward or backward → needle clears acromion
- Essential step → **bursectomy**; osteophyte removal if present
- **Routine acromioplasty** in cuff repair → no longer the rule
- Second paper → no outcome difference vs **isolated bursectomy**
- Not an essential step in cuff management
- **AAOS** (American Academy of Orthopaedic Surgeons) → routine acromioplasty not needed

High Yield

- Essential step = **bursectomy**, not acromioplasty
- **AAOS** → routine acromioplasty not needed in cuff tears or repairs
- **Coracoacromial ligament** → restraint to proximal migration; no routine release
- **AC joint** → commonest cause of impingement in Egypt

Procedures via the Subacromial Space Ask AI

- Subacromial space → many procedures beyond **subacromial bursectomy**
- **Remplissage** → engaging Hill-Sachs lesion in shoulder instability
  - Anchor in humeral head → sutures retrieved with **bird beak**
  - Tied blindly from inside joint, or tied from above in subacromial space
- **Subscapularis repair** → from inside joint or subacromial space
- **Single-row, double-row and triple-row** cuff repairs
- Removal of **calcific deposits**
- **Suprascapular nerve release**
- **Frozen shoulder release** → may need subacromial space; mostly inside joint
- **Acromioclavicular joint dislocation** → arthroscopic
- Subacromial space ≠ **subacromial impingement**
- **Orientation** → anterior, posterior, medial, lateral → more procedures possible

Operative Setup and Arthroscopic Technique Ask AI

Positioning Ask AI

- **Beach chair position** or lateral decubitus
  - Cuff surgery → beach chair, most surgeons worldwide incl. most American
  - Lateral decubitus → surgeons accustomed to it

Anaesthesia and Manipulation Ask AI

- **Manipulation under GA** → for secondary frozen shoulder from impingement or tear
  - Breaks all **adhesions**
  - Some bleeding in subacromial space + **glenohumeral joint** → beneficial
  - Without it → stiffness again six weeks after cuff repair
- **Hypotensive anaesthesia** → BP lowered within limits
  - Hypertensive at 180 → lowered to 60 or 50/30 → **brain insult** risk
  - Never endanger patient just to see well
  - BP cannot be lowered → **mini-open**

or open repair → must know both

Landmarks Ask AI

- Mark lateral acromion, **scapula**
  - Lateral end of clavicle, **AC joint**
- Fluid → shoulder swells like **balloon** → landmarks and portal sites lost
- Spend 5 minutes drawing **portals**

Portals Ask AI

Standard portals	Additional portals
Posterior	Posterolateral
Lateral	Anterolateral
Anterior	Low anterior

- Choice → according to familiarity
- Common set → **lateral portal** for viewing; anterior + posterior for instrumentation
- **Posterolateral portal** = universal portal → sees anteriorly, reaches posteriorly
- Posterior portal only → cuff **retracts** away
  - Not seen, not reached, not repaired → **switching portals** essential

Diagnostic Arthroscopy Ask AI

- Always **diagnostic glenohumeral examination** → other pathology, e.g. biceps
- Two to three minutes looking around
  - No pathology missed
- **Posterior portal** → 2 cm below and medial to posterolateral acromial edge
- Marked → clavicle, acromion, AC joint, **coracoid process**

Subacromial Bursectomy Ask AI

- Arthroscopic entry into **subacromial space**
  - **Adhesions** released
  - Camera driven into acromial undersurface → breaks it
- Fluffy, cotton-like material = **bursa** → clear well
- Spend half an hour to an hour and a quarter → clear, **shave**, all portals clear
- Bursa may **mask the tear** beneath
- **Marking stitch** → partial tears seen from glenohumeral joint
  - Cannula or needle → **Prolene** suture passed into tear
  - Subacromial view → blue suture leads to tear

Acromioplasty and AC Joint — Arthroscopic Views Ask AI

- Burring acromial undersurface
  - **Periosteum** removed with shaver, not acromionizer → time-consuming
  - Periosteum off → **burr** begins
- Rounded structure = **lateral end of clavicle** → at AC joint
- Needle through **AC joint** → between two bones
  - Lateral end (medial border) of acromion
  - Lateral end of clavicle
- Most **impinging osteophytes** → this area
- **Acromioplasty** → like sawing bone
- Mark intended **depth** with shaver or acromionizer → burr osteophytes

Calcific Tendinitis — Arthroscopic Removal Ask AI

- Needle localizes **calcific tendinitis** from joint surface
  - Site marked with **Prolene**
- Calcifications extruded like **toothpaste**
- Repeated needling → widen → extract all
- May leave cuff **defect** → repair forced
- Calcifications → inflamed shoulder → **very severe pain**
- Open bursa + cuff well → remove all visible **calcifications**
- Step may take half an hour to an hour

Rotator Cuff Repair Ask AI

Goals Ask AI

- **Close the gap**
- **High initial fixation strength** → allows arm movement afterwards
- **Mechanical stability under cyclic loading** → construct stable as arm swings
- **Tendon-to-bone healing**

Open and Mini-Open Approaches Ask AI

- Options → open, mini-open or **arthroscopic**

	Open	Mini-open
Incision	Deltopectoral, 15 or 20 cm	Extension of acromial lateral portal site
Length limit	—	Maximum 5 cm distally (axillary nerve below)
Current use	Rare; learn it if only way to repair	Deltoid splitting approach

- Below 5 cm → **axillary nerve**
- Skin → deltoid → **deltoid splitting** → self-retaining retractor
- Bursa in Egyptians **very thick** → mistaken for cuff
  - Some surgeons start suturing it immediately
- Good imaging + confirmed tear → layer in front = **bursa**
- **Excise** bursa → cuff defect beneath
  - Never suture bursa to anchor as is
- Sequence
  - Lateral portal incision, 5 cm → **deltoid splitting** → identify bursa
  - Confirm bursa, not cuff → bursectomy → tear seen → **refresh the footprint**

Pitfall

- Thick bursa → mistaken for cuff → sutured to anchor
- Mini-open beyond 5 cm → **axillary nerve** at risk
- Acromionizer on footprint → **cancellous bone** → anchor pulls out

Footprint Preparation Ask AI

- Shaver, rasp → roughen to **bleeding surface**
- Acromionizer on footprint → sizeable hole → **cancellous bone** reached
  - Poor **holding power** → anchor pulls out on tensioning
- Shaver abrasion only → bleeding surface → cuff **healing** on it

Arthroscopic Repair — Requirements and Constructs Ask AI

- Requirements
  - Hypotensive anaesthesia
  - **Pressure pump**
  - Good marking; tear well known
- Old belief → cuff attaches at single point
- Since 2006 and 2007 → attachment over **surface area**
- Generations of repair
  - First → **single row**
  - Second → **double row**
  - Next → medial-row anchors; sutures crossed over cuff

→ fixed with **knotless anchors** in lateral row

- **Distribution of tension** → no reliance on single anchor or medial pair
  - Stresses transferred **laterally**
- Crossing sutures press cuff onto **footprint**
  - Helps healing; secures cuff against cutting through
- **Musculotendinous junction** → commonest site of failure
  - Too far **medially** → tear there easily; all tension transmitted medially
- **Mason-Allen** → two suture pairs crossed
  - **Mattress suture** + suture behind and over it
  - No sliding, no cut-through
- Different **types of rotator cuff repair**
- Construct by tear
  - Small → one anchor
  - Larger → two anchors; **double row** → 2 + 2 anchors, sutures in bridge
  - Less tension → **side-to-side sutures** + anchor in that area

Arthroscopic Repair — Case Steps Ask AI

- Bursa not removed → thick → tear dimensions hidden
- **Radiofrequency ablation** (cautery) → easier with bleeding
- **Anchor insertion** at footprint → to laser mark
  - Confirm seated → withdraw handle (**inserter**)
- Sutures retrieved one by one → passed through cuff
- **Lateral row** → sutures over cuff
  - Fixed by screw-like device, as in cruciate ligament reconstruction
- Arm into **internal rotation** → confirm fixation
- Never repair in abduction beyond 20–30 degrees
  - More abduction → cuff nearer footprint → **failure** when arm at side
- **Lateral-row knotless anchor** → another case
  - Inserted below, taking sutures → view from inside after repair
  - All cases → same scenario
- **Additional portals** → e.g. reaching posterior aspect

Partial Articular-Surface Tear Ask AI

- **Partial articular-surface tear** → bursal side may look intact
- View from below → clearly torn
- Options
  - **Transtendon suture** from inside
  - Convert to full-thickness tear → repair

Clinical Pearl

- Never repair in abduction beyond 20–30 degrees → **failure** at side
- Arm into **internal rotation** → confirm fixation
- **Musculotendinous junction** → commonest failure site
- Bursal side intact ≠ no tear → inspect **articular surface**

Biceps Tenosynovitis Ask AI

- **Biceps tendon** red and frayed
- Treatment → **tenotomy**; no other solution
- New papers: healthy tendon → usable as **graft**
- Red, inflamed tendon → **pain generator** → biceps tenotomy forced

Rotator Cuff Repair — Key Points and Contraindications Ask AI

Technical Points Ask AI

- **Switch the scope** between portals
- **Cannulas** → sutures do not tangle
- **Bursectomy** → enough time for later visualisation
- Inspect cuff from **undersurface** (articular) and bursal surface
- **Footprint debridement** → little bone removed → anchor holding preserved
- Good **release** → cuff back to footprint
- Never repair in **abduction**

Contraindications Ask AI

- X-ray → **severe osteoarthritis**
- **Medically unfit**, e.g. hypertensive
  - → open surgery only; arthroscopy not possible
- **Adhesive capsulitis** → capsule release first → repair after improvement
- **Failed repair** with unidentified cause
- **Fatty infiltration** → Goutallier classification on MRI predicts improvement

Subscapularis Tears Ask AI

- **Subscapularis** → often overlooked; tears usually medial
- Assistant → arm into **internal rotation** → exposes subscapularis attachment
  - Reveals otherwise unseen tears → repair needed
- Case images → **subscapularis tear** repaired

- Anchors passed → sutures retrieved sequentially, one by one
- Four sutures tensioned → pushed onto **subscapularis** → repair complete

Massive and Irreparable Rotator Cuff Tears Ask AI

Massive vs Irreparable vs Pseudoparalytic Ask AI

- Not every tear reparable → **irreparable cuff tear**
- Massive ≠ irreparable ≠ **pseudoparalytic shoulder** → three distinct descriptions
- Massive tear may be **reparable**

Feature	Massive tear	Irreparable tear
Extent	More than 2 tendons (e.g. supraspinatus + infraspinatus)	Retracted to glenoid (not every glenoid-retracted tear irreparable) *[Q1 · As spoken; massive tear is two or more tendons; the example given has only two.]*
Retraction	More than 5 cm	To glenoid
Humeral head	—	Acromiohumeral distance less than 6 mm; migrating; acetabularization
Muscle	—	Fatty infiltration Goutallier 3 and 4; muscle atrophy; weakness
Grasper test	Pulled back to footprint → reparable	—

- **Hamada classification** → rotator cuff arthropathy
- Image → acromiohumeral distance ~1 mm
- **Pseudoparalysis** → cannot actively elevate to 45 degrees *[Q2 · As spoken; pseudoparalysis is usually defined as active elevation below 90 degrees with full passive range.]*
  - Passive range presumably full

Exam Pearl

- **Massive tear** ≠ irreparable tear ≠ pseudoparalytic shoulder
- Massive → more than 2 tendons, more than 5 cm retraction
- **Irreparable** → acromiohumeral distance less than 6 mm, Goutallier 3 and 4
- **Hamada classification** → rotator cuff arthropathy

Force Couple Imbalance Ask AI

- Vertical + horizontal **force couples** → imbalance → pseudoparalytic shoulder
- Vertical mechanism → strong **deltoid** pulls head up
  - Weak supraspinatus → head not centred → **superior escape**

Imbalance	Torn / weak muscles	Result
Vertical	Supraspinatus (deltoid strong)	Isolated loss of elevation; superior escape
Horizontal — posterior	Infraspinatus + teres minor	Isolated loss of external rotation
Horizontal — anterior	Subscapularis	Isolated loss of internal rotation
Combined horizontal + vertical	Supraspinatus + infraspinatus + teres minor	Both imbalances

- Torn supraspinatus, infraspinatus, teres minor + **intact subscapularis**
  - Anterior escape blocked → **posterosuperior escape** only
- Tear in subscapularis + supraspinatus → **anterosuperior** escape

Restoring Balance Ask AI

- Options → tendon transfer or **reverse shoulder**
- Infraspinatus + teres minor deficit → **latissimus dorsi transfer**
  - Or **lower trapezius transfer** → technique of Dr. Bassem Elhassan, Mayo Clinic, America
- Subscapularis deficit → latissimus dorsi or **pectoralis major transfer**

Options for Massive Irreparable Tear Ask AI

- **Conservative** → physiotherapy
  - Elderly, cardiac, respiratory problems, interstitial lung fibrosis, any **unfitness**
- **Biceps tenotomy** and debridement → most improve greatly
  - Elevation not as with repair; severe pain improves
- **Partial reconstruction** / partial repair → cuff pulled onto head as far as possible
  - Arm not in **abduction** → repair free of tension
- **Subacromial balloon** (subacromial spacer)
  - Idea of an Israeli doctor
  - Distended with 20–30 cc fluid
  - **Cushion** between humeral head and acromial undersurface
  - Prevents superior migration; centralizes head on glenoid
  - Goal → avoid progression to **rotator cuff arthropathy**
- **Tendon transfer** → latissimus dorsi, pectoralis minor or pectoralis major *[C1 · transcript: "pectoralis minor (said twice)"]*
- New trend → **superior capsular reconstruction**
- **Reverse shoulder**

Choice by Age Ask AI

Patient	Options
Young	No joint replacement first → tendon transfer (salvage / joint preservation), partial repair, superior capsular reconstruction
Elderly, unfit	Conservative, biceps tenotomy or debridement, balloon spacer, partial repair → reverse shoulder if all fail

Subacromial Balloon Spacer Ask AI

- Cushion inserted arthroscopically
- **Coracoacromial arch** (ligament) preserved
- **Distended with saline**
- Prevents **proximal migration** → no abutment against acromial undersurface
- **Longevity** unknown → short period in use; time to bursting unknown
- Debatable → hardening = stone in shoulder
- May **migrate** within shoulder
- Marketed by companies; no **consensus** yet

Superior Capsular Reconstruction Ask AI

Concept Ask AI SCR

- **Patch** fixed on glenoid → extended to cuff footprint
- Edges sutured to rotator cuff
- **Superior capsule** → keeps head centred; prevents migration

Indications Ask AI SCR

- **Massive irreparable rotator cuff tear**
- **Superior migration** of humeral head
- Subscapularis intact or reparable → superior part only reconstructed
  - Anterior or posterior defect → **horizontal balance** affected → SCR not possible
  - SCR replaces **supraspinatus** + infraspinatus only
  - Subscapularis + teres minor → intact or reparable
- No **severe rotator cuff arthropathy**
  - Present → **reverse shoulder**
- **Pseudoparalytic shoulder** → no active forward elevation → motion-improving procedure
- Previously **failed arthroscopic repair** → poor tissue quality → graft needed

Graft Options Ask AI

SCR

- **Mihata** (Japanese inventor) → harvest fascia lata from thigh
- **Acellular dermal allograft** → synthetic / manufactured *[Q3 · As spoken; acellular dermal allograft is processed human dermis, not a synthetic graft.]*
- **Long head of biceps** → if healthy, not frayed, not torn
- LHB attached at **glenoid** → taken onto two anchors → acts as SCR
- **Minimum thickness** 8 mm → functions easily
- Fascia lata → folded, reportedly 3 or 4 times
- Egypt → started, not wide scale → **expensive**
  - At least 4 or 5 anchors → 2 on glenoid, 3 on head
  - Shown configuration → 2 on glenoid, others on head
  - Head → two + two → **six** total
- Thigh opening + 4–5 anchors → not yet **acceptable** to patients

Graft	Advantages	Disadvantages
Fascia lata	No purchase price (three or four thousand dollars); helps greatly	Thigh incision; donor-site complications
Acellular dermal matrix (dermal patch)	No donor site morbidity; histology resembles cuff tendon	Strength much weaker than fascia lata; cost 3–4 thousand dollars
Long head of biceps	No donor site morbidity; no cost; attached to glenoid → no glenoid anchors; 5–10 minutes arthroscopically; good mechanical strength	—

Overall Status Ask AI

SCR

- Worldwide idea → constantly **modified**
- **Joint preservation** → no replacement; no futile repair of irreparable cuff
- Studies → **short follow-up**; no perfect graft
- Fascia lata + long head of biceps → **promising results** so far

High Yield

- SCR → **massive irreparable tear** + superior migration
- Subscapularis + teres minor → intact or reparable
- **Graft thickness** → minimum 8 mm
- Three grafts → fascia lata, dermal matrix, **long head of biceps**
- Inventor → **Mihata**

Revision Rotator Cuff Repair Ask AI

Weak Tendon vs Weak Bone Ask AI

- Before revision → identify cause of **primary failure**
- Failure → weak tendon or weak bone
- **Weak tendon** → augmentation, e.g. superior capsular reconstruction
  - Not another repair bound to fail
  - Good **tendon quality**, prior single row → double row, triple row or Mason-Allen
  - More anchors → stresses distributed
  - **Suture tapes** → wider → less erosion (cut-through)
- **Weak bone**
  - **Osteoporotic** → treatment; correction takes long

→ cuff waits 6 months, at least a year, maybe two → not reliable

  - Anchors **perpendicular** to bone surface
  - **Inter-anchor distance** at least 6 mm; old suture anchors → 2 to 3 mm
  - Cement at old anchor sites → new anchor → may work with **metal anchor**

→ not with plastic (**PEEK**) anchors

  - Bone too weak → **giant needle** → transosseous suture, no drilling, no anchor

Causes of Failure Ask AI

- Problem in diagnosis, surgical procedure, cuff itself, or **patient**
- **Age** → under 55 years ~90% success
  - Success falls with age → ~10% at 70–80 years *[Q4 · As spoken; reported healing in patients over 70 is roughly 40-50%, not about 10%.]*
- **Clinical factors** → repaired tendon does not heal
  - **Diabetics**

→ poor tissue healing

  - Obese
  - Smokers
  - Poor **bone mineral density**
- **Tear factors** → size (one vs two vs three tendons, or five), area
  - Tendon **quality**
- **Surgeon factors** → skilled, high volume → much better results
- **Technical / tear factors**
  - Single row repairs
  - Single tendon vs multiple tendons
  - Full thickness vs partial thickness
  - Shape → L-shaped, trapezoidal, massive
- All → affect final result

Approach to Failed Repair Ask AI

- Good imaging, exam, fatty infiltration assessed, good tendon quality → **arthroscopic repair**
- Poor tissue → **augmentation** → superior capsular reconstruction
  - Or **synthetic patch**, if available
- **Balloon spacers**
- **Tendon transfer**
- **Reverse arthroplasty**
- Least favoured → probably arthrodesis; very difficult for patients *[C2 · transcript: "arthrolysis"]*
- Same options as **massive irreparable cuff tear**

High Yield

- Revision → identify cause first: **weak tendon** or weak bone
- Weak bone → anchors perpendicular, at least 6 mm apart
- Cement + new anchor → metal only, not **PEEK**
- **Age** → under 55 ~90% success; ~10% at 70–80

Questions and Answers Ask AI

Tendon Transfer, Tapes and Positioning Ask AI

- Q: Failed tendon transfer → repeat?
  - Few transfers available → **latissimus dorsi** best
  - Lower trapezius → problems; tendon needs lengthening to reach **greater tuberosity**
  - Transfer fails → no tendons left → **reverse shoulder**
- **Physiotherapy** → dedicated lectures later, after surgical part
- Q: Anchor sutures vs **suture tapes**?
  - No "best suture"; choice available → suture tape
  - Tape ~three times wider than anchor suture → good **bite** in tendon
  - Cheaper than anchor
  - Needs **experience** → passing tapes not easy; techniques keep changing
  - Anchor vs tape → **tape**
- Q: Position for primary repair and revision?
  - **Beach chair**
  - Lateral → **instability** → labrum at 6 or 7 o'clock, inaccessible sitting
  - Unless assistant → abduction + **posterior translation** of humeral head → labrum exposed inferiorly

Biceps Tenotomy vs Tenodesis Ask AI

- Q: **Tenotomy** or tenodesis?
  - Debatable worldwide; still discussed at conferences
  - Situation in Egypt

Patient	Choice	Reason
Woman, thick, overweight	Tenotomy	Lower cost (no anchor); muscle twitches (cramps) one or two months then disappear; arm shape unchanged
Men, especially manual workers	Tenodesis, if circumstances allow	Popeye sign (arm bulge) not acceptable
Frayed tendon	Tenotomy	—
Athlete (tennis, volleyball, water polo)	Tenodesis	Needs biceps function

- Q: Tenodesis site → in joint or **subpectoral**?
  - Most literature → subpectoral, below **pectoralis major tendon**
  - Open area → deliver tenotomized tendon → **screw** in bicipital groove
  - Some fix inside joint → lecturer does so in many patients
  - Groove **tenosynovitis** persists → cause of pain
  - Preferable → address **lever arm**; tenodesis below pectoralis major

→ strong humeral bone → stronger than anchor above + tied sutures

Scores, Margin Convergence and Portals Ask AI

- Q: **Scoring systems**?
  - Evaluate active range of motion + **muscle strength**
  - **ASES** (American Shoulder and Elbow Surgeons) score
  - **UCLA** (University of California Los Angeles) score
  - **Constant score**
  - Useful especially in **research work**
- Q: **Margin convergence**
  - Tear retracted to **glenoid** → release + pull fails
  - **Side-to-side sutures** → two layers approximate → near footprint → easy to bring
  - Pulling whole cuff from far → sometimes fails; judged arthroscopically
  - Sequence → **cuff release** → side-to-side convergence → footprint
- Q: Preferred **viewing portal**?
  - None; habit-dependent
  - Assistant rotates arm internally/externally → tear brought into view
  - Portal of experience → **suturing** easier
- Q: One portal for both **glenohumeral joint** and subacromial space?
  - None; glenohumeral joint → standard **posterior portal**
  - Redirected into subacromial space → cuff retracts away; end not seen
  - Switch scope → finish from **lateral portal**

→ **panoramic view** of whole tear

Humeral Head Cysts and Myotendinous Failure Ask AI

- Q: Tips with a **cyst** in humeral head?
  - Study imaging; locate cyst
  - Very large anchor → **6.5**, requested from company; not standard
  - Anchor placed away from **cyst**
  - Or **giant needle** prepared → transosseous repair
  - Very large cyst → anchor holding in air
  - Tensioning → anchor **pulls out** → hole + defect in bone (happened to lecturer)
  - **Backup** → giant needle or any transosseous technique
- Q: 3 cm stump attached to footprint → tendon-to-tendon or tendon-to-bone?
  - Stump on footprint → failure at **myotendinous junction**
  - Muscle-to-tendon suture → very difficult; usually **cuts through**
  - Indication for **superior capsular reconstruction**

Reverse Shoulder and Osteoporotic Bone Ask AI

- Q: New **reverse shoulder** designs → compensate rotation deficiency?
  - No → role is to **lateralize**
  - **Glenoid lateralization** or humeral lateralization → tension remaining cuff → some rotation
  - Subscapularis or teres minor deficit → **tendon transfer** with reverse
  - Main role of reverse → **active forward elevation**

→ not internal or external rotation

- Q: **Osteoporotic** proximal humerus, above ~60 years → metal or big anchor? *[C3 · transcript: "above 6 years"]*
  - **Big anchors**
  - X-ray very osteoporotic → prepare large anchors + **giant needle**

→ **backup**; problems happen a lot

  - Example: anchor insertion → **bone cracked** → loose piece removed arthroscopically
  - Alternative → two or three stitches + **cruciate ligament**-type fixation device on cortex

→ no screw, no anchor → **surface fixation**; newer technique

Anchor Angle, Poor Tendon and Older Patients Ask AI

- Q: **Angle of anchor insertion**?
  - Older view → **Burkhart** → 60-degree inclination *[Q5 · As spoken; Burkhart's deadman angle for anchor insertion is 45 degrees, not 60.]*
  - Newer view (author unclear, possibly Itoi) → **perpendicular** to surface
  - Not insertable through arthroscopy portal
  - **Separate stab incision** → needle from above beside acromion → anchor
- Q: 25-year-old, **fibrous**, laminated tendon; two footprint anchors only
  - Other options → **augmentation** with patch or fascia lata
  - Fibrous cuff at 25 → prior **collagen deficiency**? Something hidden
- Q: Over 65 years → **tendon transfer** suitable?
  - Yes, but test transferred muscle

→ confirm **functioning**, as in brachial plexus transfers

- **Very long training** needed
- Transferred muscle → 60% of its **power**
- Muscle at 40–50% → loses 40% → 20% → **useless**
- Net 20% symptom gain → another option better

MRI of the Knee

(MRI Principles and Normal and Pathological Knee Anatomy)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 155) “smaller than the lateral meniscus” → “smaller than medial meniscus”

C2 (p. 157) “right image degeneration, left image tear” → “right image tear, left degeneration”

Queried - kept as spoken, but looks wrong

Q1 (p. 157) As spoken; standard MRI grade 3 meniscal signal reaches the articular surface, i.e. a tear, not degeneration

Q2 (p. 164) As spoken; usual criterion is bow tie on 3 or more contiguous sagittal slices (normal two)

Q3 (p. 174) As spoken; contradicts earlier statement and table that OCD cartilage is not intact

MRI Principles and Signal Basics Ask AI

MRI Machine Types Ask AI

- Introduction mirrors **MRI of the shoulder** lecture → what MRI is, how sequences work
- Brief recap before knee proper
- Machine types
  - **Closed MRI** (closed coil) → patient enters box, likened to coffin
  - **Open MRI** → patients with claustrophobia
  - Image: open versus closed machine
  - **Extremity MRI** → developed later
    - Knee, ankle → suitable
    - Shoulder, hip → not suitable; shoulder needs whole patient inside machine
    - Closed type; quick; **high resolution**
  - **Dynamic MRI**

Dynamic MRI Ask AI

- Images patient **standing**
- Very important → hip, spine; important → knee
- Knee **instability** → demonstrated on dynamic MRI
- Used when findings not easily shown on **conventional MRI**

Advantages of MRI Ask AI

- **Multiplanar** imaging
- Avoids bone **artifacts**
- Good **soft-tissue characterization**
- **Superior soft-tissue contrast**

Hydrogen-Ion Principle Ask AI

- Magnetic field → **mobile hydrogen ions** → white on image
- Two main sequences → **T1**, T2
- Five kinds of hydrogen ion

Hydrogen ions	T1	T2	Examples
Moderately mobile	Seen	Seen	—
Slowly mobile	Black	White	Water, any body fluid, ascites, infusion
Quickly mobile	Black	Black	Blood inside normal vessels → too fast to capture
Non-mobile (fixed)	Black	Black	Cortical bone, ligaments, tendons, menisci, calcifications
Absent	Black	Black	Air → lung, paranasal sinuses

Gray Scale Ask AI

- Image requires **adequate number** of mobile hydrogen ions
- **Gray scale** → black, gray, white structures
  - Black = dark = low signal
  - Gray = intermediate signal
  - White = bright = high signal
- Whole image → shades of gray, darkest (black) → lightest (white)

Signal Combinations on T1 and T2 Ask AI

- Four T1/T2 permutations

T1	T2	Tissue
Low (black)	High (white)	Water → CSF, effusion
High	Low	Fat → subcutaneous fat, bone marrow
High	High	Blood → haemoglobin, haemarthrosis
Low	Low	No or non-mobile hydrogen ions — see below

- Black on both
  - **Bone cortices**, mature tissue, ligaments, tendons
    - Knee → ACL, PCL, patellar tendon
  - **Calcifications** → physiological or pathological (e.g. tumours)
  - Blood inside **blood vessels**
  - **Air** → lung, paranasal air sinuses
- Common confusion → which tissue black/white on T1 versus T2

Worked Example Ask AI

Sagittal

- Left image = **T1**
  - Infrapatellar **fat pad** → white T1, black T2
  - Joint **fluid** → black T1, white T2
  - **Bone cortices** (non-mobile hydrogen) → black T1 and T2
  - Black → menisci, PCL, ACL, patellar tendon, quadriceps tendon

Reading Knee MRI: Orientation, Planes and Checklist Ask AI

Orientation Rule for Knee Films Ask AI

- Every printed film → **patella on your left hand**
- Patella left → anterior = left, posterior = right
- Very important in knee, unlike shoulder
- Further right in image → more posterior
- Same **imaging planes** as shoulder

Imaging Planes Ask AI

- **Sagittal cuts** most important → T1, T2 or proton density (PD)

MRI plane	Equivalent
Sagittal	Lateral view X-ray — interpreted exactly like it
Coronal	Anteroposterior (AP) X-ray
Axial	CT cuts

- Sagittal cuts → taken **outside to inside**
- Coronal cuts → **front to back**

Identifying the Pulse Sequence Ask AI

- **Fat** → bright T1, dark T2
- Interpret by signal table above
- Joint **fluid** → three colours
  - Black → T1
  - White → T2
  - Gray → **proton density**
- Table recap
  - Black on both → **cortical bone**, menisci, ligaments (e.g. PCL)
  - High T1, low T2 → **subcutaneous fat**, bone marrow
  - Low T1, high T2 → **fluid**, effusion
  - High on both → **blood**, haemarthrosis

Structures Evaluated and Best Plane Ask AI

- Shoulder parallel → labrum, tendons, ligaments, capsule, bones

Structure	Best plane
Menisci, medial and lateral	Sagittal
Cruciates → ACL, PCL	Sagittal
Collateral ligaments (tear)	Coronal — most important view
Retinaculum → MPFL, lateral patellofemoral ligament	Axial
Tendons → quadriceps, patellar	Sagittal
Bones	Sagittal
Synovium	Sagittal

- **Sagittal cuts** → most important in MRI diagnosis

High Yield

- Patella on your left → left anterior, right posterior
- Sagittal = lateral X-ray; coronal = AP X-ray; axial = CT cuts
- Sagittal → most knee diagnoses; coronal → collaterals; axial → MPFL
- Fluid → black T1, white T2, gray PD

Normal Menisci Ask AI

- First, most important knee MRI topic → **meniscal tears**
- Many tear types → know **normal meniscus** first
- Start → medial meniscus

Shape Ask AI

- **Lateral meniscus** → closed letter C
- **Medial meniscus** → banana, or large C embracing small lateral C
- Banana or C-shaped → either description

Medial versus Lateral on Sagittal Cuts Ask AI

- Three parameters
  - **Horn shape** → both triangles (wedge-shaped)
    - Posterior wedge longer, thicker

→ **posterior horn of medial meniscus**

    - Anterior horn → one close to patella (patella-left rule)
    - Posterior horn ≈ twice anterior → **medial meniscus**
  - **Tibia** shape → flat, no downward curve → medial
  - **Fibular head** → appears or not
- Lateral meniscus → horns roughly **same size**
- Medial identified by → tibia shape + posterior/anterior horn size
- More sagittal cuts pass through medial than lateral meniscus

Attachments and Tear Frequency Ask AI

Medial MM

- Both menisci **wedge-shaped**
- Posterior part **firmly attached to capsule** → entrapped between femur and tibia
- Most tears → **medial meniscal tears**
  - Lateral meniscus → more mobile

Bow-Tie Sign Ask AI

Medial MM

- Two sagittal cuts
  - Yellow arrow → through horns → small anterior, big posterior → medial
  - Blue arrow → through **body** → bow-tie sign
    - Meniscus shaped like party **bow tie**
- Another image → patella left; posterior horn larger; ordinary tibia, no **fibular-head curve** → medial

The Lateral Meniscus Ask AI

- **C-shaped**
- Smaller than medial meniscus *[C1 · transcript: "smaller than the lateral meniscus"]*
- **Freely mobile** → space with popliteus tendon; not firmly attached to capsule here
  - → **less commonly injured**
- Two features distinct from medial
  - **Discoid lateral meniscus** possible
  - **Meniscal cysts** more common
- **Discoid** → complete disc, not crescent; detailed later
- **Meniscal cyst** → tear lets synovial fluid into meniscal body → cyst
  - Common MRI report finding
- Left image → medial meniscus
- Right image → lateral meniscus; **two horns** same size
- Two sagittal cuts, as for medial
  - Yellow arrow → horns → **equal** size
  - Blue arrow → body → **bow-tie sign**
- Lateral identified by → equal horns, visible **fibula**, tibia shape

Feature	Medial meniscus	Lateral meniscus
Shape	Banana / large C	Closed C, smaller
Horns	Posterior ≈ twice anterior	Equal size
Tibia	Flat, no curve	Curved to articulate with fibular head
Fibular head	Absent	Seen
Capsule	Posterior part firmly attached	Not firmly attached; popliteus tendon intervenes
Mobility	Less mobile → entrapped	Freely mobile
Tear frequency	Most tears	Less common
Special	—	Discoid; meniscal cysts more common

Normal versus Abnormal Lateral Meniscus Ask AI

- Left image → normal anterior + posterior horns
- Right image, red arrow → **posterior horn** missing
  - → ghost meniscus or **bucket-handle tear**
- Anteriorly → **two anterior horns**
  - Front one → true anterior horn
  - Second → posterior horn flipped anteriorly → **flipped lateral meniscus**
  - Detailed later
- **Double PCL sign** → bucket handle of medial meniscus
- Two anterior horns → **double delta sign** → lateral meniscus
- Cuts through body → **bow-tie sign** in middle; nothing at end

Pitfall: Popliteus Tendon Ask AI

- Black line posteriorly → white gap → rest of lateral meniscal body
- Not a tear → **popliteus tendon**
- Lateral meniscus not firmly attached to capsule → **popliteus tendon** separates
- Inflammation, synovial fluid around tendon, or **popliteus tendinitis** → false impression of tear
- Another lateral meniscus → horns ≈ same size; tibia **curved** for fibular head

> **Pitfall**
> - Gap between posterior lateral meniscus and capsule → popliteus tendon, not tear
> - Popliteus tendinitis or fluid → false lateral meniscal tear

Meniscal Degeneration and Tear Patterns Ask AI

- Meniscal tears on MRI; grades of **meniscal degeneration**

Degeneration Grades Ask AI

- Common **conflict** → report says degeneration, you see tear — or reverse
- Healthy meniscus → entirely black triangle / wedge
- Meniscal degeneration → three grades

Grade	Appearance
1	White spot / globule inside meniscus
2	Large white line, much of posterior-horn wedge
3	Larger line + erosion at edge *[Q1 · As spoken; standard MRI grade 3 meniscal signal reaches the articular surface, i.e. a tear, not degeneration]*

- All grades → **superior + inferior surfaces** intact

Degeneration versus Tear Ask AI

- Both surfaces intact → **meniscal degeneration**, no tear
- Surface disrupted + high signal → **meniscal tear**
- Left image → very high signal inside body, 2 cuts
  - Signal not breaching inferior or superior surface → **degeneration**
- Right image → high signal + **inferior articular surface** cut → tear *[C2 · transcript: "right image degeneration, left image tear"]*
  - → horizontal / **cleavage tear**
- Centres often misreport degeneration ↔ tear
- **Decisive point** → superior + inferior articular surfaces of posterior horn intact?

Must-Know

- Internal high signal + both surfaces intact → degeneration
- Superior or inferior surface disrupted + high signal → tear

Basic Tear Types Ask AI

- **Longitudinal tear** → parallel to collagen bundles (run as closed circles)
  - Extends anteriorly + posteriorly, fragment flips → **bucket-handle tear**
- **Horizontal tear** → knife opening loaf horizontally → superior + inferior flap
- **Radial tear** → cut from outside to inside
- Displaced form → **flap tear**
- Longitudinal image → continues anterior + posterior, central fragment flips → **bucket-handle tear**

Degeneration or Tear — Further Example Ask AI

- Right image → inferior articular surface cut → true **tear**
- Left image → **degeneration**; outline drawable, both surfaces intact
- Right image → superior intact, **inferior surface disrupted**; signal stops mid-part → tear

Two Criteria for Meniscal Tear Ask AI

- **Abnormal morphology**
  - **Double PCL sign** → medial meniscus
  - **Double delta sign** → lateral meniscus
  - **Absent / empty meniscus sign**
  - Loss of two normal black triangles → or one present, other gone
- **High signal** reaching (touching) surface

Displaced Patterns — Six Types Ask AI

- Three basic tears → longitudinal, horizontal / cleavage, radial
- Displaced → three further types

Basic tear	Displaced form
Longitudinal	Extends → bucket handle
Horizontal	Displaced / flips → flap tear
Radial	Small radial extends → parrot-beak tear; radial flips → flap tear

- Horizontal → knife splits superior + inferior piece, like loaf or sandwich
- Total → **six types**: three basic + three displaced

Longitudinal and Bucket-Handle Tears Ask AI

Longitudinal Tear Ask AI

- Runs along meniscus → **parallel to collagen bundles**
- Splits meniscus → outer + inner part
- Left image → top to bottom
  - **Inner fragment** → joint side
  - Outer fragment → capsule side
- Second example → outer + inner fragments

Appearance by Sagittal Cut Ask AI

- Depends on tear **extension** + sagittal cut level
  - Yellow line → posterior horn split outer/inner; **anterior horn intact**
  - Green line → same as yellow
  - Red line → body → **bow tie** split in two
    - Posterior outer piece
    - Inner piece → continuous with body and anterior horn

Flipped Meniscus Ask AI

Lateral MM

- Lateral bucket handle → called **flipped meniscus**; distinguishes from medial
- Medial → displaced longitudinal tear = **bucket-handle tear**
- Lateral → same lesion, but entire posterior horn **flips anteriorly**
  - Cause → loosely attached to capsule
- Image → longitudinal tear, then after flipping → most of posterior horn anterior
- Definition → bucket-handle tear of lateral meniscus; ruptured posterior horn flips anteriorly, sticks on anterior horn
- Anterior horn → **double delta sign** (two anterior horns)
- Anterior horn → large relative to posterior horn
  - Abnormal in lateral meniscus
- Double delta → lateral; **double PCL sign** → medial bucket handle
- Mechanism
  - Posterior horn cut → flips anteriorly → anterior + posterior horn = **double delta**
  - Posteriorly → small remnant or no meniscus

Coronal View: Intercondylar Notch Ask AI

- Coronal = **anteroposterior view** of MRI; also notch view
- Femur, tibia, fibula → left = lateral, right = medial
- Normal **intercondylar notch** → two structures only
  - Lateral side → **ACL**
  - Medial side → **PCL**
- **Third structure** → bucket-handle tear, medial or lateral
- Use → confirm flipped meniscus or medial bucket handle
- Third structure lateral or medial to ACL → **bucket-handle** / longitudinal tear, either meniscus
- Coronal findings, flipped lateral meniscus
  - Anterior horn → very large
  - **Fragment** entering notch → pushes ACL slightly
  - Posterior fragment → very small
- Diagnosis mainly **sagittal**; coronal → confirmation
- Another → anterior horn very large (abnormal); fragment entering notch; posterior horn small
  - → longitudinal / bucket-handle tear lateral meniscus → **double delta sign**
- Two deltas; horn sizes completely different → lateral horns should be equal
- Anterior horn large; posterior horn **almost invisible**
- Far right → very large anterior horn; posterior absent or tiny → **flipped lateral meniscus**
- Double delta; posterior horn almost completely absent

Why Medial and Lateral Bucket Handles Differ Ask AI

- Medial bucket handle → **displaced into intercondylar notch**
- Lateral (flipped) cannot → **ACL** strongly prevents notch entry
- Key → **ACL on lateral side**
  - Lateral fragment pushes toward notch → intact ACL blocks
  - Medial meniscus far from ACL → easy displacement into notch
- ACL blocks notch → only option: flip anteriorly → stick on anterior horn → **double delta sign**
- Lateral anterior horn → close to **ACL**
  - Posterior horn tries to enter → enters slightly, blocked → flips anteriorly → **flipped meniscus**
- Medial bucket handle → meniscus **far from ACL** → enters and flips into notch
  - Not onto anterior horn
- **Intact ACL** → prevents displacement into notch

Classic Bucket Handle Ask AI

Medial MM

- Notch → ACL lateral, PCL medial; third structure (red arrow) → **bucket-handle tear**
- Diagnosed more by **double PCL sign**
- Double PCL sign image
- Arthroscopy → posterior horn cut, flipped, in notch → **knee locking**
- Medial → posterior horn flips into notch, not anteriorly
  - Attached **anterior to PCL**
- **Double PCL sign** → intact PCL (buckled) + flipped posterior horn in front of it

	Medial meniscus	Lateral meniscus
Name	Bucket-handle tear	Flipped meniscus
Fragment goes	Into intercondylar notch	Anteriorly, onto anterior horn
Sign	Double PCL	Double delta
Reason	Meniscus far from ACL	Intact ACL blocks notch

High Yield

- Longitudinal tear appearance → depends on sagittal cut taken
- Displaced lateral → flipped meniscus → double delta sign
- Displaced medial → bucket-handle tear → double PCL sign
- Intact ACL → blocks entry into intercondylar notch

- ACL absent or torn → lateral **bucket handle** enters notch instead of flipping

Horizontal Tears and Meniscal Cysts Ask AI

Horizontal (Cleavage) Tear Ask AI

- Knife cutting sandwich across → **superior flap** + inferior flap
- Opening → **synovial fluid** enters meniscal core
- Fluid cuts **collagen fibres** → meniscal cyst
- Marker
  - Meniscal cyst, lateral or medial → likely **horizontal cleavage tear**
  - Meniscus must open superiorly + inferiorly for fluid entry

Surgical Implication Ask AI

- Two procedures
  - **Arthroscopic** → debride tear from inside → suture superior flap to inferior flap
    - Distinct arthroscopic suturing technique
  - Open from outside → **excise** synovial cyst
- Two questions
  - Horizontal cleavage tear → **meniscal cyst** or not?
  - Cyst present → open excision from outside first → then arthroscopic suture
- Horizontal tear → superior + inferior fragment → cyst forms inside

Meniscal Cyst — Three MRI Criteria Ask AI

- **Horizontal tear**
- **Fluid accumulation** inside meniscus
- **Flattening** at capsular edge
- Image → horizontal tear line → cyst → **lateral edge** separated from capsule
- Cyst between capsule and meniscal body → **open excision** first, then arthroscopic suture
- Left → **horizontal cleavage tear** across whole meniscus; coronal + sagittal
- Longitudinal (vertical) tear → **superior to inferior**; horizontal → across
- Not degeneration → **inferior articular surface** disrupted
- Further sagittal and coronal examples

Clinical Pearl

- Meniscal cyst → look for horizontal cleavage tear
- Treatment → open cyst excision + arthroscopic suture of flaps

Radial and Root Tears Ask AI

Radial Tear Ask AI

- Cut **across** meniscus along its radius → from centre outward
- Green line → anterior + posterior horns intact
- → **most difficult tear** to diagnose; needs very skilled reader
  - No cut passes through tear itself
- Orange line → anterior horn intact triangle; posterior horn **incomplete triangle**
  - Abnormal morphology → suspicion → check coronal
- Coronal, blue line → **bow-tie disruption**
- Coronal bow-tie disruption + sagittal abnormal horn → **radial tear**
- Missing radial tear → understandable; needs precision, experienced eye

Diagnostic Criteria Ask AI

Radial

- Apex tip of triangle **disrupted** → partial triangle → suspicion
- Coronal → **bow-tie disruption** (yellow arrow)
- Axial → intact lateral meniscus versus missing part → **radial tear**
- Sagittal T1 → **empty meniscus**
  - Posterior part not seen → tear took whole posterior part

Root Tears Ask AI

- Subcategory of **radial tears**
- Radial + root → meniscus **non-functioning**; all collagen fibres cut across
- Treatment → **suture** to restore normal shape
- **Two worst tears** → radial, root
  - Meniscus → anterior + posterior fragment
  - Not superior/inferior (horizontal); not inner/outer
- **Empty meniscus sign**
  - Fragments completely separated → posterior horn not seen (blue arrow) → radial tear
- Fragments displaced outwards → **meniscal extrusion**
  - Especially coronal → likely radial or root tear
  - Seen or reported → **re-review** cuts
- Root tear → empty meniscus + extrusion
- Coronal → no meniscus, or meniscus **extruded** outside joint
  - Not between **medial femoral condyle** and medial plateau
- Empty meniscus + extrusion → two signs of radial tears

Meniscal Root Tear Ask AI

- Arthroscopy → meniscus **floating**, unattached posteriorly
- → extrusion out of joint
- Repair → arthroscopic **bony tunnel** → suture through root of posterior horn
  - Sutures passed down, tensioned → re-fixed onto surface
- Unrepaired → flap flying in joint → **instability**
- Definition → **radial tear** at meniscal root
- Imaging → empty / absent meniscus sign
  - = **ghost meniscus**: shadows of remaining posterior horn
  - → root or radial tear

Three Signs of Root Tear Ask AI

- Coronal → **disruption of body** at root, with meniscal extrusion
- Sagittal → **empty / absent / ghost** meniscus
- PCL cut → **posterior horn absent** anterior to PCL
  - Posterior horn normally attaches anterior to PCL
- Root tear → type of radial tear
- Normal → posterior horn fills area anterior to PCL
  - Black triangle anterior to PCL → **root intact**
  - Absent + extrusion + empty/ghost meniscus → **root tear**
- Coronal → red arrows = sites of **root disruption**
- Yellow arrow → meniscus **extruding**; plateau not fully covered
- Example → **medial meniscal root tear**
- Another → yellow arrows; PCL present, posterior horn triangle absent → **empty meniscus sign**
- Complete empty / ghost / absent meniscus → **radial or root tear**; confirm extrusion on coronal

Must-Know

- Root tear signs → posterior horn absent anterior to PCL
- Empty / absent / ghost meniscus on sagittal
- Body disruption at root + extrusion on coronal
- Meniscal extrusion → suspect radial or root tear

Other Meniscal Lesions Ask AI

Complex Meniscal Tears Ask AI

- Cleavage tear across + longitudinal tear along
- Two tear types together → **macerated meniscus** / complex tear
  - Radial + longitudinal
  - Longitudinal + horizontal cleavage
- Most difficult to suture → **meniscal repair** harder
- Combinations → horizontal + longitudinal (vertical), radial + vertical, radial + horizontal
- Image → top to bottom and across → **complex tear**: vertical + horizontal

Vacuum Phenomenon Ask AI

- Confusing; not required in depth; more in **discoid lateral meniscus**
- Joint **nitrogen** → vacuum appearance on X-ray
- Mimics absent meniscal body → impression of tear
- Possible **MCQ** point
- Nitrogen gas from **negative pressure** of patient position
  - → false impression of lateral meniscal tear

Discoid Lateral Meniscus Ask AI

- Almost **complete disc** versus normal meniscus
- **Squaring** / flattening → lateral plateau + lateral femoral condyle
- Commonest → boys aged 9 or 10 years; **clicking** (popping) on walking
  - Click → kneeling, praying, kicking
- **Plain X-ray**, long-standing → squaring of lateral femoral condyle + lateral plateau
  - Complete disc between bones, not crescent
- Sagittal diagnosis
  - Normal **bow-tie sign** → three views
  - Discoid → bow tie in at least five views
- X-ray → **squaring** of lateral femoral condyle + lateral tibial plateau
- Young boy, clicking knee → click felt at **lateral joint line**
  - X-ray → lateral femoral condyle **squared**, not semicircular
- Tears in discoid → **radial**, cleavage, vertical — like any meniscus
- Intact discoid → knees still click (age nine)
- Clinic → child walks → **clicking** audible
- Age 8–9 + click → discoid → MRI shows tear → surgery even at 7–8
- Youngest operated → six years; **taekwondo**; torn discoid lateral meniscus
- Arthroscopic appearance
  - Central space absent → **complete ring** (disc), not crescent

Exam Pearl

- Vacuum phenomenon → joint nitrogen → false lateral meniscal tear
- Discoid → bow-tie sign in at least five sagittal views (normal three) *[Q2 · As spoken; usual criterion is bow tie on 3 or more contiguous sagittal slices (normal two)]*
- Discoid X-ray → squaring of lateral femoral condyle + lateral plateau

Red-Zone Tear and Ramp Lesion Ask AI

- **Red-zone meniscal tear**
  - Confused with **ramp lesion**; ramp tear diagnoses rising fast
- **Debate** → red-red zone tear versus complete peripheral detachment
- Red zone, **posterior horn of medial meniscus** (longer than anterior)
  - Longitudinal / vertical tear, several images → red-red zone; some call it ramp lesion
- Medial meniscus → **banana-shaped**
- Lateral meniscus → letter C
- Site → capsular reflection + **meniscal attachment**
- Name → slipping off pavement kerb → down-slope = **ramp**
- First described 1988 → **Strobel** (German)
- Debate → red zone versus **meniscocapsular attachment**
- Image → tear not in meniscus itself → **between meniscal attachment and capsule**
- Classification **type 2** → red-red / red vascular zone tear; indistinguishable → debate

Ramp Lesion — Danger Ask AI

- Often **not diagnosed** → meniscus viewed from above; tear may be on inferior surface
  - Seen only if complete vertical red-zone tear, or on **probe** traction
- ACL reconstruction re-ruptured → suspect **undiagnosed ramp lesion** as cause of failure
- Ramp lesion image
- On MRI → looks like red / **vascular zone** tear
- Naming irrelevant → diagnosis matters

Clinical Pearl

- Failed ACL reconstruction → suspect missed ramp lesion
- Ramp tear → may sit on inferior surface; probe traction reveals it

Meniscal Extrusion Ask AI

- Meniscus out of joint
- → suggests **meniscal root tear**
- Danger → out from between femoral condyle and tibial plateau → **osteoarthritis**
  - Meniscus non-functioning
- Complete radial / root tear → anterior + posterior fragment
  - Collagen fibres fail → **hoop stresses** lost → meniscus effectively absent
- Image → first blue line = **joint-line** end; second → meniscus fully outside

Segond Fracture Ask AI

- ACL injury finding; arthroscopists look for it
- = **lateral capsular tear** → twist injury + internal rotation
- Strongly indicates **ACL** injury
- Clinic X-ray → lateral side, just above **fibular head** → shadow in ~70%
- → predict ACL tear before MRI in 70%; impresses patient
- Acute **haemarthrosis** → examination very difficult
  - Anterior drawer, pivot shift hard → tense, painful knee

- Plain X-ray, lateral side
  - Segond → 75% probability of **ACL tear**
- Associated **meniscal tear**, medial or lateral → 60–70%
- Image → Segond + ACL absent from notch → **empty notch sign**
- Sagittal → ACL **torn**, sagging down

High Yield
- Segond → lateral capsular avulsion; twist + internal rotation
- Shadow just above fibular head on plain X-ray
- 75% probability of ACL tear; 60–70% associated meniscal tear

Meniscal Cysts — Sites and Presentation Ask AI

- Follows **horizontal tear**
- Sites
  - **Posterior horn** → medial meniscus
  - **Anterior horn** → lateral meniscus
- Rule → lateral anterior; medial posterior
- Presence → confirm **horizontal cleavage tear**; route for synovial fluid
- Image → **meniscal cyst** + horizontal tear
- Presentation → no locking; **lump** at joint side, marble-like
  - Lateral → anterior; medial → posterior
- Cyst itself may bring patient, not tear
- Further images → **meniscal cyst** + horizontal cleavage tear

Cruciate Ligaments Ask AI

Normal Cruciates Ask AI

- Major ligaments → intact ACL, PCL, MCL, LCL
- ACL + PCL in same cut → **very rare**
- Sagittal → find **PCL** first → ACL in cut in front or behind
- Same-image example → possible but uncommon

Meniscofemoral Ligaments Ask AI

- Intact PCL → two **black dots**, one anterior, one posterior
- = **meniscofemoral ligaments**
  - Anterior → **ligament of Humphrey** (H)
  - Posterior → **ligament of Wrisberg** (W)
- Origin **lateral meniscus** → insertion medial femoral condyle
- Sagittal → fibula + lateral meniscus → strand arises
- Insertion → **medial femoral condyle**
- Pitfall → don't mistake dots for **posterior horn**
  - Posterior horn → behind tibial spine
  - Dots → anterior to tibial spine
- Posterior horn attaches in this area → **double PCL sign**

Normal ACL Ask AI

- Coronal → **ACL** intact
  - Tibial spine → **medial wall of lateral femoral condyle**
- Identify PCL first → ACL in adjacent anterior or posterior cut
- Intact ACL → **taut**
  - **Parallel** to intercondylar notch roof
  - Fibres not sagging onto plateau; not stuck posteriorly onto PCL

Empty Notch Sign Ask AI

- Known arthroscopic sign → also seen on **MRI**
- Gray/black band along roof → ACL **intact**
- Absent or very high signal → **torn**; coronal and sagittal
- **Fibular head** → lateral side → lateral femoral condyle
- Notch contents
  - **ACL** → medial border of lateral femoral condyle
  - **PCL** → lateral border of medial femoral condyle
- ACL torn → no black / dark-gray line at attachment
  - **Fluid signal**; fibres not reaching lateral wall → empty notch sign
- Sagittal → notch roof line versus ACL line → **not parallel**
  - Fibres sag → ACL fallen onto plateau

Associated Findings Ask AI

ACL

- **Bone bruises** → lateral femoral condyle + lateral tibial plateau → support ACL tear
- Plain X-ray → **Segond fracture**
- **Unhappy triad of O'Donoghue**
  - ACL tear
  - **MCL tear**
  - Medial meniscal tear

Torn ACL Lying on PCL Ask AI

- Yellow line = notch roof → ACL should parallel it
- ACL falls onto plateau, then runs upward
  - Torn stump stuck onto **PCL** → continues with PCL vertical part
- Mimics **vertical segment** reaching femoral condyle → still ACL tear
- **Empty notch**
- Left image → intact ACL → definite **black band** on medial border of lateral femoral condyle
- PCL → lateral wall of medial femoral condyle
- Right → clear **empty notch sign**

ACL Ganglion Cyst Ask AI

- Uncommon → lecturer saw about one case
- **Ganglion cyst** on ACL
- May cause **locking** in notch; ACL intact
- Cyst extends up toward **insertion**
- Treatment tricky → **remove cautiously**
  - Shaver burst → risk of cutting ACL; cyst lies behind ACL
  - Clear carefully with **probe**

PCL Tears Ask AI

- Normal sagittal → **cord-like black structure**
- Absent + fluid signal → **empty notch sign of PCL**
- Very high fluid signal filling notch → **PCL tear**
- **Midsubstance** disruption
- Commonly **avulsion** with piece of posterior tibial plateau
  - Arthroscopic fixation → **TightRope**, or tunnel + sutures
- **Partial tear**
  - Heals very well → **conservative treatment**
- Operate only → complete sag + **complete disruption**
- High **healing power** → large posterior tibial surface area, good blood supply

Collateral Ligaments and Posterolateral Corner Ask AI

Medial Collateral Ligament Ask AI

- From **medial femoral epicondyle** → 7 cm below articular surface
- Average length 9–11 cm; width 1.5 cm
- Normal image → **inferior medial geniculate** artery + vein; seldom so clear
- **MCL** strongly adherent to medial meniscus
  - Total knee → cutting medial meniscal body → scalpel slip → **MCL cut**
- MCL tear
  - Torn from **tibia**; remnants seen
  - Criteria
    - Not in **anatomical site**
    - High fluid signal → **haematoma** / haemarthrosis
    - Meniscus **extruding** → sign of MCL tear
  - Detached from **tibial site**

LCL and Posterolateral Corner Ask AI

- Structures
  - **Fibular collateral ligament** (FCL)
  - **Popliteus tendon**
  - **Biceps femoris** → letter V with FCL
- FCL → **lateral femoral epicondyle** → fibular head
- Biceps femoris joins FCL → **letter V**
- Deep → **popliteus tendon**
- **Posterolateral corner injury**
  - Popliteus
  - Popliteofibular ligament
  - Fibular collateral ligament
  - Biceps femoris insertion into fibular head
- Sagittal → FCL + biceps → V → insert together into **fibular head**
- Collaterals diagnosed on **coronal**, not sagittal
- Coronal → lateral femoral epicondyle, fibular head, FCL
- White arrow → FCL **avulsion** from lateral femoral epicondyle
- Intact FCL, both attachments + surrounding high signal → **sprain**
- Right image → separated from **femoral epicondyle**
- **Posterolateral corner injury**
  - Biceps femoris discontinuous at fibular head insertion
  - Very high signal intensity
  - → PLC likely torn
- FCL **disrupted**; not attached to fibular head
- Sagittal → V discontinuous; high fluid signal
  - Neither biceps femoris nor FCL descends to attach

Knee Dislocation Ask AI

- **Fibular collateral** torn
- Notch → **ACL + PCL** both torn
- Green arrow → **MCL** torn

Extensor Mechanism and Patellar Retinaculum Ask AI

Quadriceps and Patellar Tendons Ask AI

- Two black structures on **patella**
  - **Quadriceps** → three layers → upper pole
  - **Patellar tendon** → from inferior pole
- Both black; definite **thickness**
- **Gap** / disruption in deep layer
  - → **partial quadriceps tendon rupture**; palpable gap or depression
- Complete gap
  - Finger from skin hits femur → **complete quadriceps tendon rupture** (blue arrow)

Jumper's Knee and Patellar Tendon Tears Ask AI

- **Jumper's knee** / patellar tendinopathy → young athletes, jumping sports
  - Tendon **thickened**; posterior border sometimes not seen
  - **Fibre disruption** possible
- Thickened tendon; **posterior fibres** discontinuous; volume increased
- Hanging on anterior band only → **partial patellar tendon** tear
- **Complete disruption** from inferior pole
  - Black line discontinuous; high fluid signal → sagittal views

Patellar Sleeve Avulsion Ask AI

- **Patellar sleeve avulsion** → athletes, young children training heavily
- Needs **suspicion**; commoner in gymnastics
- Little on **X-ray** → suspicion → MRI
  - Found in several cases
- **Cartilage** disruption + thin fracture line, inferior pole cartilage
- Missed → passive rest + knee support one month
  - Early return to play (one week–ten days) → **complete disruption** of patellar tendon
  - Seen in boy aged 8

Pitfall

- Gymnast child, inferior pole pain, little on X-ray → MRI for sleeve avulsion
- Missed sleeve avulsion + early return → complete patellar tendon disruption

Patellar Retinaculum and MPFL Ask AI

- **Axial cuts** → exactly like CT cuts
- Patella, medial side, lateral side
- Medial versus lateral → no **fibular head** on axial
- Sign → **lateral patellar facet** = longer articular surface
  - Longer facet → lateral side
  - Shorter facet → medial side
- Lateral + **medial patellofemoral retinaculum**
- Patellar dislocation, sports, contact → **disruption** common
- Orientation needed → medial versus lateral; torn from patella or **medial epicondyle**
- **MCL** → directly behind MPFL
  - Black line/spot after gap → **MCL**, not MPFL extension
- MPFL tear → two criteria
  - Not in **anatomical site** → as any ligament: ACL, PCL, MCL, LCL
  - **High fluid signal** at site
- Example → black line not continuous down to spot (MCL); high signal
  - → torn from **femoral attachment**
- Torn from **patellar attachment** → patellar dislocation
  - Not reaching **medial border of patella**; high signal on axial
- Plane summary
  - **Axial** → MPFL
  - **Coronal** → collaterals; helps when in doubt
  - **Sagittal** → almost everything

Bone Marrow and Osteochondral Lesions Ask AI

Bone Marrow Ask AI

- Marrow mostly **fat** → light T1, dark T2
- Normal → in **metaphysis**; respects physis
  - Never extends into physis or epiphysis
- Odd tibial marrow pattern → suspect **haemophilia**, haemosiderosis, leukaemia
- Marrow descends into / takes whole tibial physis → **leukaemia**
- Marrow filling physis → **haemosiderosis** case
- Both abnormal → compare colour of **physeal line** + epiphysis with marrow
  - Marrow fills and descends → **marrow disorder**

Three Entities to Distinguish Ask AI

- Cartilage + **subchondral bone** conditions
  - **Avascular necrosis** (AVN)
  - **Osteochondritis dissecans** (OCD)
  - **Stress fractures**
- Each has own sign; often confused

Avascular Necrosis Ask AI

- Abnormality → **subchondral bone** (white line); cartilage above
- Pathology extends into bone; **subchondral cysts**
- Cartilage → **normal** or minimal disruption
- **Crescent sign** → fluid between subchondral bone and cartilage
- → AVN, not OCD → cartilage normal
- **Wedge-shaped infarction** → here lateral femoral condyle
- Diagnosis → intact articular cartilage + **crescent sign**

Stress (Insufficiency) Fracture Ask AI

- **Line** appears; no fluid signal; colour change only
- Elderly, cannot bear weight; **osteoarthritis** signs → fracture often missed
- Athletes, jumping / violent sports → **passive rest** 6 weeks

Osteochondritis Dissecans Ask AI

- **Cyst** at lesion base
- Fluid between bone and cartilage → cartilage **not intact**
- Crescent location
  - AVN → directly **under cartilage**
  - OCD → between bone and bone, colour changed
    - Cartilage edge involved throughout
- Diagnosed by **X-ray** more than MRI
- MRI role → lesion **stable or unstable**
- Image → cysts; fluid signal separating piece with cartilage + subchondral bone
- AVN → cartilage alone, **fluid signal** beneath
- OCD → **cartilage layer** + piece of subchondral bone separate from bone
- AVN → **cartilage** separates from subchondral bone
- Stability → apparently judged by fluid beneath fragment
  - Fluid present → likely **unstable**; none → likely stable
- Both examples → fluid present → most likely unstable

	AVN	OCD	Stress fracture
Cartilage	Normal / minimal disruption	Not intact; edge involved	—
Fluid	Crescent directly under cartilage	Between bone and bone	None; colour change + line
Separation	Cartilage from subchondral bone	Cartilage + subchondral bone piece	—
Other	Subchondral cysts; wedge infarct	Cyst at base	Elderly; athletes → rest 6 weeks
Imaging	MRI	X-ray mainly; MRI → stability	MRI; often missed

Questions and Answers Ask AI

Ramp Tear and Failed ACL Ask AI

- **Ramp lesion** → risk factor in ACL patients
- Failed ACL reconstruction → careful look at revision → **ramp tear**

Partial ACL Tear Ask AI

- Signal must be very clear → one **intact bundle**, one not
- Continuous from tibial spine to femoral condyle, not **full thickness** → partial tear
- Partial versus complete → more accurate **arthroscopically**; most accurate on examination
- **Chronic ACL** tear → sticks onto PCL
  - Limits motion on examination → incomplete pivot shift; anterior drawer not easy
- MRI → continuous bundle, tibial spine → femoral condyle
  - Reduced thickness, or high signal with intact part → **partial ACL** tear

Cut Spacing and Field Strength Ask AI

- Request **cuts every 4 mm**, 1 mm spacing → nothing skipped
- Larger gaps → findings lost between cuts
- Machine → 1.5, 2, 4 or 3 Tesla; **3 Tesla** outstanding

Fusiform ACL Ask AI

- Incomplete **midsubstance / interstitial tear** → synovial fluid enters → fusiform shape
- Continuous above + below; bulging in middle
- Type of **partial ACL tear**; significant injury

MR Arthrography Indications Ask AI

- Previous ACL reconstruction or meniscal repair + suspected **re-tear**
- **PVNS** / localized PVNS → tumour versus PVNS
  - MRA → clear image
- **OCD** → flap attached or separated from bed

Cartilage Ulcers Ask AI

- Not covered → needs **classification** first; later lecture
- Overlap with other entities → not easy
- Not diagnosed much on MRI; **axial cuts**
  - Posterior patellar cartilage < 1 mm + **longitudinal cracks** → likely chondromalacia patellae
  - Also seen in **lateral patellar subluxation**

Timing of MRI in Acute Trauma Ask AI

- Day of injury, or after a week → **haematoma** partly resolved
- **Haemarthrosis** high signal → may hide intact ACL

MRI with Implants Ask AI

- Depends on **screw** fitted
- Old **stainless steel** screws → marked artifact
  - Graft fixed even with 6.5 cancellous screw
- Current screws
  - **Biodegradable** / PEEK → MRI-friendly
- Lecture → **general introduction** to knee MRI reading; each lesion detailed in later lectures

OCD versus AVN — Clarification Ask AI

- OCD → **fluid signal** between rest of bone and subchondral bone; overlying cartilage intact *[Q3 · As spoken; contradicts earlier statement and table that OCD cartilage is not intact]*
- AVN → **cartilage** separated from subchondral bone; fluid directly beneath

Segond and the Unhappy Triad Ask AI

- Segond → avulsed piece of **capsule**
- Some → site of **anterolateral ligament** disruption
- "Horrible triad"
  - = **unhappy triad of O'Donoghue**

Internal Derangement of the Knee — Part 1

(Internal Derangement - Collateral and Cruciate Ligament Injuries)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 182) “in ACL reconstruction” → “in PCL reconstruction”

C2 (p. 183) “ACL anterolateral bundle” → “PCL anterolateral bundle”

C3 (p. 183) “7 to 10 cm” → “7-10 mm”

C4 (p. 191) “PCL injuries only 28% isolated” → “PLC injuries only 28% isolated”

C5 (p. 192) “grade 2 opens 16 mm” → “grade 2 opens 6-10 mm”

C6 (p. 192) “grade 2 dial test 16 degrees” → “grade 2 dial test 6-10 degrees”

C7 (p. 194) “no autografts in Egypt” → “no allografts in Egypt”

C8 (p. 195) “Larson: tunnel in the tibia” → “Larson: fibular tunnel”

C9 (p. 197) “bifurcate ridge: appropriate point for PCL” → “bifurcate ridge: ACL femoral point”

C10 (p. 198) “PCL tears give false-negative Lachman” → “PCL tears give false-positive Lachman”

C11 (p. 198) “pivot is the posteromedial bundle” → “pivot is the posterolateral bundle”

C12 (p. 202) “part of the popliteal muscle” → “part of semimembranosus”

Queried - kept as spoken, but looks wrong

Q1 (p. 185) As spoken; most series report midsubstance or femoral-sided PCL tears more common than tibial bony avulsions.

Q2 (p. 190) As spoken; popliteus femoral insertion is anterior-distal to FCL; also 5 mm below stated earlier.

Internal Derangement of the Knee Ask AI

Scope of the Lecture Ask AI

- Rarely a major exam topic; frequent in **MCQs**
- Lecture in 2–3 parts; this part → collateral ligaments and cruciates
- Each topic worth a full lecture → ACL alone ≈20 lectures, PCL another 20

Definition Ask AI

- **Internal derangement**: injury/incident → disturbed anatomy of normal knee joint
  - Analogy: room turned upside down
- Disturbed anatomy → **intermittent signs and symptoms**
  - Pain
  - Instability
  - Abnormal mobility of the knee
- Group of disorders disrupting function of ligaments, cartilage and menisci
- Three topics
  - **Injured ligaments** → ACL, PCL, MCL, LCL
  - **Loose bodies** → bone/cartilage pieces, e.g. osteochondritis dissecans
  - **Torn menisci** → flipped, bucket-handle, degenerative tear, meniscal cyst

Clinical Features Ask AI

- **Locking** / recurrent locking → pain → pseudo-paralysis of hamstrings and quadriceps
- Sudden loss of motion → usually **full extension**
- Sudden loss of function
- Some patients **self-reduce** the knee
  - Bucket-handle tear → patient reduces it
  - Loose body → patient flicks it out of the way
  - Patient senses something moving / not right
- Clinical evaluation → under each individual injury

The Three Pillars of the Knee Ask AI

- Knee stands on 3 pillars → **lateral, central, medial complexes**
- Central complex → **intra-articular** structures
- Main four lateral components → form **posterolateral corner**
  - Other lateral components exist; these four most important

Lateral complex	Central complex	Medial complex
Lateral collateral ligament (= fibular collateral ligament)	ACL	MCL — superficial and deep
Iliotibial band (tract)	PCL	Posteromedial capsule
Biceps femoris	Medial meniscus	Semimembranosus
Popliteus	Lateral meniscus	Pes anserinus
—	—	Others, e.g. posterior oblique ligament

Posterior Cruciate Ligament — Anatomy Ask AI

Importance Ask AI

PCL

- PCL → **main pillar** of the knee
- Torn → addressed first; rest of knee arranged around it
- Done wrong at surgery → every procedure built on it also wrong

Attachments and Dimensions Ask AI

PCL

- **Origin**: posterior tibial sulcus → ≈17 mm below posterior articular surface
- **Insertion**: crescent on anterior aspect of lateral surface of medial femoral condyle
- Fans out like a **crescent** ("hilal"); ACL → clearly defined band
- Length < 4 cm; thickness ≈13 mm
- **30% larger than ACL** → frequent MCQ

Components Ask AI

PCL

- 2 bundles + **meniscofemoral ligaments** → 3 components
- **Anterolateral bundle**
  - More important → posterior stability at 90°
  - **Stronger** of the two — unlike ACL bundles
- **Posteromedial bundle**
  - Tight in extension
  - Less important
- Single-bundle **PCL reconstruction** → anterolateral bundle only *[C1 · transcript: "in ACL reconstruction"]*
  - Carries stability against posterior translation at 90° flexion
- Meniscofemoral ligaments → surround PCL front and back
  - **Humphrey** → anterior meniscofemoral ligament
  - **Wrisberg** → posterior meniscofemoral ligament
  - Course: posterior horn (probably lateral meniscus) → blend with PCL → medial femoral condyle
- Strength ≈3000 N → **strongest ligament in the knee**; axis of whole knee
  - Broad, large, very strong
- **Incidence** of PCL injury not high → high-energy trauma tears it easily

High Yield

- PCL 30% larger than ACL
- Strength ≈3000 N → strongest ligament in the knee
- Anterolateral bundle → stronger; reconstructed in single-bundle technique
- Humphrey → anterior; Wrisberg → posterior meniscofemoral ligament

Footprints Ask AI

PCL

- AP view: **anterolateral bundle** → trochlear point (mid-trochlea) to medial arch point *[C2 · transcript: "ACL anterolateral bundle"]*
  - ≈2–3 o'clock on lateral surface of medial femoral condyle
- Anterior meniscofemoral ligament → in front
- Posterior meniscofemoral ligament → behind
- Femoral insertion close to **articular cartilage**
  - PCL → anterior, on lateral surface of medial femoral condyle
  - ACL → posterior, on medial surface of lateral femoral condyle
- Figure: ACL cut away → PCL fully visible
- **Champagne-glass drop-off point** → ≈7–10 mm down into posterior tibial sulcus; PCL attachment *[C3 · transcript: "7 to 10 cm"]*
- Site of arthroscopic **guide pin** for PCL reconstruction
- **Killer turn**: PCL curves; ACL straight
  - → difficult surgery; higher failure if set up wrongly
- Vital structures directly behind → needs experience and technical care
  - **Popliteal vessels** ≈0.5 cm away
- PCL footprint: anterolateral bundle very large
- Trochlear point → medial arch point; principal **reconstruction** target
- Beneath: posteromedial bundle, anterior and posterior meniscofemoral ligaments
- Anterolateral bundle → **closest to articular cartilage** of medial femoral condyle
- **Bifurcate ridge** on medial femoral condyle → separates anterolateral from posteromedial bundle
  - Compare ACL footprint → bifurcate ridge + resident's ridge

Tibial Attachment and Medial Meniscus Root Ask AI

PCL

- Axial view of tibia → ridge = **posterior tibial sulcus**, PCL attachment
- Shiny white fibers → root of **posterior horn of medial meniscus**
- Tunnel entry/exit at articular level → injures **meniscal root** → ruins everything
- Root very close to PCL
- Distance ≈8 mm
- Tunnel exit → **below articular surface** of tibia

Pitfall

- Tunnel exit at tibial articular surface → injures medial meniscus posterior root
- Root ≈8 mm from PCL → exit below articular surface
- Popliteal vessels ≈0.5 cm behind PCL tibial attachment

PCL Injury — Mechanism, Grading and Diagnosis Ask AI

Mechanism of Injury Ask AI

PCL

- Needs very hard, substantial blow
- **Dashboard injury** → most common mode
  - Car accident → knee hits dashboard → tibia pushed posteriorly → PCL tears
  - Usually **isolated PCL** injury
  - Violent → corners + remaining ligaments + ACL → **knee dislocation**
- Fall on **flexed knee**, foot in plantar flexion → tibia driven backwards
- **Hyperextension** injury
- Associated **femoral fractures** (likely shaft)
  - → very high incidence of PCL injury
- Summary
  - Direct blow to anterior tibia (dashboard)
  - Hyperextension
  - **Hyperflexion** with foot in plantar flexion

Cornerstone of the Knee Ask AI

PCL

- PCL = **cornerstone** → not rebuilt → knee never stable
- **Central stabilizer** → must be addressed
- PCL + quadriceps tendon → **main dynamic stabilizers** in sagittal plane
- Video: football player → **hyperextension** → PCL injury

Grading Ask AI

PCL

- **Posterior tibial translation** → measured relative to femoral condyles
  - Normal: medial tibial plateau 10 mm anterior to femoral condyles
- Grade 1: tibia back 5 mm → still 5 mm **anterior**
- Grade 2: tibia back 10 mm → anterior tibia **flush** with femoral condyle
- Grade 3: tibia **falls behind** femoral condyles

Grade	Posterior tibial translation	Tibia vs femoral condyles	Injury
1	1–5 mm	Still anterior	Partial
2	5–10 mm	Flush	Complete; most likely isolated PCL
3	> 10 mm	Behind	+ ACL + posterolateral corner (figure)

- Figure, upper left: normal → tibia ≈10 mm anterior
- Back 5 mm → 5 mm anterior; back 10 mm → flush, PCL completely torn
- > 10 mm with ACL + posterolateral corner → **posterior sag sign**
- Femoral condyle now **anterior** to tibia

Bony (Tibial) Avulsions Ask AI

PCL

- **Bony avulsions** → most common PCL injury *[Q1 · As spoken; most series report midsubstance or femoral-sided PCL tears more common than tibial bony avulsions.]*
  - Type 1 → separated, in place
  - Type 2 → minimal displacement
  - Type 3 → complete separation

Exam Pearl

- Dashboard injury → most common mode of PCL injury
- Tibial bony avulsion → most common PCL injury
- Femoral fractures → very high incidence of PCL injury

Clinical Examination Ask AI

PCL

- PCL injury **commonly missed** → sagged tibia not noticed
- Examiner pulls tibia forward → subconscious assumption of ACL injury
- Sagged tibia + anterior drawer → **false-positive anterior drawer**
  - Tibia seems to move forward; not truly
- First confirm **tibia not sagged** → then examine
- Many such patients → taken to ACL surgery
- Arthroscopy: ACL **lax** → wrongly judged partially torn
  - Lax because tibia sagged; tibia reduced → laxity disappears
- → unnecessary **ACL reconstruction** from wrong examination
- Especially after **dashboard injury**
- Or **hyperextension** (football player in video) → check sag first
- **Varus/valgus stress** → collateral ligaments
  - Opens at 0° → MCL or LCL + **PCL** injury
  - Opens at 30° only → MCL or LCL only; **PCL intact**
- **Posterior sag sign** (Godfrey posterior sag test)
  - Supine, knee 90°; examiner stands away, looks **tangentially**
  - → posterior shift of tibia; normal medial plateau 10 mm anterior to medial femoral condyle
  - Sag not reduced → false-positive anterior drawer and **Lachman** → mimics ACL injury
- **Posterior drawer test**: knee 90° → posterior force → posterior tibial translation
  - Assess > 10–12 mm
  - > 15 mm vs other knee → combined **posterolateral corner** ± ACL
- **Quadriceps active test** → most specific
  - Quadriceps contraction → tibia pulled anteriorly
  - Tibia returns to **normal position** → very characteristic of PCL injury
- **Dial test** → more specific for posterolateral corner / posterolateral rotatory instability
  - > 10° external rotation at 30° and 90° → **PCL + posterolateral corner**
  - > 10° at 30° only → **isolated posterolateral corner**
- Figure: sag → femoral condyles and **patella** protrude beyond anterior tibial edge
- Figure: quadriceps active (active drawer) test → tibia moves anteriorly
- Figure: **valgus test**
  - Stresses **medial collateral ligament**
- Figure: **posterior drawer** → like ACL anterior drawer

- Torn PCL → whole tibia moves **backwards** with examiner's force
- **Quadriceps active test** technique
  - Knee 90° → patient tightens quadriceps → sagged tibia pulled forward, visible
  - Modification: knee at 45° and 90°
  - Knee beyond **quadriceps neutral angle** ≈60–70°
    - Above → contraction translates tibia anteriorly
    - Below → probably translates tibia posteriorly
  - Specific for PCL → tibia reduces **anteriorly** when PCL torn
- **Dial test** technique → supine or prone both possible
- Original (**Cooper**) → prone
  - Knee 30°; grasp feet
  - Externally rotate both feet → **side-to-side difference** > 10°?
  - Repeat at 90° → isolated posterolateral corner vs **posterolateral corner + PCL**

Pitfall

- Sagged tibia → false-positive anterior drawer and Lachman
- Lax ACL at arthroscopy → may be sagged tibia, not ACL tear
- Result → unnecessary ACL reconstruction
- Rule → reduce sag before any drawer or Lachman

Imaging Ask AI

PCL

- **Plain X-ray**
  - Posterior tibial avulsion of PCL
  - Posterior tibial subluxation → uncommon
- **Stress views** / kneeling stress radiographs → clearer
- **Stress kneeling view**
  - Patient kneels, knee on pillow
  - Line along posterior border of tibia
  - Normal: line 5–6 mm anterior to posterior **Blumensaat line**
  - Measure **perpendicular distance** to posterior-most Blumensaat line
- Example: normal side +5.7 mm; subluxed side −5.9 mm
  - **Side-to-side difference** ≈11.6 mm → grade 3
- Classification → posterior drawer grade vs **kneeling stress view**
- Grade 2 → complete PCL; grade 3 → **PCL + posterolateral corner**

Posterior drawer grade	Clinical	Kneeling stress posterior translation	Meaning
1	5 mm of plateau still anterior	≈7 mm	—
2	5–10 mm; plateau flush with condyle	≈8–11 mm	Complete PCL
3	Tibia behind condyle	≥ 12 mm	PCL + posterolateral corner

- X-ray → **tibial bony avulsion**
- MRI: normal PCL → thick, black, strong, easily followed; tear seen in figure

PCL — Clinical Scenarios and Management Ask AI

Clinical Scenarios Ask AI

PCL

- **Acute isolated PCL** → uncommon, easily missed, very mild symptoms
- **Acute PCL + posterolateral corner**
  - Examine **common peroneal nerve**
  - **Marked instability** on examination
- **Chronic isolated PCL** → patient functions on it
  - Only 50% complain of **instability**
  - Tibial sag → ↑ contact pressures on patella and trochlea → **patellofemoral arthritis**
- **Chronic PCL + posterolateral corner**
  - Severe instability → **triple varus**
  - Varus types → primary, secondary (double), triple

Management Algorithm Ask AI

PCL

- Algorithm

Injury	Finding	Treatment
Acute tibial avulsion fracture	Large fragment	ORIF via modified posteromedial approach
	Small fragment, grade 1–2 laxity	Brace, conservative
	Small fragment, grade 3 laxity	PCL reconstruction
Isolated PCL	Grade 1–2	Conservative
	Grade 3 (giving way), athlete	PCL reconstruction
	Grade 3, non-athlete	Conservative
Combined ligament injuries	< 3 weeks	Surgical repair and reconstruction

- **Isolated** → no posterolateral corner, ACL, LCL or MCL injury
- Non-athlete grade 3 → broad, thick PCL → **potential to heal**
- **PCL brace** ≠ ACL brace
  - Posterior calf pad → pushes **sagged tibia** forward
  - Also protects early after PCL surgery
- **Surgical indications**
  - Grade 3
  - Associated injuries, e.g. posterolateral corner
  - Bony avulsions
  - Chronic → usually **osteotomy**, not PCL reconstruction
    - Triple varus → symptomatic **severe varus deformity**

Surgical Techniques Ask AI

PCL

- **Bony avulsion repair** → endobutton, screws; arthroscopic or open
  - Many techniques
- **PCL reconstruction** → two main techniques
  - **Tibial inlay** → open; trough in posterior tibia at sulcus
  - Graft placed in trough → fixed with screw
  - **Transtibial drilling** → as ACL, posteriorly onto champagne drop-off point
    - Care → **popliteal vessels**
- Controversies
  - Single vs double bundle
  - Autograft vs allograft
- **PCL graft longer** than ACL graft
  - Tunnel exits ≈1 cm below tibial articular surface
  - Wraps posteriorly → enters knee back to front → fixed in femoral condyle
  - PCL → ≥ 11–12 cm; ACL → 9.5–10 cm acceptable, not for PCL
- Chronic instability / chronic PCL deficiency → **medial opening-wedge HTO** for varus malalignment
  - Trick → **increase posterior tibial slope** → reduces tibial sag
- **Grafts**
  - Patellar tendon
  - Quadriceps tendon or hamstrings (as in ACL) → **most common**
  - Allografts → not available in Egypt

Arthroscopic PCL Reconstruction Ask AI

PCL

- Standard portals
  - High anterolateral
  - High anteromedial
  - Low anterolateral
  - **High posteromedial** → avoids working blindly
- Viewing behind tibia → needs **70° lens**; uncommon in Egypt, harder orientation
- No 70° lens → **posteromedial portal** to protect vessels
  - Needle first → confirm direction toward PCL
  - Entry behind femoral condyle → posteromedial aspect
  - Opened with blade / arthroscope
- **Femoral tunnel** → target anterolateral bundle
  - Trochlear point → medial arch point; ≈12 o'clock to 2–3 o'clock (side-dependent)
  - ≈7–8 mm behind **articular cartilage**
- **Tibial tunnel**
  - **PCL guide** ≠ ACL guide → part sits behind tibia
  - High posteromedial portal preferred while drilling
  - Tip visible; **curette** pushes capsule and vessels back → protects from guide pin
- Fixation → screws, as ACL
- Figure: pin exits away from shiny white fibers (**medial meniscus posterior root**)
  - Ideally 1 cm lower; at least behind tibia, not articular surface
- Figure: **double-bundle reconstruction**

Rehabilitation Ask AI

PCL

- Differs from ACL
  - **Gravity** stretches graft → mostly prone, or brace pushing tibia forward
  - **Much slower** than ACL
  - **Return to sport** → not before 1 year (professional footballer)
  - **Individualised** → physiotherapist understands protocol and gravity effect

Posterolateral Corner — Anatomy and Function Ask AI

Overview Ask AI

PLC

- "**Jungle of the knee**" / "dark side of the knee" → many structures, confusing
- Anatomy → essential focus
- Three most important structures
  - **Fibular collateral ligament** → just behind lateral epicondyle
  - **Popliteus tendon** → 5 mm behind and below; deeper, lower on lateral femoral condyle
  - **Popliteofibular ligament** → popliteus → fibular head
- Main three → FCL, popliteofibular ligament, **popliteus tendon**
- Figure: specimen
- **Static stabilizers** → 3 major (remember for exam) + 3 others

Major static ( )	Other static ( )	Dynamic ( )
LCL (fibular collateral ligament)	Arcuate ligament	Iliotibial band (tract)
Popliteus tendon	Fabellofibular ligament	Popliteus muscle (not tendon)
Popliteofibular ligament	Posterolateral capsule	Biceps femoris
—	—	Lateral head of gastrocnemius

- **Fabellofibular ligament** → attaches at fibular head beside FCL
- Total → **6 static, 4 dynamic** stabilizers
- Confusing for everyone → hence "jungle"

Femoral Attachments Ask AI

PLC

- Figure: attachments
- **FCL** → ≈4–5 mm posterior to lateral epicondyle
- **Popliteus tendon** → ≈1 cm below FCL; also 5 mm posterior to lateral epicondyle
- Key figure → **reconstruction** built on it
  - Lateral epicondyle → FCL directly behind, 5 mm
  - Popliteus → 10 mm below, 5 mm posterior *[Q2 · As spoken; popliteus femoral insertion is anterior-distal to FCL; also 5 mm below stated earlier.]*
- **PLC reconstruction** → based on FCL + popliteus tendon
- Posterior view → **arcuate complex**
  - = FCL + arcuate ligament + popliteus tendon
  - **Arcuate ligament** → 2 limbs: lateral, medial
  - FCL → shown green
- Still most important three → FCL, popliteus tendon, **popliteofibular ligament**

Layers of the Lateral Knee Ask AI

PLC

- Divided by **layers**, not structures

Layer	Contents
1 (superficial)	Iliotibial band; biceps femoris behind it
Between 1 and 2	Common peroneal nerve
2	Lateral patellar retinaculum
3 (deep)	LCL, fabellofibular ligament, arcuate ligament, popliteus tendon, popliteofibular ligament, capsule

- **Common peroneal nerve** → directly beneath biceps femoris
- PLC exposure via curved incision → find **biceps femoris** first → nerve beneath
- Unknown nerve location → easy **common peroneal nerve injury**

> **Clinical Pearl**
> - Opening the posterolateral corner → find biceps femoris first
> - Common peroneal nerve → directly beneath biceps femoris
> - Nerve lies between layers 1 and 2

Function and Associations Ask AI

PLC

- PLC resists **external rotation** and **varus** → basis of dial test
- Only 28% **isolated** *[C4 · transcript: "PCL injuries only 28% isolated"]*
- Associated → **more often with PCL than ACL** → MCQ
- Missed PLC → famous cause of **ACL reconstruction failure**
  - Returns after 1–2 months → knee opens
  - Leg → **varus + recurvatum**

Posterolateral Corner Injury — Diagnosis Ask AI

Mechanism Ask AI

PLC

- Figure: **failed ACL reconstruction** → missed PLC injury
- Blow to **anteromedial** knee
- **Varus stress** on flexed knee
- **Hyperextension** → American football player → PCL + PLC injury
- **External rotation** twisting injury
- **Knee dislocation**
- **Common peroneal nerve** injury → 30% (≈3 in 10)
- Nerve status not documented + postoperative **foot drop** → cannot prove preexisting → criminal liability

High Yield

- PLC → resists external rotation and varus
- Only 28% isolated → more often with PCL than ACL
- Common peroneal nerve injury → 30%; document before surgery
- Missed PLC → failed ACL or PCL reconstruction

Grading Ask AI

PLC

- Based on **varus stress** opening + dial test rotation

Grade	Varus stress opening	Dial test rotation	End point
1	≈5 mm	≈5°	—
2	6–10 mm (X-ray)	6–10°	— *[C5 · transcript: "grade 2 opens 16 mm"] [C6 · transcript: "grade 2 dial test 16 degrees"]*
3	> 10 mm	> 10°	None

- Dial > 10° → **PLC + PCL**
- Grade 3 → everything torn; usually **PCL** involved

Examination Ask AI

PLC

- Same tests as PCL
- Key → **watch patient walk**
  - Chronic → **varus thrust** → knee buckles from lateral side
  - Standing → **varus**
- **Varus stress**: opens at 0° → combined cruciate injury; 30° only → isolated PLC
- Dial test → > 10° at 30° only vs at 30° and 90°
- **External rotation recurvatum test** → lift big toe → leg falls into ER + recurvatum
  - Very characteristic of PLC injury
- **Posterolateral drawer test** → posterior drawer with leg in 15° external rotation
  - Foot fixed → push tibia back → tibia moves backwards
- **Reverse pivot shift test** → PLC; opposite of ACL pivot shift
  - Knee 90°, external rotation, **valgus** on tibia
  - Knee **extended** (ACL test → flexed)
  - Tibia sagged posteriorly → reduces at ≈20° flexion → reduced in **full extension**
  - Mechanism → **iliotibial band** changes from flexor to extensor
- Figure: external rotation recurvatum test
- Figures: dial test; **reverse pivot shift** — video technique
  - Supine, knee 90° → ipsilateral hand externally rotates tibia
  - Contralateral hand on proximal lateral tibia → valgus; **axial load** via distal arm or iliac crest
  - Posterolateral rotatory instability → **lateral tibial plateau** subluxes posteriorly
  - Passively extend → maintain external rotation, valgus, axial load

Imaging Ask AI

PLC

- **Stress X-rays**
  - Side-to-side 2.5–4 mm → **isolated LCL**
  - > 4 mm → **posterolateral corner**
- **Long-standing radiographs** → chronic PLC → deformity type
- Summary
  - Plain X-ray → **avulsion** of fibular head or femoral condyle
  - Stress radiographs → **joint line opening**
  - Long-standing X-ray → **alignment**
  - MRI → see knee MRI lecture

Types of Varus Knee Ask AI

- Three types of **varus malalignment**

Type	Components	Cause
Primary (bony)	Tibiofemoral varus malalignment	Loss of medial meniscus + articular cartilage, as in osteoarthritis
Double (secondary)	Primary + stretched lateral structures	FCL torn / soft tissues stretched
Triple	Double + external rotation + recurvatum	PLC completely torn; whole lateral ligamentous complex

- Triple varus → not only FCL → **whole lateral ligamentous complex**
- PLC protects against **external rotation and recurvatum** → lost in triple varus
- Double varus → **soft tissues** stretched, or FCL alone torn
- Treatment → **wedge osteotomy**

Imaging Examples Ask AI

PLC

- Figure: **normal alignment**
- Figure: **mechanical alignment deviation**
- Plain X-ray: avulsed **fibular head** fragment → FCL or whole PLC avulsed
- **Stress view**: joint line opens with varus → ≈12 mm, 16 mm
- **Long-leg film** → primary vs double vs triple varus
- MRI: normal **LCL**; popliteus tendon
- LCL separation from **fibular head**
- **Popliteus tendon** injury

Posterolateral Corner — Management Ask AI

Conservative vs Operative Ask AI

PLC

- **Conservative** → usually no role; considered if
  - Grade 1 → opening < 4 mm
  - Isolated → no ACL, PCL, MCL injury
- PLC → **surgery** very commonly needed
- **Acute** → PLC repair
  - Rarely alone → usually with ACL; some form of **knee dislocation**
  - No allografts in Egypt → **autografts** *[C7 · transcript: "no autografts in Egypt"]*
    - Semitendinosus and gracilis harvested → one for PCL
    - Semitendinosus → ACL; **gracilis** → LCL
    - No other grafts → harvest from **healthy knee**
  - < 2 weeks → tissues good → **anchor** in fibular head → repair
  - Weak repair → **hybrid reconstruction and repair**
- **Chronic** (≥ 3 weeks) → PLC + ACL + PCL reconstruction
  - ± **High tibial osteotomy** if triple varus

Reconstruction Techniques Ask AI

PLC

- Operative options → **fibular-based** or tibial- and fibular-based

Technique	Tunnels	Femoral fixation	Reconstructs
Larson	Fibular	1 point	FCL only
Modified Larson	Fibular	2 screws	Popliteus + FCL
Tibial + fibular, single point	Tibial + fibular	1 point (femoral epicondyle)	Popliteus + FCL
LaPrade	Tibial + fibular	2 screws	Popliteus + FCL

- **Larson** → usual; fibular-based
  - Fibular tunnel → semitendinosus or gracilis → both limbs fixed at one point *[C8 · transcript: "Larson: tunnel in the tibia"]*
  - Reconstructs FCL only
- Popliteus not reconstructed → use if rest of PLC **intact**
- **LaPrade** → full reconstruction
  - Popliteus + FCL
  - Fibular tunnel + tibial tunnel
- Specimen, posterior view → 2 **femoral fixation** points → popliteus, FCL
- **Modified Larson** → two separate points, still fibular-based
  - Limbs crossed → one as popliteus (own screw), one as FCL
- Another technique → **tibial tunnel**
  - Graft around back of tibia → acts as popliteus
  - + **fibular tunnel** → two limbs taken up
  - All fixed at femoral epicondyle → one point
- **LaPrade**: 2 tunnels below (tibial, fibular) + 2 screws above
  - One screw → popliteus; one → FCL
- **Larson modification**: posterior limb → anterior, as popliteus
  - Anterior limb → posterior, more proximal, as **FCL**; two different screws

Complications Ask AI

PLC

- High incidence of **arthrofibrosis**
- Missed PLC → failure of **ACL or PCL reconstruction**
- **Peroneal nerve injury** ≈30% → always examine → medico-legal problems

Anterior Cruciate Ligament — Anatomy Ask AI

ACL

Bundles and Dimensions Ask AI

- 2 bundles → **anteromedial**, posterolateral
- Named by **tibial attachment** sites
- Length ≈3–3.5 cm
- Width ≈7–12 mm; average ≈9–10 mm
- **30% smaller** than PCL
- **Anteromedial bundle** → main surgical focus; anterior drawer
  - Resists **anterior tibial translation**
- **Posterolateral bundle** → tight in extension
  - Mainly **rotational stability** → pivot shift

Feature	ACL	PCL
Size	30% smaller	30% larger
Length	3–3.5 cm	< 4 cm
Width / thickness	7–12 mm (average 9–10 mm)	≈13 mm
Bundles	Anteromedial + posterolateral	Anterolateral + posteromedial (+ meniscofemoral ligaments)
Tight in extension	Posterolateral bundle	Posteromedial bundle
Key bundle	Anteromedial → anterior translation	Anterolateral → posterior stability at 90°; stronger
Femoral insertion	Posterior, medial surface of lateral femoral condyle	Anterior, lateral surface of medial femoral condyle
Course	Straight	Killer turn
Graft length	9.5–10 cm acceptable	≥ 11–12 cm
Rehabilitation	—	Much slower; return to sport ≥ 1 year

Femoral Attachment Ask AI

ACL

- Landmarks → **resident's ridge** (intercondylar ridge), bifurcate ridge
- Debated graft position
  - Isometric → at this site
  - **Eccentric** → behind bifurcate ridge
  - Direct fibers → all arise and attach at one place
  - Least tension → at this site
- Latest paper → **IDEAL point** (black dot)
  - Flush with intercondylar (resident's) ridge
  - Flush with, but behind, **bifurcate ridge**
- IDEAL → combines five debated advantages

Must-Know

- I → Isometric
- D → Direct fibers
- E → Eccentric
- A → Anatomic
- L → Low tension

Tibial Attachment Ask AI

ACL

- **Tibial landmarks**
  - Posterior border of anterior horn of **lateral meniscus** → guide line
  - Just lateral to, flush with, medial tibial spine
  - 7–9 mm anterior to **PCL**
- ACL **stump** present → big help
- Stump removed (shaver, burr, vaporizer) → rely on landmarks
  - 9 mm anterior to PCL
  - Middle of notch → pull **arthroscope** back to confirm centre
- **ACL remnant** → aim guide at its centre for drilling
- Old method → bifurcate ridge as femoral point *[C9 · transcript: "bifurcate ridge: appropriate point for PCL"]*
- Now → behind bifurcate ridge → **IDEAL point**

ACL Injury — Mechanism and Diagnosis Ask AI

Mechanism Ask AI

ACL

- **Twisting** injury with hyperextension
- **Hyperextension** alone → ACL, PCL or both
- **Hyperflexion** injury
- **Pop** → patient hears snap ("cucumber snapped") → very characteristic
- Classic position → **valgus** + internal tibial rotation + slight flexion
  - → tibia translates **anteriorly** (vector analysis) → ACL tears

Clinical Evaluation Ask AI

ACL

- **Lachman test** → most sensitive
  - Grade 1 → 3–5 mm; grade 2 → 5–10 mm; grade 3 → > 10 mm
  - PCL tear → **false-positive Lachman** → reduce sagged tibia first *[C10 · transcript: "PCL tears give false-negative Lachman"]*
  - Tibia 10 mm anterior to **femoral condyle** → then test
- **Pivot shift test**
  - Pivot = **posterolateral bundle**; torn → shift transferred to iliotibial band *[C11 · transcript: "pivot is the posteromedial bundle"]*
  - Iliotibial band → **reduces subluxated anterior tibia** backwards
  - Requires **intact MCL** → valgus stress impossible if torn
- Figure: **anterior drawer**
- Demonstration: from full extension → knee to **30°** flexion
- Outer hand → slight **external rotation** of tibia
- Then translate tibia **anteriorly**
- **Pivot shift technique** (video)
  - Supine; hip flexed and abducted 30°
  - Grasp calcaneus from outside → **internal tibial rotation**; ± axial compression
  - Other hand → slight valgus at fibula level
  - Torn ACL → tibia **subluxes anteriorly** in extension
  - Move leg from extension → **flexion**

Imaging Ask AI

ACL

- No fibers on **lateral wall of notch**
- **Empty notch sign** → coronal/notch view; normally PCL + ACL, here only round shape
- Sagittal → torn fibers **sagging** at bottom
- Figure: **ACL tear**
- Axial → **fluid signal** along medial wall of lateral femoral condyle → empty notch
- X-ray → **tibial avulsion** of ACL
- Intact vs torn ACL
  - Normal → parallel to, or steeper than, **Blumensaat line**
  - Torn → fallen; abnormal fluid signal
- **Avulsion of tibial insertion**
  - Seen in many patients nowadays
- **Chronic ACL signs**
  - Deep **sulcus terminalis** on lateral femoral condyle
  - **Segond fracture** → capsular avulsion at ALL (anterolateral ligament)
  - **Notching** of intercondylar notch → narrowed, descends on lateral view
- **ACL avulsion classification**
  - Type 1 → small crack; lifting, not separated
  - Type 2 → separated, not fully displaced
  - Type 3 → completely separated; fluid signal beneath
  - Type 4 → **complete dissociation**

ACL Management Ask AI

Options Ask AI

ACL

- **ACL reconstruction**
- **ACL repair**
- **ACL revision** → failed previous surgery
- Usually **not isolated** → look for
  - **Ramp lesion**
  - MCL injury
  - Meniscal injury
  - Chondral injury, ulcer, **OCD**
  - PCL / posterolateral corner injury
- Associated varus or valgus → osteotomy
  - **High tibial** or distal femoral osteotomy if chronic
- Figure: normal reconstruction
- **Repair** → acute **flush separation** from femoral footprint
  - Suture → two anchors in medial wall of lateral femoral condyle
  - Pulled down with **internal brace**
- Technique possible but **not common**
- Avulsion → many methods, e.g. **buttons**
  - Through fragment → out below → tied over tibial tunnel medially

Reconstruction Variables Ask AI

ACL

- ACL reconstruction → worth 1–3 lectures
- **Graft** → bone–patellar tendon–bone, quadriceps tendon, quadruple or triple hamstring
- **Single** vs double bundle
- **Transtibial** vs anteromedial portal vs all-inside
- **Fixation** → screws, suspension, staples, screw + washer, anchors, press-fit (tibial bone plug)
- **Remnant preservation** or not

Pediatric ACL Ask AI

Indications Ask AI

Pediatric

- **Pediatric ACL** → more common now; children in sport; cases ≈weekly
- **Open physes** until ≈14 years (boys and girls)
- Surgery → athlete; complete tear with significant trauma; refuses to leave sport

Techniques Ask AI

Pediatric

- **Intra-articular, physeal-sparing** → all drilling in epiphysis, below physeal line
  - No crossing → no **growth arrest**
- **Transphyseal** → boys 13–16 years, girls 12–14 years
- **Partial transphyseal** → transtibial without femoral side
  - Or transfemoral without touching **tibial physis**
- **Combined** intra- and extra-articular

- Boys 15–16 years / girls 14–15 years → complete **adult reconstruction**
- Results → **no difference** between physeal-sparing, transphyseal, partial transphyseal
  - 6–7 mm drill → no significant **physeal injury** if reaming slow and gentle
  - No screws → screws induce **physeal arrest**
- **Soft-tissue grafts** → few problems

Technique	Physis	Notes
Epiphyseal (intra-epiphyseal)	Spared	All drilling within epiphysis, below physeal line
Extraphyseal	Spared	Drill and fix below physis, and above physis
Partial transphyseal	Crossed on one side	Transtibial only, or transfemoral only
Transphyseal	Crossed	Boys 13–16, girls 12–14
Adult reconstruction	—	Boys 15–16, girls 14–15

- **Factors ↑ physeal injury**
  - Tunnel diameter > 12 mm
  - **Oblique** tunnels
  - **Interference screws** across physis
  - High-speed drilling or reaming
  - **Extra-articular tenodesis** → at lateral femoral physeal line → high lateral growth arrest
  - Suturing near **tibial tubercle**
- Figures: pediatric reconstruction methods
- Iliotibial band strip → left attached to **Gerdy's tubercle**
  - → **over-the-top** point → back through notch
  - → under **intermeniscal ligament** → re-sutured to proximal tibial periosteum
- Lateral view → strip from **iliotibial tract**
  - Attached to Gerdy's tubercle → over-the-top position
- = **MacIntosh** technique → very old, for ACL
  - Under intermeniscal ligament → proximal tibia → **sutured to periosteum**
- Other options
  - Transphyseal → usual way
  - **Extraphyseal** → drill and fix below physis; drill and fix above physis
  - **Epiphyseal** → within epiphysis, away from physeal line
- Extraphyseal + epiphyseal → **physeal-sparing**; transphyseal → jeopardises growth
- Most avoid screws → **suspension systems** (endobuttons)
  - Fixation far above and far below, or screw + washer from above

ACL Avulsion Repair and Complications Ask AI

Avulsion Repair Ask AI

ACL

- **Avulsion** repair options
  - Suture-passing device → **lasso loop** around stump → pulled down via tibial tunnel
  - Multiple sutures within stump
  - Fixed with **anchors** → may need 3–4 → costly
  - Sutures through stump + bone fragment → **bony tunnel**
  - Sutures on anchors
  - Sutures around stump and bone
  - Two tunnels in fragment + two **tibial tunnels** → tie over bone bridge
- **Lasso** catches ACL from behind
  - Suture retrieved with **grasper** via anterolateral portal; two tibial tunnels

Complications Ask AI

ACL

- **Intraoperative**
  - **Graft–tunnel mismatch** → graft larger or smaller than tunnel
  - **Posterior wall blow-out** of femur while drilling → screw-only plan fails
    - Always prepare **screws + suspension**
    - Rescue → screw + washer from outside, or suspension on lateral femoral cortex
  - Femoral or tibial **tunnel malposition** → footprint / bony landmarks ignored
  - **Fixation failure** → screw breaks on insertion; endobutton breaks on tensioning
- **Postoperative**
  - **Infection** / septic arthritis → not rare
  - Loss of motion, **arthrofibrosis**
  - **Tunnel osteolysis** → CT: 8 mm tunnel → 14 mm
    - Causes → **high-speed reaming**, fluid tracking into tunnels, others
  - **Cyclops** lesion
  - Vascular or nerve injury

Clinical Pearl

- Prepare screws and suspension before every ACL
- Posterior wall blow-out → screw + washer outside, or cortical suspension
- Pediatric ACL → suspension fixation; no screws across physis

Medial Collateral Ligament Ask AI

Anatomy Ask AI

MCL

- **Reconstruction** rarely needed → superficial MCL broad, thick, 2 strong attachments
  - Complete injury → uncommon
- Layers → named as on lateral side
- **Superficial MCL** → layer 2
  - Femoral: epicondyle, 1 cm anterior and 1 cm distal to **adductor tubercle**
  - Tibial: ≈4.5 cm below articular surface, deep and posterior to **pes anserinus**
  - Pes anserinus → tendons harvested in ACL surgery
- **Deep MCL** → layer 3; capsular thickening
  - Meniscofemoral + **meniscotibial** ligaments → part of medial joint capsule

Feature	Superficial MCL	Deep MCL
Layer	2	3
Femoral attachment	Epicondyle, 1 cm anterior + distal to adductor tubercle	—
Tibial attachment	≈4.5 cm below articular surface; deep + posterior to pes anserinus	—
Composition	Broad, thick	Meniscofemoral + meniscotibial ligaments
Function	Main stabilizer against valgus, all angles	Secondary stabilizer against valgus

Stabilizers Ask AI

MCL

- **Superficial MCL** → main valgus stabilizer at all angles
- **Deep MCL** → secondary valgus stabilizer
- Other stabilizers (as on lateral side)
  - **Posterior oblique ligament**
  - Oblique popliteal ligament → probably part of semimembranosus *[C12 · transcript: "part of the popliteal muscle"]*
  - Posterior capsule
  - Dynamic → **semimembranosus complex**

Mechanism Ask AI

MCL

- **Contact** or non-contact injury → high load
  - **Valgus stress** → MCL tears
- Valgus → MCL; **varus** → LCL

Classification Ask AI

MCL

- **American Medical Association** → valgus stress at 30° flexion
  - ≤ 3 mm → **physiological laxity**
  - Grade 1 → up to 5 mm
  - Grade 2 → 6–10 mm
  - Grade 3 → > 10 mm
- Same scale as PCL / posterolateral corner → valgus here; varus / posterior there

Grade	ACL (Lachman)	PCL (posterior translation)	PLC (varus opening / dial)	MCL (valgus at 30°)
1	3–5 mm	1–5 mm	≈5 mm / ≈5°	Up to 5 mm
2	5–10 mm	5–10 mm	6–10 mm / 6–10°	6–10 mm
3	> 10 mm	> 10 mm	> 10 mm / > 10°	> 10 mm

- Figure: possible **Stener-like lesion** of MCL
  - MCL injury shown
- MCL tear dragging meniscus → **meniscal extrusion**
- Tear off **medial femoral epicondyle** → MRI confirms

Management Ask AI

MCL

- **Majority conservative**
  - Broad ligament
  - **Extracapsular**
  - Very good blood supply
  - → heals easily → **orthosis** 4–6 weeks + protected weight bearing
- Operative
  - **Repair** → anchors or internal brace
  - **Reconstruction** → semitendinosus or gracilis
- Figure: MCL reconstruction → epicondyle to below articular surface
  - **Posterior oblique ligament** → adductor tubercle to posterior tibia
- Fixation → **anchors** or staples
- Figure: with **ACL reconstruction**
  - Internal brace → anchor + synthetic suture repair

High Yield

- Superficial MCL → main valgus stabilizer at all angles
- MCL majority conservative → broad, extracapsular, good blood supply
- Orthosis 4–6 weeks + protected weight bearing
- Valgus at 0° → cruciate + posteromedial capsule + MCL torn

Questions and Answers Ask AI

PCL as a Dynamic Stabilizer Ask AI

- Why is PCL a **dynamic stabilizer** with quadriceps?
  - **Force couple** → both hold tibia in sagittal plane
  - Position + **wrap** around back of knee → more freedom than usual ligament
  - → dynamic stabilizer in flexion/extension; bundles move more than ACL's

Measuring Laxity Ask AI

- Millimetres measured clinically? → no
  - Millimetres from **stress view X-rays**
  - Varus or valgus stress → measure **joint space opening**
  - By hand → opening felt; 2 mm vs 4 mm indistinguishable

Sulcus Terminalis Ask AI

ACL

- **Sulcus terminalis** → chronic ACL sign
  - ACL-deficient knee → excess stress / load → **deep sulcus terminalis**

Bracing and Newer Tests Ask AI

- Bracing in MCL injury? → yes; most conservative → braced
  - Isolated or with associated injury → usually conservative
  - Exception → **frank grade 3** (joint line > 10 mm) → collaterals with cruciates
- Newer ACL tests, e.g. Lelli test?
  - Pivot shift + Lachman done correctly → PCL excluded → more **reliable**, better recognised

Rehabilitation Precautions Ask AI

- **PCL** → posterior sag problem
  - Brace pushing tibia forward; do not rush; follow protocol
- **ACL** → range of motion; hamstring **donor site**

Pediatric ACL Follow-up Ask AI

Pediatric

- Age 10–12 + **transphyseal** → parents told of possible growth disturbance
  - New literature → no big difference in physeal arrest vs physeal-sparing
- Review **every 6 months** even if well → malalignment, growth arrest
- **Healing slower** → no rigid fixation
  - Buttons (suspensory fixation) → screws, strongest fixation, not used in child
- **Return to sport** → not before 9 months

PCL Healing Ask AI

- **Spontaneous PCL healing** → recorded cases
  - Broad, close to capsule → good **blood supply**
  - May heal if avulsion or proximal injury
  - **Mid-substance** tear → no spontaneous healing

Valgus Stress at 0° Ask AI

- Opens at 0° → **cruciate** + posteromedial capsule + MCL torn

ACL

Isometric vs Low Tension Ask AI

- Femoral ACL point → **isometry** and low tension are different concepts
  - Isometric → graft length constant through **range of motion**
  - Low tension → ACL fibers statistically under **lowest tension**
  - Avoid high early graft tension → prevents **failure**

Lachman and Pivot Shift Technique Ask AI

- **Lachman in obese patients**
  - Hold limb well; patient **relaxed**; large hands help
  - Pivot harder than Lachman
  - Pivot reproduces **giving way** → patient resists
  - Best pivot shift → under **anaesthesia**
  - Lachman → relaxed muscles; patient not watching
- Pivot shift → which structure moves?
  - **Iliotibial band** → flexor to extensor → reduces anteriorly subluxed tibia
  - MCL must be **intact** → counter-stress
  - Iliotibial band → shifts tibia; **pivot** of the test

Intermediate PCL Grade Ask AI

- No intermediate grade → tibia back 5 mm and holds, or goes **all the way**
- Some **stiff** knees → chronic adherent PCL, or partial ACL tear

Internal Derangement of the Knee — Part 2

(Meniscal and Articular Cartilage Injuries)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 222) “6 to 10 mm in radius” → “6, 8, 10 mm diameters”

Queried - kept as spoken, but looks wrong

Q1 (p. 211) As spoken; transitional (middle) zone is usually thickest (40-60%); deep zone about 30%

Q2 (p. 211) As spoken; water content is highest in the superficial zone and lowest in the deep zone

Q3 (p. 214) As spoken; Segond is a lateral tibial rim avulsion; the LFC defect is the lateral femoral notch sign

Q4 (p. 230) As spoken; three 6 mm plugs cover about 0.85 cm^2, not a 2 cm^2 defect

Articular Cartilage: Basic Science Ask AI

The Knee as a Room Ask AI

- Four pillars → **ACL**, PCL, medial collateral, lateral collateral
  - Covered last lecture with cruciate and collateral injuries
- Floor / carpet → **menisci**
  - Whole lecture dedicated; numerous types
  - Management carries great debate
- Floor and ceiling → **articular cartilage**
- Commonly confused entities
  - **OCD** (osteochondritis dissecans)
  - Osteonecrosis of the condyles
  - **SONK** → spontaneous osteonecrosis of the knee
  - Differentiating signs covered for each

Five Types of Cartilage Ask AI

- **Hyaline** → articular cartilage
- **Fibroelastic** → menisci
- **Fibrocartilage** → tendon and ligament attachments
  - Where they approach and grip bone
- **Elastic** → trachea
- **Physeal** → growth plate
- Today → hyaline; next lecture → fibroelastic (menisci)

Zones of Articular Cartilage Ask AI

- Top → bottom
  - Superficial zone → transitional zone → deep zone → tidemark → subchondral bone → cancellous zone
- Layer picture underpins **classification**, management and prognosis of cartilage injury

Superficial Zone Ask AI

- **Type II collagen** → main collagen of articular cartilage
- Fibres **parallel to surface** → sliding / gliding of joint surface
- Non-parallel fibres → friction and resistance to motion
- **Highest collagen**, lowest proteoglycan concentration

Transitional and Deep Zones Ask AI

- Transitional zone → fibres **haphazard** / randomly arranged
- Deep zone
  - **Thickest layer** of articular cartilage *[Q1 · As spoken; transitional (middle) zone is usually thickest (40-60%); deep zone about 30%]*
  - Type II fibres **perpendicular to surface** → run to and cross tidemark
  - **Highest proteoglycan** concentration
    - Proteoglycans hydrophilic → attract water
    - → **largest water content** *[Q2 · As spoken; water content is highest in the superficial zone and lowest in the deep zone]*

Zone	Collagen fibres	Key feature
Superficial	Type II, parallel to surface	Highest collagen, lowest proteoglycan
Transitional	Haphazard / random	—
Deep	Type II, perpendicular; cross tidemark	Thickest; highest proteoglycan and water

Tidemark and Subchondral Bone Ask AI

- **Tidemark** → separates superficial uncalcified true cartilage from deeper calcified cartilage
  - Cartilage separation injuries → whole piece separates at tidemark
  - Mobile region linking uncalcified and calcified layers
  - Present only in joints → these injuries occur at joint level
- Five layers (four main layers)
- **Subchondral bone** → directly beneath
  - Injury reaching it → all cartilage layers breached
- **Cancellous bone** → beneath subchondral bone

Function, Composition and Healing Ask AI

Functions Ask AI

- Reduce **friction**
- **Load distribution**
  - Shared with menisci
  - Intact cartilage + intact meniscus → loads even and equal over whole surface
- **Stress shielding** of solid matrix components
  - Via high water content + collagen–proteoglycan organisation → structure not disrupted

Composition Ask AI

- **Cells** + extracellular matrix
- **Chondrocytes** → lie in lacunae
  - Synthesise and maintain matrix → secrete collagen and proteoglycans
- **Matrix** → water, type II collagen, proteoglycans
- **Type II collagen**
  - 10–20% of total cartilage mass
  - Framework → **tensile strength** (resists pulling)
  - Framework housing the proteoglycans
- **Proteoglycans**
  - Attract water → **compressive** behaviour
  - ↓ Proteoglycans → impaired compression → ↓ water retention
  - → cartilage **ageing**
- **Water** → third matrix component
  - ↓ With normal ageing
  - ↑ In **osteoarthritis** (counterintuitive)
    - → ↑ permeability, loss of proteoglycan retention
    - → ↓ strength, ↓ elasticity
    - → cartilage damage → worn joint → replacement
- Basic science → unlikely essay; likely **MCQ** in master's / doctorate

Exam Pearl

- Water ↓ in ageing but ↑ in osteoarthritis
- Deep zone → highest proteoglycan and water
- Superficial zone → highest collagen, parallel fibres
- Type II collagen → tensile strength; proteoglycans → compression

Poor Healing Potential Ask AI

- Five reasons
  - **Avascular** → no blood vessels
  - **Aneural** → no nerve endings
  - **Alymphatic** → no lymphatic drainage
  - **Hypocellular** → few cells relative to role
  - **Nourishment** external → synovial fluid or subchondral / cancellous bone
- → healing very difficult

Healing by Depth Ask AI

- **Tidemark** → dividing point between two healing patterns
- Superficial laceration → takes three cartilage layers, spares tidemark
- Deep laceration → healing by **fibrocartilage**
- Fibrocartilage cannot function as **hyaline** cartilage

Feature	Superficial laceration	Deep laceration
Layers	Superficial, transitional, deep	Through tidemark
Chondrocytes	Retain ability to proliferate	—
Healing	Some, imperfect (avascular)	Fibrocartilage
Tissue	Hyaline	Stiff; altered stiffness, elasticity, wear properties
Outcome	Not a problem	High risk → arthritis

- Superficial vs deep → always relative to **tidemark**

Classification of Cartilage Injury Ask AI

- Two international classifications
  - **Outerbridge** → older
  - **ICRS** → International Cartilage Repair Society
- Initially thought different → side by side, minimal difference

Grade	Outerbridge	ICRS
0	Normal (arthroscopic)	Normal
1	Softening and swelling	Nearly normal; superficial lacerations
2	Superficial fissures	Lesion <50% of cartilage depth
3	Deep fissures	Lesion >50% of cartilage depth
4	Exposed subchondral bone	Severely abnormal; reaching subchondral bone

- ICRS grade 1 → simple longitudinal **cracks**; not deep, not entering substance, small area
- Outerbridge grade 2 → close to ICRS grade 1
- Common end point of grade 4 → **exposed subchondral bone**

ICRS on the Drawing Ask AI

ICRS

- Normal → all layers intact; subchondral and cancellous bone beneath
- Grade 1 → **nearly normal**; superficial fissures / lacerations
- Grade 2 → <50% of thickness
  - Superficial + transitional layers only
  - Deep layer = thickest → remainder <50%
- Grade 3 → **severely abnormal**; >50%, enters deep layer
  - Subgrades → blisters / bulge over subchondral bone
  - Subchondral bone not yet reached (despite drawing)
- Grade 4 → severely abnormal; enters **subchondral bone**
  - Subtypes A and B → reach cancellous bone beneath

Outerbridge on the Drawing Ask AI

Outerbridge

- Grade 0 → normal cartilage
- Grade 1 → **softening**
  - Arthroscopic probe sinks in; cartilage apparently intact
- Grade 2 → **superficial fissures**
- Grade 3 → **deep fissures**
- Grade 4 → subchondral bone reached; all cartilage layers lost

Cartilage Defects: Location and Aetiology Ask AI

Location: ACL vs OCD Ask AI

- Defect site → helps **differentiate** ACL injury from OCD

Feature	ACL-associated defect	OCD
Femur	Anterior lateral femoral condyle	Posterior medial femoral condyle
Tibia	Posterolateral tibial plateau	—
Side	Lateral	Medial

- ACL defect → where knee twists; at **Segond fracture** site (MRI and ACL lectures) *[Q3 · As spoken; Segond is a lateral tibial rim avulsion; the LFC defect is the lateral femoral notch sign]*
- All ACL-related cartilage injuries → **lateral** side
- Posterolateral plateau fracture → **bony pivot** injury / fracture
  - Same site where pivoting of examination test occurs
- Anterior + lateral condyle in ACL ↔ **posterior** + medial condyle in OCD

Aetiology Ask AI

- Three pathoaetiologies
  - **Acute trauma** → e.g. American football player; collateral + cartilage injured together
  - **Chronic repetitive trauma** / chronic overload
    - Cause → **malalignment** (varus or valgus knee)
    - Abnormal mechanical axis → continuous cartilage wear → total knee
  - **Idiopathic** → hormonal, vascular or traumatic
    - OCD → no definite cause known

Clinical Evaluation and Imaging Ask AI

History Ask AI

- **Trauma** history → usually absent unless associated cruciate / collateral injury
- Some patients **asymptomatic** → esp. OCD
  - OCD often **bilateral** → image other knee
  - → silent / stable OCD in contralateral knee
- Or **localised pain**, ± trauma
- Typical OCD → **teenager** 13–14 years
  - Pain with every training session
  - Mild swelling after training
- **Stable OCD** → effusion + incomplete range of motion
- Detached / loose fragment → **locking**, catching, instability

Examination Ask AI

- Look for
  - **Ligamentous laxity**
  - Malalignment
  - **Compartment overload** → esp. post-meniscectomy
- Post-meniscectomy patient → pain on stairs, tashahhud position, cross-legged sitting
  - → **compartment overload**: meniscus (equal load distribution) lost
- Malalignment / cartilage / OA patient → examine **standing**
  - Not on couch alone
  - Exposure → mid-thigh (or pelvis) down to heels
  - → reveals **varus** or valgus

Imaging Ask AI

- Radiology → lesion site and appearance
  - Arthritis present? bony defects?
- Alignment → **long-leg film**
- Views → AP, lateral, **Merchant** / skyline
- **CT** → quantify bone loss
- **MRI** → most important; **stable vs unstable** lesion
  - Fluid signal beneath cartilage + separation → **unstable**, poor prognosis

Decision-Making Factors Ask AI

- Athlete's questions → operation? return to sport?
- Two groups → patient factors, lesion factors

Patient Factors Ask AI

- **Age** → younger / skeletally immature → better prognosis
  - Stable lesion → stop sport + protective **brace**
  - Or **casting** 6 or 8 weeks → normal healing potential
- **BMI** → <25 vs >40–45 → completely different management
- **Symptoms** → exertion-related?
  - Giving way? relation to walking and standing?
- **Occupation** → high demand (professional athlete) vs
  - Recreational sport → daily walking, jogging
- **Meniscectomy** history
- **Alignment** → varus, valgus or normal
- **Ligament** injury → ACL, PCL, LCL, MCL
  - Any pillar lost → imbalance / instability
- Summary → age, weight, **skeletal maturity**, work, symptoms, pathology

Defect Factors Ask AI

- **Location** → femur, tibia, patella
  - **Kissing ulcer** → opposing femur + tibia
  - Each site → own problems in decision-making
- **Number** → single vs multiple
  - Single OCD ≠ 55-year-old woman with 3–4 MRI defects
  - Age also differs
- **Size** → papers differ
  - Small → <2 cm^2
  - Large → >2.5 cm or 3 cm^2
- **Depth** → tidemark spared vs **subchondral bone** reached
  - Depth of penetration → key
- **Containment**
  - Contained → **vertical perpendicular walls** around bone graft, osteochondral allograft, mosaicplasty
  - Uncontained → lost cartilage edge → empty space beside defect
- **Opposing surface** → often neglected at arthroscopy
  - Femoral condyle lesion → inspect tibia beneath
  - Kissing ulcer → very poor **prognosis**
- 55-year-old woman after microfracture, debridement or lavage
  - Counsel → likely start of road to **total knee**
  - No false hope of perfect knee → returns dissatisfied

Pitfall

- Femoral lesion seen → tibia not inspected → missed kissing ulcer
- Over-promising after lavage in older patient → dissatisfied patient
- Alignment examined on couch → patient must stand

Treatment Categories and Algorithm Ask AI

Three Categories plus Primary Repair Ask AI

- **Palliative** → arthroscopic debridement and lavage ("two small incisions", knee / joint washout)
  - Specific indications only
  - Worn knee, TKA candidate refusing TKA → arthroscopy promising cure = deceiving patient
  - Indicated → multiple **loose bodies**, debris, synovitis, severe pain unfit for surgery (comorbidity)
  - Counsel → not **definitive** management
- **Reparative** → **marrow stimulation**: awl, K-wire or drill bit
- **Restorative** → fill defect
  - **Mosaicplasty** / osteochondral grafting
  - **ACI** or newer **MACI** → cell-based management
- **Primary repair** → fix stable OCD with Herbert or bioabsorbable screw; specific indications

Category	Procedures
Palliative	Arthroscopic debridement, lavage
Reparative	Marrow stimulation (awl, K-wire, drill bit)
Restorative	Osteochondral grafting / mosaicplasty; ACI, MACI
Primary repair	Herbert or bioabsorbable screw fixation

Algorithm: Femoral Condyles Ask AI

Femoral

- Exclude first
  - **Malalignment** → HTO? distal femoral osteotomy? varus or valgus? patellar level (with PFJ)
  - Prior **meniscectomy**
    - Abroad → meniscal allograft transplantation or synthetic meniscal scaffolds
    - Not available locally
    - Very expensive → ~100–120 thousand Egyptian pounds
  - Missed **ACL** / PCL tear → mosaicplasty or ACI
    - Knee gives way again → operation ruined
- Then → **lesion size**

Lesion size	Low demand	High demand
<2 cm² (2–3)	Marrow stimulation or OATS	Marrow stimulation or OATS
>2–3 cm²	Marrow stimulation, OATS, osteochondral allograft or ACI	Osteochondral allograft or ACI
Failure (second line)	ACI or osteochondral graft, if not done	ACI or osteochondral graft, if not done

- **OATS** → osteochondral autograft
- Borderline → 2 (some papers 2–3); >2 → **large lesion**
- Large lesion, high demand → marrow stimulation fails; **OATS** insufficient
  - Trochlear donor sites → 3–4 plugs at most
  - 4 cm² (2 × 2) lesion → mosaicplasty insufficient

Algorithm: Patellofemoral Joint Ask AI

PFJ

- Confirm **patellofemoral alignment** first
  - No lateral patellar subluxation
  - No malalignment
  - No internal tibial torsion
  - No other PFJ problems
- Patellar surface → articulates with **trochlea**
  - Inclination irregular → high in one place, low in another
  - → mosaicplasty, OATS, allograft not feasible
- **ACI** / MACI → usual choice, low or high demand
  - Patellar thickness and cartilage shape completely different
- Large lesion, low demand (elderly) → light **drilling** or ACI
- High demand (jumper, scaffold worker, athlete) → **ACI** or osteochondral allograft
  - Counsel → patellar prognosis usually poor
- Second line → ACI or osteochondral allograft
  - **ACI** superior for PFJ

Management by Location Ask AI

Location	Management
Femoral condyle, very large defect	Osteochondral allograft
Patellofemoral	ACI (best)
Tibia, small / kissing ulcer	Awl drilling, if not full thickness
Tibia, large	Retrograde osteochondral autograft

- Femoral condyle allograft → **not available** locally; bone bank → femoral heads only
  - Worn LFC → distal femoral / LFC allograft, fashioned, fixed with screws or plate
  - → **near-anatomical** reconstruction
- PFJ → ACI best; lesions usually small
  - Concave / convex surface → **structural grafts** (autograft, allograft) very difficult
- **Tibia** → awkward, very difficult site
  - Small lesion or kissing ulcer → **awl** drilling, if not full thickness
  - Large → osteochondral autograft, **retrograde**
    - Like tibial plateau elevation → window + pusher from below
    - **K-wire** marks defect → cannulated reamer → defect piece removed completely
    - Prepared plug inserted from below upwards; not from side
  - Tibial management → most demanding

Associated Conditions Ask AI

- Key factors → location, number, size, depth, **associated conditions**
- **Malalignment** → concomitant or staged osteotomy
  - Osteotomy first → cartilage procedure later; or both together
  - Options → HTO, distal femoral osteotomy, **tibial tuberosity anteromedialization** (PF malalignment)
- **Ligament insufficiency** → concomitant or staged reconstruction
  - Restores **stability** → prevents re-injury
- **Meniscal deficiency** → meniscal allograft transplantation; not available locally
- PF lesion → ACI or marrow stimulation + **realignment** (tibial tuberosity anteromedialization)

Primary Repair Ask AI

Indications Ask AI

- **Acute osteochondral fragment**
  - Patellar dislocation striking LFC → fragment off LFC or patella
  - Seen twice → entire patellar articular surface sheared off from anterior part
- Osteochondral fragment separated but **in situ**
- **Stable** lesion not yet separated → fix early

Principles Ask AI

- Steps
  - Unstable / open → open defect, **debride fibrous base**
    - Fibrous tissue / fibrocartilage → blocks healing
  - **Microfracture** to subchondral bone → enhance bleeding / blood supply
  - Subchondral cyst → **bone grafting**, packed well
  - Finally → **rigid fixation** under compression (Herbert screw)
- **Herbert screw** → headless → fully buried
  - Even flush → usually some protrusion → removal operation
  - Counsel → arthroscopic **removal** at 2–3 months
- **Bioabsorbable screws** (France, Germany) → degrade like ACL screws
  - Non-metal, headless → usually no problems
  - Briefly in Egypt; agency closed; costly
- Disadvantages
  - **Healing** very poor >18 years; better in skeletally immature
  - Compression screw → removal needed in younger patients

Clinical Pearl

- Debride fibrous base → microfracture → graft cyst → compress
- Herbert screw → headless, yet often protrudes
- Plan removal at 2–3 months
- Poor healing beyond 18 years

Debridement and Marrow Stimulation Ask AI

Arthroscopic Debridement / Lavage Ask AI

Lavage

- Removes loose cartilage flakes and chondral fragments
  - → improves **mechanical symptoms** (locking, giving way)
- Merits
  - Simple arthroscopic procedure
  - **Rapid rehabilitation**
- Other procedures → prolonged non-weight bearing
- Problem → exposed **subchondral bone** remains; no expected progressive improvement

Procedure	Weight bearing after surgery
Lavage	Immediate
Primary repair	None before 8 weeks
Mosaicplasty, ACI	Non-weight bearing 2–3 months (average)
OATS	None for ~2–2.5 months
Marrow stimulation	Partial, crutches 6–8 weeks (~6 weeks non-weight bearing)

Marrow Stimulation (Chondroplasty) Ask AI

Marrow

- Aim → bring **marrow elements**
  - → stimulate reparative tissue (fibrocartilage) → close defect
- Technique
  - Arthroscopic **curette** → stable edges
    - Cartilage separated from subchondral bone → remove; no value
    - Target → stable cartilage, good **vertical walls**
  - Awl, drill bit or K-wire → **multiple drills** 3–4 mm apart
  - Example area → ~15–20 holes
- Names → **microfracture**, abrasion chondroplasty, osteochondral drilling
- Lesions <2 cm
  - No cost
  - **Single stage**, arthroscopic
  - Good results → acute **contained** lesions <2 cm
- Limitations
  - Very poor results → **larger areas**
  - Bone defect not addressed
  - **Partial weight bearing** on crutches 6–8 weeks (vs immediate after lavage)
- Mechanism → microfracture → **fibrin clot** → fibrocartilage fill

Osteochondral Grafting Ask AI

Mosaicplasty Technique Ask AI

OATS

- **Sizer** → measure defect
- **Preparation knife** → cut recipient
- **Harvester** → plug flush with surface, arthroscopic
- Donor site → defect
- 3–4 plugs → maximum area 2 × 2 cm
- Knife diameters → 6, 8, 10 mm *[C1 · transcript: "6 to 10 mm in radius"]*

Procedure Selection by Lesion Ask AI

- Small, high demand → **mosaicplasty** (osteochondral autograft)
- Larger → **allograft**
- Shallow → **ACI**; stable floor for periosteal patch
- Deep → osteochondral allograft or mosaicplasty
- **Marrow stimulation** requirements
  - Vertical holes + contained lesion → protection against shear and compression
  - Holes **perpendicular** to bone surface, 3 mm apart
  - ~6 weeks non-weight bearing

OATS Principles Ask AI

OATS

- Transfer one or more **osteochondral plugs** → cartilage + underlying bone
- Harvest depth → 10–12 mm
  - **Graduated** knife → depth known
- Recipient bed ~2 mm shorter than plug → **impaction**

  - Harvest 12 mm → bed 10 mm
- Harvest site → **non-weight-bearing** periphery of femoral trochlea, or intercondylar notch
  - All-arthroscopic → **notch** best
  - Mini-arthrotomy (lateral / medial) → trochlear harvest; complete arthroscopically
- Limitations
  - Limited **donor tissue**
  - **Donor-site morbidity** → frequent complaints
  - Initial healing failure → no incorporation → **loose body**
  - No weight bearing ~2–2.5 months
- Harvesting knife → graft taken and filled
- Very good results → **small lesions**, high-demand persons
- **Mega-OATS** (mega osteochondral autograft) → lesions >4 cm
  - Donor → **posterior femoral condyle**; difficult site
  - e.g. 25-year-old, large cartilage loss → arthroplasty not considered
  - Big, somewhat risky; allograft easier abroad; unavailable locally
- Approach → arthrotomy, or arthroscopic harvest from intercondylar notch edges
  - Notch harvest → avoid **cruciate attachments**
- Harvester → **perpendicular** to surface
- Depth → 12–15 mm
  - Recipient socket ~2 mm shorter → impaction
- **Chondrocytes** sensitive to impaction → insert gently, no banging
  - Plastic cushion in set → gentle final tamping → flush

High Yield
- Donor → trochlear periphery or intercondylar notch
- 3–4 plugs → maximum 2 × 2 cm
- Plug ~2 mm longer than socket → impaction
- Harvester perpendicular; tamp gently
- Mega-OATS → posterior femoral condyle, lesions >4 cm

Osteochondral Allograft Ask AI

Allograft

- Mini-arthrotomy → prepare graft
  - Imported femoral condyle / trochlear **allograft** → mark, prepare bed, insert (sometimes by finger)
- Indicated → **large lesions** >2.5 cm with substantial bone loss
- Second line → after failed marrow stimulation, or prior lavage
- Limitations
  - **Availability**; high cost ~2000 dollars per piece (USA)
  - **Disease transmission** → HIV, hepatitis

Autologous Chondrocyte Implantation Ask AI

Procedure Ask AI

ACI

- Arthroscopic cartilage **harvest** → ~200 mg
- Laboratory → chondrocytes proliferate, **expanded** in culture
- Defect → cleaned thoroughly
- **Periosteal patch** from proximal tibia → sutured over defect
- **Fibrin glue** → airtight
- Cultured cells → injected beneath patch

ACI vs MACI Ask AI

Feature	ACI	MACI
Name	Autologous chondrocyte implantation	Matrix-induced autologous chondrocyte implantation
Cover	Periosteal patch, proximal (medial) tibia	Matrix / scaffold carrying cells
Fixation	Sutured + fibrin glue; cells injected beneath	Single patch stuck in; fibrin glue at edges
Stages / cost	Two stages	Two stages; expensive

- **MACI** → cells on matrix; one patch; fibrin glue at edges
- ACI → thesis of Amgad Halim (USA); results largely very good

Indications Ask AI

ACI

- **Unipolar** lesions → not kissing / bipolar
- Chondral or osteochondral defects 2–10 cm; preferably <6 cm
- **Second line** of treatment

Technique Ask AI

ACI

- First stage → arthroscopy
  - Cleaning, preparation, sizing; **cartilage biopsy**
- Second stage → lab expansion → defect preparation → patch from proximal medial tibia → cell injection
- **No tourniquet**
  - Floor must not bleed excessively unnoticed
  - Under tourniquet → patch, fibrin glue, 6-0 Vicryl, cells
  - Release → rich **subchondral** blood → lifts patch → all lost
  - → avoid tourniquet
- Patch → **medial proximal tibia**
- Patella case → **everted**; patch sutured 6-0 Vicryl, sealed with fibrin glue; chondrocytes beneath
- **Patella** → among most successful sites for ACI
- MACI → **scaffold** / matrix base instead of periosteal patch
  - Expensive, two-stage procedure

Pitfall

- ACI under tourniquet → release → subchondral bleeding lifts patch
- Kissing / bipolar lesion → not an ACI indication

Osteochondritis Dissecans Ask AI

- Cartilage lesion subtypes
  - **OCD**
  - Avascular necrosis of the condyles
  - **SONK** syndrome

Definition and Location Ask AI

OCD

- **Pathological lesion** → articular cartilage + subchondral bone
  - Variable clinical patterns
- Knee site → **posterolateral** aspect of medial femoral condyle
  - Also → capitellum, talus
- Contrast → ACL defect: anterior LFC or posterolateral tibial plateau

Types Ask AI

OCD

- **Juvenile** → open physes; majority of patients
  - 10–15 years; knee pain, effusion, sport, daily fatigue
- **Adult** → very poor prognosis
- Site → medial femoral condyle, posterior aspect

Aetiology Ask AI

OCD

- Unknown; theories
  - **Idiopathic**
  - **Genetic** predisposition → siblings, father, first-degree relatives
  - Trauma
  - Adult form → **vascular insult** to subchondral bone
    - → avascular necrosis → overlying cartilage separates
  - **Endocrine** / glandular problems
  - Abnormal **ossification**

Pathology Ask AI

OCD

- **Softening** → cartilage appears intact
- → separation without detachment
- → partial **detachment**
- → complete osteochondral separation + **loose body**

Prognosis Ask AI

OCD

- **Juvenile** form much better
  - Open physes with 3–4 years before closure → better
- **Lateral femoral condyle** and patella lesions → poor
- X-ray **sclerosis** at subchondral base → poor
- MRI **synovial fluid** beneath lesion → poor
- **Adult** form → worst

Classification Ask AI

OCD

Type	Clanton	MRI	Stability (Clanton)
1	Depressed osteochondral fracture	Cartilage thickening, low signal	Stable
2	Attached by osseous bridge; edge separation begun	Cartilage breached; signal beneath fragment (darkening → fibrous tissue)	Stable
3	Detached, not displaced	Cartilage eroded; fluid between fragment and underlying bone	Unstable
4	Detached, displaced / loose body	Loose fragment	Unstable

- **Clanton** → 1–2 stable; 3–4 unstable
- Types 1–2 → better prognosis
  - **Fixation** (Herbert / biodegradable screw) → much better results
- Types 3–4 → other cartilage procedures
- **MRI** classification → similar to Clanton

Clinical Features Ask AI

OCD

- **Pain**
- Mechanical symptoms → **locking**, if separated (grade 3 or 4)
- **Recurrent effusion**
- Examination → swelling, tenderness, stiffness in range of motion
- **Wilson test**
  - Tibia internally rotated, knee extending 90° → 30° → pain
  - External rotation of tibia → complete relief
- Radiology → as discussed

Management Ask AI

OCD

- **Non-operative** → restrict weight bearing, brace; some cast
  - Stable lesion, 8–9 or 10 years, open physes
  - Asymptomatic adult lesions
- **Operative** → symptoms + failed conservative treatment
  - **Diagnostic arthroscopy** → healing chance, stable vs unstable
    - Now shown on MRI → scope unnecessary
  - Lesion enlarging on follow-up or failed non-operative management
  - **Subchondral drilling** → debatable
    - Antegrade → from inside knee, stable lesions
    - **Retrograde** → avoids diseased cartilage; C-arm or arthroscopic view, ACL C-guide
      - Multiple drillings → ↑ subchondral **vascularity**, cartilage spared
  - **Fixation** → Herbert, cannulated or bioabsorbable screw
  - **Mosaicplasty** / chondral resurfacing → large lesions, or small in high demand
  - Periosteal patch → ACI / MACI
  - **Knee arthroplasty** → ~60 years, arthritic knee; usually vascular insult
- End of OCD

High Yield

- OCD site → posterolateral medial femoral condyle
- Juvenile → good prognosis; adult → worst
- Clanton 1–2 stable → fixation; 3–4 unstable
- Wilson test → IR pain 90° → 30°; ER relieves
- Poor prognosis → LFC, patella, sclerosis, fluid beneath lesion

Osteonecrosis of the Knee Ask AI

- Two forms → **secondary** osteonecrosis; spontaneous (SONK)

Secondary Osteonecrosis Ask AI

Secondary

- Usually **women**; one compartment
- 80% **bilateral**
- Risk factors
  - Alcohol
  - **Marrow-replacing** diseases → Gaucher's
  - Sickle cell anaemia
  - Hypercoagulable states
  - **Steroids**
  - Systemic lupus, rheumatoid disease
  - Viral infection
  - Trauma
- Pathophysiology → **subchondral insufficiency fracture**
  - Due to osteoporosis, weight, risk factors
  - Linked to **meniscal root tears** → more common in women
- Radiology
  - AP + lateral → **wedge-shaped** lesion
  - Sagittal view → wedge-shaped
  - Wedge shape → distinctive sign of secondary osteonecrosis
  - **MRI** → most diagnostic; T1 dark, T2 bright / bone marrow oedema
- **Ficat staging**
  - Stage 1 → normal X-ray
  - Stage 2 → condylar **sclerosis**
  - Stage 3 → **crescent sign**
  - Stage 4 → subchondral **collapse**

Differential Diagnosis Ask AI

Secondary

- Discriminators → age, sex, site, joints involved, trauma

Entity	Distinguishing features
OCD	Posterolateral medial femoral condyle; boy 15–20 years
Osteonecrosis	Woman 40s–50s, obese, steroids, Gaucher's, viral infection
Transient osteoporosis	Men 40–50; uncommon in knee; multiple joints (hip + knee marrow oedema); self-limiting
Occult fracture / bone bruise	Trauma; CT → clear fracture line

Management Ask AI

Secondary

- Same lines as cartilage → **diagnostic arthroscopy**
- Treated like hip AVN → **core decompression**
  - ± osteochondral allograft → avascular piece removed
  - → replaced with similar graft
- Very large area → **arthroplasty**

Spontaneous Osteonecrosis of the Knee (SONK) Ask AI

- **Middle-aged**; women > men
- One joint → unlike **transient migratory osteoporosis**
- Site → condylar **epiphysis**
- Two distinguishing signs
  - **Sudden severe pain** → screaming; fine the day before
  - **Crescent** lesion → not wedge-shaped
    - Unlike ordinary osteonecrosis

Feature	Secondary osteonecrosis	SONK
Patient	Usually women	Middle-aged; women > men
Extent	One compartment; 80% bilateral	One joint
Lesion shape	Wedge	Crescent
Onset	Risk factors	Sudden severe pain

- **Differential diagnosis** → OCD, transient osteoporosis, bone bruise, idiopathic osteonecrosis
- Management
  - Often **arthroplasty** → severe pain; conservative treatment often fails
  - Young + malalignment → **HTO** to relieve pain

Questions and Answers Ask AI

- **Q: Explanation for ACL defect location?**
  - Knee rotates on this site during injury
  - Same site where ligament tears at attachment
- **Q: Cast duration in OCD?**
  - **Two months** → reduce joint motion
  - Young → high healing potential
  - Older patient → do not cast; usually will not heal
- **Q: Hyaluronic injection in cartilage defects?**
  - Patients awaiting **total knee**
  - Keep ≥3 months from surgery
  - <3 months before TKA → ↑ arthroplasty risk
  - Pain relief only; **literature not supporting**
- **Q: Mosaicplasty in large OCD lesions?**
  - Grafting knife → 3–4 sites by defect size

  - 2 cm² defect → 3 plugs of 6 mm
    - 6 + 6 + 6 → ~18 mm coverage *[Q4 · As spoken; three 6 mm plugs cover about 0.85 cm², not a 2 cm² defect]*
  - Donor → **edge of trochlea**, from above
  - Plug → cartilage + tidemark + subchondral bone
  - Bone included → better **integration** and healing
- **Q: Localisation for retrograde drilling in OCD?**
  - Arthroscopic preferred over C-arm alone
  - **ACL C-guide** → lateral or medial condylar aspect, per lesion
  - Drill like outside-in femoral tunnel
  - C-arm → articular cartilage not breached
  - Arthroscope → drill not inside knee
- **Q: PRP injection?**
  - PRP / stem cells (adipose, iliac crest) → proven in **grade 1–2 osteoarthritis**
  - Not proven in cartilage defects → surgical procedures above preferred
  - Supervised, published thesis → patients did not improve much
- **Q: 50-year-old, ACL tear 6 months, lesion >2 cm?**
  - ACL reconstruction limit → **40 years**
  - Sport exceptions → up to 49 years
    - Tennis player keen to return → arthroscopy first
    - Good knee → ACL reconstruction; degenerative / ulcers → no
  - Confirm **alignment**
  - High-grade **pivot shift** → justifies ACL
  - At 50 → preferably not
- **Q: Microfracture in exposed cancellous bone?**
  - No → aim is marrow inflow to cover area
  - Large lesion (half condyle) → **marrow stimulation** likely insufficient
  - 2 cm lesion → well covered
  - Whole condylar articular surface → **total knee** later
- **Q: Spacing between microfracture holes?**
  - ~3–4 mm
  - Imported **awls** → graduated tips → depth known
  - Trick → stop irrigation, open **suction**
    - Pressure relieved → yellow marrow / bleeding emerges
    - → confirms **bleeding healthy bone** reached
  - No need for 5–6 cm K-wire depth → just pass subchondral **sclerotic** area
- **Q: Acute meniscal root tear repair at same sitting?**
  - Yes → do it
  - 6–8 weeks protection needed anyway
- **Q: Bone scan in OCD?**
  - Reported for fragment **viability**
  - **MRI** most accurate
  - Bone scan → viability unclear
- MRI findings question → deferred to MRI lecture (Dr. Ahmed)
- **Q: Mechanism of Wilson test?**
  - Tibial **internal rotation** → impinges on lesion

  - External rotation → relieves area
- **Q: MRI and arthroscopy for every osteoarthritis patient?**
  - Check **BMI** first
  - OA grade on **standing X-ray**
  - Average weight, grade 1–2 → MRI
  - 60–70 years, 120 kg , grade 4 → MRI adds nothing
    - Even with a meniscal tear

Patellofemoral Instability — Part 1

(Anatomy and Biomechanics of the Patellofemoral Joint)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 242) "between adductor tubercle and gluteal tuberosity" → "between adductor tubercle and medial epicondyle"

Patellofemoral Joint — Overview Ask AI

Lecture Series Plan Ask AI

- Topic: **patellofemoral joint** and patellofemoral instability
- Large subject → tedious unless properly understood
  - Tests known; concepts such as **patellar tilt** poorly understood
- Anatomy poorly grasped: **MPFL** known, whole concept lacking
- One session too heavy → **three sessions**
  - Session 1: **anatomy and biomechanics**
  - Session 2: **radiology** — very important
    - One of the most important diagnostic tools
  - Session 2 also: **pathology**
    - Trochlear dysplasia, patella alta, increased femoral anteversion
    - Touched briefly today; main discussion next time
  - Session 3: application in **clinical examination**
    - Identify cause of **patellofemoral pain** or instability in the patient
    - Management / treatment

Prerequisites for a Healthy Joint Ask AI

- Three prerequisites for healthy, properly functioning joint:
  - Patella **located in the trochlear groove** — no instability symptoms
    - Typical patient: girls / young women, 13–14 years
    - No complaint: knee giving way, kneecap slipping out → back, dislocating → relocating
    - Examination: no lateral exit → return of patella
    - Most important prerequisite
  - Properly functioning **extensor mechanism**
    - Healthy quadriceps, healthy **VMO**, whole mechanism functioning
  - No **patellofemoral pain**
- Seated patella: neither laterally displaced nor proximally displaced (**patella alta**)

Why the Joint Is Different Ask AI

- Unlike easily understood joints
  - e.g. **shoulder joint**
- One of the most heavily published topics; dissection-based anatomy changing constantly
  - **European society** book: *Patellofemoral Joint Instability* — new, released May
  - Much of this series taken from it
- Many concepts changing
  - **TT–TG distance** no longer main parameter in calculations
  - → lecture planned as three sessions
- Reasons it is different:
  - Articulation between a **sesamoid bone** (patella) and a groove
    - Not a true joint in usual sense — not tibiofemoral; part of knee joint
    - Sesamoid bone moving on **trochlear groove**
  - **Multiple contact points**

        - Posterior patella and trochlear contacts change with every degree of flexion
        - 0° ≠ 30° ≠ full **squat position**
        - → different **joint reaction forces** at each flexion degree
    - **Numerous soft tissues** attached to patella
        - Superior: **quadriceps**
        - Both sides: **retinacula**
        - Inferior: patellotibial and **patellomeniscal (meniscotibial)** structures
        - Each: attachment, anatomy, role in biomechanics and symptoms
    - **Joint reaction force** varies with:
        - Degree of knee flexion
        - Degree of quadriceps contraction
        - Foot planted on ground or not
        - **Flat foot** → ↑ Q angle → ↑ lateral quadriceps pull on patella
            - → ↑ JRF → ↑ tendency to subluxation / dislocation
    - **Thickest articular cartilage** in body — 7 mm
        - Function: dissipate JRF from quadriceps contraction
        - Hypothetical 1–2 mm cartilage + quadriceps contracting with every movement:
            - Rising from chair, descending stairs, entering car, prostration in prayer → very high JRF
            - → **arthritis** → severe pain and disability
- **Patellofemoral arthritis**: large part of knee OA in our country
    - One of three compartments; very common here
    - Varus / valgus (tibiofemoral) arthritis less common than patellofemoral OA
- Summary — joint different because:
    - **Complex**
    - Multiple contact points
    - Numerous soft tissues attached to patella
    - JRF varies with flexion, muscle contraction, **lower limb alignment**, foot on ground
    - Articular cartilage thickness

Osseous Anatomy Ask AI

Patella Ask AI

Patella

- **Plane joint**: posterior patellar surface ↔ trochlear groove of distal anterior femur
- Posterior patella ≠ vertical ridge + medial facet + lateral facet
  - **Median ridge** present
  - **Lateral facet** → superior, middle, inferior (numbered 1, 2, 3)
  - **Medial facet** → superior, middle, inferior
  - **Odd facet** — medial side
    - Only contact point when sitting on floor — **tashahhud** position
    - Last contact point of the joint
- Each facet ×3 + odd facet → **seven facets**
  - MCQ form: number of facets — two, five or seven?
  - Instinctive answer two → wrong; **seven**
- Upper two-thirds → **articular surface**
- Inferior part (apex) → attachment
- Shape: **inverted triangle**
  - Base above → muscle attachment
  - Apex below → **patellar tendon** attachment

Exam Pearl

- Posterior patella: **seven facets** — not two
- Lateral and medial facets → superior, middle, inferior each
- **Odd facet** (medial) — sole contact in full flexion (tashahhud)
- Upper two-thirds articular; apex → **patellar tendon**

Trochlea Ask AI

Trochlea

- **Trochlea**: two femoral condyles + groove between
  - Lateral facet and medial facet of distal femur
  - Articulates with posterior patellar surface
- **Lateral groove**: ridge / groove within articular cartilage
  - Visible on careful look at a skeletal bone
  - Separates **lateral femoral (trochlear) facet** from lateral femoral condyle
- Lateral facet **projects more laterally** → strut against lateral patellar subluxation
  - Joint stability depends only on patella engaged in groove
  - Lateral buttress absent → lateral subluxation very easy
- **Trochlear dysplasia**: absent sulcus
  - Lateral facet sometimes not higher, longer, more proximally projected than medial
  - → **instability** — with pathology of patellofemoral instability
- Lateral line (red): lateral facet | lateral femoral condyle
- Medial line (light blue): **medial facet** | medial femoral condyle
  - Often overlooked in anatomy; very important
- Lateral trochlear facet ↔ **lateral patellar facet**
- Medial trochlear facet ↔ **medial patellar facet**
- **Lateral femoral condyle proper** — with cartilage line / groove
- **Medial femoral condyle** — with cartilage line / groove
- Posterior: **intercondylar notch** — ACL and PCL attachments
- **Femoral sulcus** (femoral condyle)

Trochlear Groove and Sulcus Angle Ask AI

- **Trochlear groove (sulcus)**: concave surface; lateral + medial facets
  - Medial + lateral sulcus lines → intersection = **deepest point** → groove depth
  - Lateral facet larger, projects more proximally → **buttress** against lateral subluxation
- **Sulcus angle**: average 138°–139° (140° in some books)
  - Wider → loss of depth: 145°, 150°, 160° → **flatter** sulcus
  - = form of **trochlear dysplasia** — Dejour, four types: A, B, C, D (later)
  - ↓ depth → ↑ liability to patellar subluxation / dislocation

> **High Yield**
> - Normal **sulcus angle**: 138°–139° (140° in some books)
> - Wider (145°–160°) → flatter sulcus → **trochlear dysplasia**
> - Dejour classification: **four types** — A, B, C, D
> - Lateral facet projects more proximally → **lateral buttress**

Soft-Tissue Stabilizers — Overview Ask AI

Why Soft-Tissue Stabilizers Are Needed Ask AI

- **Shallow joint**: trochlear groove + patella only
  - Without surrounding soft tissue → subluxation / dislocation
- Not perfectly congruent — high **incongruency**
  - → soft-tissue stabilizers hold patella in **trochlear sulcus** of distal anterior femur

Medial and Lateral Static Stabilizers Ask AI

- Figure: upper part = lateral side; lower part = medial side
- Medial–lateral **balance** lost → **patellar maltracking** → patellofemoral instability
- **Lateral side** — four structures:
  - Lateral patellofemoral ligament
  - Joint capsule
  - Iliotibial band
  - Lateral retinaculum
- **Medial side** — four structures; two most important:
  - **MPFL** — familiar reconstruction done reflexly for every instability case
  - **Medial patellomeniscal ligament**
- Even easier operation: **lateral release**

Clinical Case — Repeated Lateral Releases Ask AI

- Patient: three prior **lateral releases**; patella still dislocating
  - Alignment and **rotational profile** never assessed
  - **Distal femoral derotation osteotomy** → patella stabilized; MPFL untouched
  - History: **malaligned limb**; three operations
  - Third look: no graft, no screws — lateral release only
- Lateral release: easiest operation — **not for everyone**
  - Focus on diagnosis and on type of operation
  - Otherwise 3–4 operations, all ending badly

Pitfall

- Repeated **lateral release** without assessing alignment → persistent dislocation
- Check **rotational profile** before operating
- Malaligned limb → **derotation osteotomy** stabilized patella; MPFL untouched

Medial Patellar Restraints Ask AI

Anatomical Illustration Ask AI

- Patella, medial side, **MPFL**, medial quadriceps tendon–femoral ligament
- **MQTFL**: part of MPFL attaching to quadriceps tendon
- Below: **medial patellomeniscal ligament**, medial meniscus, medial femoral condyle, epicondyle
- **Vastus medialis**
  - **VMO**: part where colour, fibre shape and fibre direction change
- **Quadriceps tendon**
- Anterior knee viewed from medial: patella, medial border, lateral border
  - **Apex** and base of patella
- MPFL, MQTFL, **medial patellofemoral capsule**, VMO
- **Sagittal view**:
  - **Adductor magnus** → adductor tubercle
  - MPFL
  - Medial patellomeniscal ligament
  - Patella, medial side
- Understanding this anatomy → operations and management easier

Proximal and Distal Groups Ask AI

- Medial side first — very important for **stability**
- Four restraints, two groups:
  - **Proximal**: MPFL, MQTFL
  - **Distal**: medial patellotibial ligament (MPTL), medial patellomeniscal ligament (MPML)
- **MPTL**: attaches below articular surface
- **MPML**: arises from medial meniscus
- Proximal group: most important — MPFL and MQTFL
- **Medial patellofemoral restraint**: resists lateral displacement 0°–30°
  - → before patella engages trochlear sulcus
- Clearer figure: **proximal** — MPFL, MQTFL; distal — MPTL, MPML

Medial Attachment Sites Ask AI

- Lateral-view figure — attachment sites of all medial structures
- Medial side confusing — many attachments

Structure	Femoral attachment
Adductor magnus tendon	Adductor tubercle
Superficial MCL	Medial epicondyle
Medial head of gastrocnemius	Gastrocnemius tubercle
MPFL	Midway between adductor tubercle and medial epicondyle *[C1 · transcript: "between adductor tubercle and gluteal tuberosity"]*

- MPFL point debated; not exactly the **Schöttle point**
  - Still standard → all studies rely on it
  - Detailed in radiology session
- Some structures divided into parts; some belong to distal femur

Medial Patellofemoral Ligament (MPFL) Ask AI

Anatomy and Function Ask AI

MPFL

- Most important restraint → basis of main operation for **patellar instability**
- Runs along descending vastus medialis; **not attached to adductor tubercle**
- **Origin**: between medial epicondyle and adductor tubercle
- **Insertion**: upper half — superomedial aspect of patella
  - Expansion into quadriceps tendon → **MQTFL**
    - Part of MPFL; separate name for quadriceps tendon attachment
- **Fan-shaped**: wide base at patella ≈ 2.5 cm
  - Narrows to femoral insertion ≈ 12 mm
- **Length** ≈ 5.5 cm
- **Function**: 50%–60% of patellar stability against lateral translation at 0°–30°
  - Early flexion → before patella enters and stabilizes in **trochlear sulcus**

Femoral Attachment Ask AI

- **Medial epicondyle** → MCL
- **Adductor tubercle** → adductor magnus
- **Gastrocnemius tubercle** → medial head of gastrocnemius
- MPFL: **midway** between epicondyle and adductor tubercle
  - Opinions differ
  - **Most common** site in current studies

High Yield

- **MPFL** femoral attachment: midway, medial epicondyle ↔ adductor tubercle
- Insertion: superomedial patella + **MQTFL** expansion
- Fan-shaped: 2.5 cm at patella → 12 mm at femur; length 5.5 cm
- 50%–60% of restraint to **lateral translation** at 0°–30°

Quadriceps Tendon Turndown Flap Ask AI

- Upper MPFL → **quadriceps tendon**; other part → directly into patella
- → **turndown flap**:
  - Strip from central or medial quadriceps tendon
  - Flipped, turned medially → sutured to patella → fixed at femoral condyle
  - Avoids **semitendinosus / gracilis** harvest (ACL graft site)
  - Much easier — nearly same incision
  - Medial strip secured with **two sutures**
    - → no tearing, no downward slipping
  - Fixed in distal femur at **Schöttle point**
    - Defining two lines → radiology session

Distal Medial Restraints Ask AI

- Proximal two done; **distal two**: MPML, MPTL

Medial Patellotibial Ligament (MPTL) Ask AI

MPTL

- **Layer**: second layer of medial aspect
- **Patellar attachment**: inferomedial border — below MPFL attachment
  - ≈ 3–4 mm from distal (inferior) pole
- **Tibial attachment**: medial tibial tubercle — ≈ 1.5 cm (13 mm) below articular surface
- **Length** ≈ 40–50 mm (about 4–5 cm)
- **Width** ≈ 7–9 mm
- **Function**: secondary stabilizer against lateral translation
  - Extension → ≈ 26% of medial stability
  - 90° flexion → ≈ 50%
- Recap: **inferior pole** → medial tibial tubercle

Cadaveric Dissection Ask AI

- **Cadaveric dissection** — structures truly exist, not theory
- **Gerdy's tubercle** ◎ iliotibial band
- **Medial tubercle** ◎ MPTL
- **Medial meniscus** ◎ MPML
- Reflected medially:
  - MPML → **medial meniscus**
  - MPTL → **medial tibial tubercle**
  - MPFL → **medial aspect of femur**
- Further figure: vastus medialis, MPML, patella, **MCL**, MPTL, MPFL
- **Pes anserinus** — semitendinosus and gracilis tendons
- Medial collateral ligament

Medial Patellomeniscal Ligament (MPML) Ask AI

MPML

- **Third layer** — deeper than MPTL (second layer)
- **Patellar insertion**: inferomedial patella, just proximal to MPTL
- **Meniscal origin**: anterior horn, or anterior horn–body junction, medial meniscus
- **Function**:
  - Prevents lateral subluxation in **terminal extension**
  - Resists **lateral patellar tilt** at 90° flexion

Feature	MPTL	MPML
Layer	Second	Third (deeper)
Patellar attachment	Inferomedial border, 3–4 mm from inferior pole	Inferomedial aspect, just proximal to MPTL
Distal attachment	Medial tibial tubercle, 1.5 cm below articular surface	Anterior horn / horn–body junction, medial meniscus
Role	Secondary restraint to lateral translation: 26% extension, 50% at 90°	Lateral subluxation in terminal extension; lateral tilt at 90°

Relative Contribution to Stability Ask AI

- **Cadaveric study**: contribution of each medial structure to stability
- **MPFL**: highest score at all flexion degrees, in all cadaveric studies
  - → most important → basis of reconstruction

Lateral Patellar Restraints Ask AI

- **Iliotibial band** → Gerdy's tubercle
- **Vastus lateralis**
- Lateral counterparts: patellofemoral, patellotibial, patellomeniscal ligaments
- Rule: replace "medial" with "lateral"
  - Exception: **no MQTFL equivalent** (MQTFL = part of MPFL)

Medial	Lateral counterpart
MPFL	Lateral patellofemoral ligament
MQTFL	None
MPTL	Lateral patellotibial ligament
MPML	Lateral patellomeniscal ligament
—	Iliotibial band → Gerdy's tubercle
—	Vastus lateralis

- Figure: **lateral patellofemoral ligament**
  - Lateral patellotibial ligament
  - Iliotibial band — descends to attach
  - Vastus lateralis
- **Three structures** with lateral counterparts
- Medial and lateral: importance and commonest attachments covered

Static and Dynamic Stabilizers Ask AI

- **Static**: passive structures — no muscle contraction
- **Dynamic**: structures involving muscle contraction
- **Static stabilizers**:
  - **MPFL** — most important
  - Medial patellotibial ligament
  - Lateral retinaculum
  - **Trochlear geometry** — shape of trochlea
  - Shape of patella
  - **Patellar height** — alta or baja
  - **Limb alignment** — Q angle, femoral anteversion
- **Dynamic stabilizers**: muscles surrounding joint
  - **Quadriceps**, biceps femoris, pes anserinus
  - Contractions → correct **patellar alignment**
  - **VMO**: mechanical advantage → medial stabilizing force against lateral subluxation

Functions of the Patella Ask AI

- Classic memorized teaching: patella has **three functions**
- **Pulley**: bony pulley within quadriceps tendon
  - Changes direction of quadriceps extension force (pull vector) throughout **ROM**
- **Bony shield**: buffer / barrier protecting trochlear groove from impact
  - Trochlear cartilage very weak → easily worn → **patellofemoral arthrosis**
- **Friction prevention**: between quadriceps tendon and distal femur
  - Patella not always applied against femur; quadriceps tendon attaches to it
- Not merely a **sesamoid bone**

Static Alignment and the Q Angle Ask AI

Determinants of Static Alignment Ask AI

- **Static alignment** (no muscle contraction) depends on:
  - **Femoral sulcus depth**
  - Height and projection of **lateral femoral condyle**
    - Projects more proximally → buffer against lateral subluxation
    - Deficient → **patellar subluxation**
  - Shape and height of **patella**

Patella in Full Extension Ask AI

- **Full extension**: patella not in trochlear sulcus — midway between condyles
  - Superior to trochlea
  - **Least contact** patella–trochlea → most mobile
- → examination: supine, knee extended — not in flexion
  - **Patellar glide**, apprehension test, patellar tilt — all in full extension
  - Only position with patella above trochlea, not in sulcus → mobile → proper examination
  - Least **contact area**

Clinical Pearl

- Full extension: patella above trochlea → **least contact**, most mobile
- Glide, **apprehension test**, tilt → supine, knee fully extended

Q Angle Ask AI

- Very important point
- Lines: **ASIS** → patella centre; tibial tuberosity → patella centre
  - Intersection = Q angle
- Represents **lateral vector** of quadriceps tendon pull
  - Not just a line → **force line** of quadriceps tendon
  - Line of pull = **lateralizing force** on patella
- **Normal**: ≈ 10° males; ≈ 15° females
- ↑ Q angle → excessive lateral pull → **subluxation** / dislocation

Patellar Height Ask AI

- Middle image normal
- Higher → **patella alta**
- Lower → **patella baja** (patella infera)
- Assessed on **lateral-view X-ray**, knee 20°–30° flexion — radiology session
  - **Apex** (lower pole) normally at or slightly above joint line
- Several indices; most famous: **Insall–Salvati**

Insall–Salvati Index Ask AI

- Weakness: depends on **patellar tendon length**
  - **Osgood–Schlatter**: tendon end on tibial tuberosity unclear on X-ray
  - **Inferior pole** broad: attachment here, higher, or wrapping anteriorly onto periosteum?
- Criticisms: Osgood–Schlatter; inferior pole attachment
- Still in use
- **Ratio**: patellar tendon length ÷ patellar length (superior → inferior pole), at 30° flexion

Insall–Salvati	Interpretation
1	Normal (red ÷ yellow distance)
< 0.8	Patella infera (baja) — tendon shorter than patellar height
> 1.2	Patella alta — superior position

- Cut-offs: index below 0.8 → baja; above 1.2 → **alta**
- **Patella alta** → no early entry into trochlear sulcus with flexion
  - → sits high; flexion begins before sulcus entry → ↑ lateral subluxation liability

Caton–Deschamps Index Ask AI

- Distance **D**: lower end of patellar articular surface → upper tibial surface
- Not dependent on whole patellar tendon
- Detailed next session

High Yield

- **Insall–Salvati**: tendon length ÷ patellar length at 30° flexion
- Normal 1; < 0.8 → **patella baja**; > 1.2 → patella alta
- Weakness: Osgood–Schlatter, broad **inferior pole**
- Caton–Deschamps: articular surface → tibia; **not** whole tendon
- Q angle: ≈ 10° males, ≈ 15° **females**

Patellar Movements and Tracking Ask AI

Patellar Tilt Ask AI

- Normal: **balanced** — equal pull medial, lateral, superior, inferior
  - Centred → no rotation, superior–inferior or medial–lateral
- Lateral tilt = **lateral patellar compression syndrome**
  - Key term → recurs under patellofemoral pain and instability
  - Very strong lateral pull → medial side elevated → medial border higher than lateral

Patellar Rotation Ask AI

- Depends on distal attachment → **tibial tuberosity**
- Rotation requires tuberosity movement → **external (lateral) tibial torsion**
- Patella no longer centred above
  - Lateral pull from below → laterally displaced
  - Medial pull → medially displaced
- Governed by **patellar tendon** attachment to tibial tuberosity

Determinants of Patellar Tracking Ask AI

- **Dynamic kinematics** = patellar tracking — very important
- Patellar movement depends on:
  - Magnitude of **quadriceps contraction**
  - Soft-tissue **extensibility** and medial–lateral balance
  - **Geometry** of patella and trochlear groove
- **Patellar maltracking** = abnormal tracking — next session

Four Patellar Movements Ask AI

- **Superior–inferior glide**
  - Superior glide = **patellar extension** ↔ tibiofemoral extension
  - Inferior glide = **patellar flexion** ↔ tibiofemoral flexion
- **Medial–lateral glide**
- **Medial / lateral tilt**
  - Lateral tilt → lateral patellar compression syndrome
- **Medial / lateral rotation**
  - Patellar tendon on tibial tuberosity → tuberosity moves laterally in **external tibial torsion**
- Tilt, glide and rotation must be visualized
  - Otherwise **clinical examination** and tests useless

Tracking in the Open Chain Ask AI

- **Open chain**: patella follows tibia — patellar tendon on tibial tuberosity
  - Knee flexion → **inferior glide**
  - Knee extension → superior glide
  - → superior / inferior glide governed by **tibiofemoral motion**

Contact Mechanics and Joint Reaction Force Ask AI

Contact Points Through Flexion Ask AI

- Figure: **distal anterior femur** and patella
  - Patella marks: 120° (superior aspect), 90°, 60°, 30°
  - Trochlear sulcus marks: 30°, 60°, 90°, 120°
- **Inverse relationship**

Flexion	Patellar contact	Trochlear contact
30°	Most inferior posterior patella — engages first	Most superior sulcus
60°	60° point, lower posterior patella	60° point
90°	Near highest part	Lowest part

- Extreme 120°–130° → **highest two areas** of posterior patella
- ↑ flexion → contact points change with every degree
- ↑ flexion → **upwards on patella**, downwards on trochlear sulcus — opposite
- Contact area: **proximal on patella**; inferior and posterior in sulcus
- ↑ flexion → ↑ contact area; ↑ **JRF** from quadriceps contraction
  - → 7 mm cartilage dissipates stresses
  - 1–2 mm cartilage, as other joints → early wear → **patellofemoral arthritis**

Contact Area and Joint Stress Ask AI

- ↑ ROM / flexion → ↑ **contact area**
- ↑ contact area → ↓ joint stress
  - **Joint stress = JRF ÷ contact area**
- Patellar contact: proximal → more proximal
- Trochlear contact: more inferior and posterior
- 135°: **odd facet** — last; sole contact in full flexion (tashahhud)
  - Contacts lateral aspect of **medial femoral condyle**

Patellofemoral Joint Reaction Force Ask AI

PFJRF

- **PFJRF**: resultant compression force on joint
  - From muscle (quadriceps) contraction + flexion angle
- **Actual stress** = JRF ÷ surface area
  - ↑ knee flexion → ↑ contact area → ↓ actual stress
- ↑ patellofemoral joint stress → **patellofemoral pain**

Causes of Increased Joint Stress Ask AI

- **Abnormal anatomy** = trochlear dysplasia (next session)
- Abnormal **patellar tracking** = loss of medial–lateral balance
- **Kinetic chain**
  - Disturbed muscle contraction / balance around joint
- **Overuse**: many stairs, prolonged standing, walking on rough ground
- Abnormal **alignment (malalignment)**:
  - Genu valgum
  - Increased femoral anteversion
  - Pes planus (flatfoot)
  - Position → knee into valgus
- Very high JRF → **patellofemoral pain syndrome**

Must-Know

- **Joint stress** = JRF ÷ contact area
- ↑ flexion → ↑ contact area → ↓ stress
- Full flexion: **odd facet** only → small area → ↑ stress
- Highest-stress range: 30°–90°
- Causes: trochlear dysplasia, **malalignment**, abnormal tracking, kinetic chain, overuse

Questions and Answers Ask AI

- Q: stress maximal in flexion despite wide contact?
  - End flexion: contact only at **odd facet**
  - Bears all surrounding muscle force → ↑ JRF
  - Maximum flexion: odd facet alone = **small area**
  - Highest-stress range: 30°–90°
- Q: tilt vs glide vs rotation?

Movement	Definition
Glide	Superior–inferior and medial–lateral movement in same horizontal plane; no lifting
Tilt	Medial or lateral border lifted off joint — plane changed; flipping
Rotation	Inferior pole turns medially or laterally

- Tilt: grasp both borders → lift medial (or lateral) border
  - Lifting / flipping → **tilt**, not glide
- **Glide test** (examination): move patella by quadrants
  - Same horizontal plane — no lifting
- Rotation: depends on **patellar tendon** attachment to tibial tuberosity
  - Tibial torsion → tuberosity rotates laterally (externally) → inferior pole follows
- Next session: following week

Patellofemoral Instability — Part 2

(Epidemiology, Clinical Assessment and Radiology)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 262) “trochlear depth more than 3 mm = dysplasia” → “trochlear depth less than 3 mm = dysplasia”

Queried - kept as spoken, but looks wrong

Q1 (p. 257) As spoken; lecture later gives 145° for the same sulcus angle cut-off (standard >145°).

Clinical Presentation and Epidemiology Ask AI

- Part 3 of the series → **management**
- Typical patient → **girl**, 13–19 years; males also present
- Onset
  - **Trauma** → typically while playing on the field
  - Or **voluntary** dislocation by the patient herself
- Epidemiology
  - **Young active females** → especially second decade of life
  - **Prevalence** → 6 to 77 cases per 100,000
- Prevalence not high, but now seen much more often
  - Expanded **children's sport** → specialized schools, private lessons → more injuries
- Past → low awareness among coaches and team doctors
  - **Patellar dislocation** reduced on the field by coach or team doctor
  - → never properly diagnosed → patient continues to suffer consequences

Stabilizers of the Patellofemoral Joint Ask AI

- Same slide as previous lecture → **static** and dynamic stabilizers

Static Stabilizers Ask AI

- **MPFL** (medial patellofemoral ligament) → most important
- **Medial patellotibial ligament** → also important
- **MQTFL** (medial quadriceps tendon femoral ligament) → in addition to MPFL
- **Patellofemoral ligament** → second important stabilizer
- **Lateral retinaculum**
- **Trochlear geometry**
  - Deep trochlea? → sulcus deep enough for patella in flexion and extension
  - **Lateral trochlear facet** → normal size and position?
  - = **osseous defense** (osteochondral buttress) against lateral subluxation or dislocation
- **Patellar shape and height** → patella alta, patella baja or normal
- **Q-angle** → especially with genu valgum or forefoot pronation
  - Forefoot pronation / **flat foot** → very high Q-angle, like genu valgum
  - Increased **femoral anteversion** → disturbed tibial rotation → tibial external rotation
  - **External tibial torsion** → same effect
    - Tibial tuberosity moves laterally → ↑ Q-angle → ↑ dislocation tendency

Dynamic Stabilizers Ask AI

- **Extensor mechanism** with its components
- **VMO** (vastus medialis obliquus) → most important and strongest for the PFJ

Morphological Abnormalities and Predisposing Factors Ask AI

Three Major Morphological Abnormalities Ask AI

- **Trochlear dysplasia**
  - Dysplastic = did not develop properly
  - Deep groove replaced by **flattened** surface
  - Sometimes the opposite of deep → **convex**
  - Details → classification of patellar instability (below)
- **Patella alta** → patella higher than normal
  - Fails to **engage** trochlear sulcus at the expected 20° or 30° flexion
  - Stays above groove even at 20°, 30°, 40° flexion → lateral dislocation tendency
  - Biomechanics → previous lecture
- **Lateralized tibial tuberosity**
  - External tibial torsion
  - **Genu recurvatum**
  - Increased femoral anteversion

Other Predisposing Factors Ask AI

- **Ligamentous laxity**
  - Ehlers–Danlos syndrome, **Marfan syndrome**
  - Abnormal ligaments, connective tissue, **collagen** → highly prone to stretching
  - Lax MPFL and related structures → greater dislocation tendency
- **Abnormal muscle tone** → last factor
  - Muscle tone problems, or **severe contraction** of the quadriceps tendon
  - **Force vector** → centre of patella to anterior superior iliac spine
    - → very strong **lateral vector**
    - Severe or **eccentric contraction** of quadriceps → patella dislocates very easily

Summary of Factors Ask AI

- Three major factors
  - Abnormal **trochlear groove** (sulcus)
  - High patella → no early engagement at 20° or 30° flexion → lateral dislocation
  - **Tibial tuberosity** not in correct position
- Other factors
  - Increased **femoral anteversion**
  - Laxity → Ehlers–Danlos, Marfan syndrome
  - Abnormal muscle tone

High Yield

- Three major abnormalities → trochlear dysplasia, patella alta, lateralized tibial tuberosity
- Patella alta → no engagement at 20°–30° flexion → lateral dislocation
- Laxity → Ehlers–Danlos and Marfan syndrome

Mechanism of Dislocation Ask AI

- Most common mechanism → **foot planted** (as in picture)
- Femur → **internal rotation**; tibia → external rotation
- Foot fixed on ground, knee at 20–30° flexion → patella dislocates
- Mechanism on left of slide → **most common** mechanism of patellar dislocation
- Alternative → **direct trauma**
  - Fall on kneecap → impact on ground → lateral dislocation
- Another position → foot fixed on ground
  - Knee flexed; thigh internal rotation + tibia external rotation simultaneously
  - → tears **medial retinaculum**

Dejour Classification of Trochlear Dysplasia Ask AI

- **Dejour** → French; best-known author on trochlea and patellofemoral instability
- Trochlear dysplasia → **four types**
  - Type A → normal shape, but **shallow** trochlea
  - Type B → completely **flat**, or **convex trochlea**
  - Type C → **facet asymmetry**
    - Very high lateral facet
    - Hypoplastic medial facet
  - Type D → as type C + **vertical link** between facets → **cliff pattern**
- Slide also shows crossing sign, supratrochlear spur, double contour → see radiology

Types on Images Ask AI

Dejour

- Type A → trochlear groove present, but **shallow**
  - **Sulcus angle** beyond 141° → wider angle → shallower groove → ↑ instability *[Q1 · As spoken; lecture later gives 145° for the same sulcus angle cut-off (standard >145°).]*
- Type B → completely **flat** surface, no groove; some cases **convex**
- Type C → **lateral trochlear facet** (lateral femoral condyle) more convex
  - Medial facet → **hypoplastic**
  - → patella rests on lateral trochlear facet → ↑ dislocation tendency
- Type D → as type C; lateral facet more convex and more **prominent**
  - Medial facet → **cliff** (steep drop beside a mountain), not just hypoplastic
  - No trace that a medial facet ever existed
  - Sudden drop → patella about to fall off the mountain side
- Type C → smooth descent; type D → sudden drop off a **ledge**

Type	Trochlear morphology
A	Groove present but shallow; sulcus angle beyond 141°
B	Flat, or convex
C	High, convex lateral facet; hypoplastic medial facet
D	As C + vertical link → cliff; no evidence of medial facet

Plain X-Ray Work-Up Ask AI

- Patellar instability → initial views
  - **AP view**
  - **Lateral view**
  - **Axial views** → skyline, Merchant or sunrise, at different degrees
- Plain X-ray role → exclude **loose body** or fracture
  - Patella dislocates laterally out of trochlear groove
    - → breaks cartilage of patella (medial patellar facet)
    - → or fragment breaks off **lateral femoral condyle**
    - → fracture or loose bodies
  - Most common source → **medial patellar facet**

AP View Ask AI

- **Whole lower limb alignment** X-ray, standing view
- Genu varum or **genu valgum**? → genu valgum matters here
- **Femoral anteversion** problem?
- **External tibial torsion**?
- Tibial tuberosity → **lateralized** or not?

True Lateral View Ask AI

- Lateral view → most important view
- Demand a **dead true lateral** from the radiographer
  - Main view for diagnosing **trochlear dysplasia**
  - Reject oblique film; reject non-superimposed **two condyles**
- Diagnostic signs relied on
  - **Crossing sign**
  - **Double contour sign**
  - **Supratrochlear spur**
- Importance
  - Diagnosis of trochlear dysplasia
  - **Patellar height** → alta or baja
    - Many methods; most important → Insall–Salvati and Caton–Deschamps
- Methods → previous lecture

Method	Measurement
Insall–Salvati	Whole patella to inferior pole; inferior pole → tibial tuberosity
Caton–Deschamps	Patellar articular surface; highest tibial plateau point → lowest articular point

- **Caton–Deschamps** more accurate than Insall–Salvati
  - Patellar apex (lower pole) varies between individuals
  - **Osgood–Schlatter** or tuberosity fragmentation → highest tuberosity point unclear

> **Clinical Pearl**
> - Caton–Deschamps more accurate than Insall–Salvati
> - Patellar apex varies; Osgood–Schlatter obscures tuberosity landmark

Axial View Ask AI

Merchant

- Sunrise or **Merchant view** shows
  - **Lateral patellar tilt**
  - Lateral patellofemoral angle
  - **Congruence angle**
  - **Sulcus angle** (previous lecture)
- Views at different degrees of flexion
- Egypt → mostly **Merchant view** → radiographers used to it
  - Other views difficult, especially in patellar instability
  - Prone patient, **cassette** under knee, aimed beam → difficult
- Different degrees → assess **patellar tilt**
- Congruence angle and sulcus angle shown
- Congruence angle → line along lateral facet + line along medial facet → intersection

Rejecting an Inadequate Lateral Film Ask AI

- Radiographer's "lateral view" → often not one
- Patella and femur visible, condyles not superimposed → not a **true lateral**
- True lateral criteria
  - **Two femoral condyles** superimposed
  - Posterior cortex → single line
  - Anterior cortex of femur → single line
  - Patellofemoral space visible
- Inadequate views rejected → no help in interpreting the problem
  - Cannot show crossing sign, double contour or **supratrochlear prominence**

Pitfall

- Oblique or non-superimposed condyles → reject film
- Only a dead true lateral shows crossing sign, double contour, spur

Lateral Radiograph Signs of Trochlear Dysplasia Ask AI

Normal Lateral Radiograph Ask AI

- **Blumensaat line** (red) → continues as **sulcus line** → rises upward
- **Yellow line** → lateral femoral condyle (lateral trochlear facet)
  - → meets **anterior cortex** of femur → continues upward
- Normal knee → trochlear groove and lateral condyle lines do not **cross**
- Trochlear dysplasia → **crossing sign**
  - Trochlear groove line crosses, touches or joins lateral femoral condyle line
- Second image → not a perfect dead lateral, but reasonable
  - Lateral femoral condyle outline → meets **anterior femoral cortex**
  - Trochlear groove → neither touches nor crosses lateral condyle
- Third image → no crossing → **normal trochlear groove** on lateral view

Dejour's Three Points Ask AI

- Dejour's drawing of a **normal trochlea**
  - **Green point** → lateral (trochlear) facet → highest
  - **Blue point** → floor → deepest point of trochlear groove
  - **Red point** → medial facet → second-highest
- Axial concepts applied to lateral view
  - Blue point (deepest groove) → does not cross green line (lateral facet)
  - Red line (medial facet) → hard to bring out on X-ray
  - Lateral facet → **most anterior**; deepest groove point → furthest behind
- These images → normal deep trochlea

Crossing Sign Ask AI

- Trochlear dysplasia → sulcus shallow, or **flat**
- Flat trochlea → green, red and blue points on the same line
  - No lateral facet prominence; sulcus not deep
- Lateral view → three points **superimposed** → best shown by crossing sign
- Normal → red and green lines do not touch; blue and red do not touch
- Dysplasia → blue line (trochlear groove) **intersects** green and red lines
- Normal → trochlear sulcus point lies away from lateral femoral condyle point
- Crossing → sign of trochlear dysplasia (**flat trochlea**)

Types C and D: Double Contour Ask AI

- Lateral facet much higher; trochlea flat or **convex**
- **Medial femoral condyle** → hypoplastic, or absent → **cliff**
- Green (lateral facet) + blue (deepest sulcus) on same line → crossing sign
- Trochlear line + lateral femoral condyle line cross → crossing sign
- Additional line more deep / **more posterior** than the crossing point
  - = hypoplastic (medial) femoral condyle → **double contour**
- First radiological sign of trochlear dysplasia → **crossing sign**
  - Trochlear groove line touches and crosses lateral and medial condyle lines
- Types C and D → red point (**medial trochlear facet**) recedes backward
  - → not superimposed → double contour

Crossing Sign: Definition Ask AI

Crossing

- True lateral X-ray → trochlear groove line crosses **anterior border** of femoral condyle
- **Predictor** of trochlear dysplasia
- → high probability of patellar dislocation or subluxation
- Image → Blumensaat, trochlear sulcus, lateral condyle lines → crossing begins

Double Contour Sign Ask AI

- Hypoplastic medial femoral condyle → line appears behind
  - Should run with the same lines → **recedes** backward → double contour
- Right radiograph → Blumensaat, trochlear sulcus and lateral condyle lines
  - Crossing sign → **flat trochlea**
  - Clearer receding line behind → **double contour sign**
  - **Bump** → as the cliff in type D
- Definition → sign of trochlear dysplasia on true lateral X-ray
  - **Double line** at anterior aspect of the condyles
  - Cause → hypoplastic medial femoral condyle
  - Hypoplastic medial facet → **posterior and distal** to crossing line
  - Sometimes somewhat more distal
- Examples → double contour in **type D** trochlear dysplasia

Supratrochlear Spur Ask AI

- **Proximal and lateral osseous protuberance**
- Indicates
  - Lateral femoral facet higher than normal
  - Trochlear groove below → flat
  - No **medial facet**

Dejour Classification: Radiological Correlation Ask AI

- Types A and B → shallow or flat trochlea → **crossing sign**
- Types C and D
  - Lateral femoral condyle **more prominent**
  - Trochlear groove flat or convex
  - Hypoplastic medial facet
  - → double contour + crossing sign
- **Cliff** → adds supratrochlear spur → crossing + double contour + spur
- Image → groove line intersects condyle line; second line behind → **double contour**

Review by Type Ask AI

Dejour

- Type A (type 1) → trochlear sulcus present, but **shallow**
  - Plain lateral X-ray → crossing sign
  - MRI → **sulcus angle** more than 145° → shallow trochlea
  - Lateral X-ray, MRI and explanation shown
- Type B → flat trochlea + supratrochlear spur + crossing sign
- Type C → femoral condyle (lateral facet) **more prominent**
- Type B slide → flat trochlea, supratrochlear spur, hypoplastic medial facet
  - **Lateral convexity** → bump or spur
- Type C → double contour + crossing sign
- Type D → double contour + supratrochlear spur + crossing sign → **cliff pattern**
- Cliff not often seen; **types C and B** more common
- Sulcus angle → about 145° or more than 145°
  - Cliff (drop) → appears as **supratrochlear spur** or supratrochlear bump
- Summary on plain X-ray and MRI

Type	Morphology	Crossing sign	Supratrochlear spur	Double contour
A	Shallow groove	Yes	—	—
B	Flat (or convex)	Yes	Yes	—
C	Prominent lateral facet; hypoplastic medial facet	Yes	—	Yes
D (cliff)	As C + cliff	Yes	Yes	Yes

Must-Know
- Crossing sign → all four Dejour types
- Supratrochlear spur → types B and D
- Double contour → types C and D
- Cliff pattern = type D; types C and B more common

MRI after Patellar Dislocation Ask AI

- **MPFL** starts to detach and tear
  - From **medial femoral epicondyle**
  - Or from medial border of patella
- Lateral X-ray → very clear crossing sign and double contour
- Sequence → femur internal rotation + tibia external rotation → lateral patellar dislocation

Role of MRI Ask AI

MRI

- **Loose bodies** → osteochondral bone fragments
  - Most common source → medial patellar facet
  - Others → lateral trochlear facet, lateral femoral condyle
- **MPFL tear site**
  - Majority → from medial femoral condyle
  - Sometimes → detached from **medial border** of patella
  - Medial epicondyle attachment → **multiple lamina** → MPFL disruption begun
- **Hemarthrosis** or effusion → certainly present
- **Bone marrow edema** or contusion of lateral femoral condyle → from the dislocation

Exam Pearl
- Commonest loose body source → medial patellar facet
- MPFL tear → majority from medial femoral condyle
- Bone marrow edema → lateral femoral condyle

MRI Measurements of Trochlear Dysplasia Ask AI

- **Lateral trochlear inclination**
  - Lateral trochlear facet vs **posterior condylar tangential line**
  - Normal more than 11°; less than 11° → trochlear dysplasia
  - Posterior condylar tangent transferred to facet line → intersection angle
- **Trochlear depth**
  - Trochlear floor → tangent joining most anterior points of medial and lateral facets
  - Less than 3 mm → trochlear dysplasia *[C1 · transcript: "trochlear depth more than 3 mm = dysplasia"]*
  - Tangent touches **two femoral condyles** anteriorly → distance beneath
- **Ventral trochlear prominence**
  - Anterior cortex of distal femur → most anterior cartilaginous point of groove
  - More than 8 mm → trochlear dysplasia
- **Medial condyle trochlear offset**

  - Medial condyle → tangent through trochlear groove
  - Tangent parallel to posterior femoral condyles
- **Facet asymmetry**
  - Ratio → medial to lateral trochlear facet length
  - Less than 40% → trochlear dysplasia

Measurement	Defined between	Dysplasia
Lateral trochlear inclination	Lateral facet and posterior condylar tangent	Less than 11° (normal more than 11°)
Trochlear depth	Floor and anterior facet tangent	Less than 3 mm
Ventral trochlear prominence	Anterior femoral cortex and most anterior groove cartilage	More than 8 mm
Medial condyle trochlear offset	Medial condyle and groove tangent parallel to posterior condyles	—
Facet asymmetry	Medial : lateral facet length	Less than 40%

CT: TT–TG and TT–PCL Distances Ask AI

- Before MRI measurements → **CT** most famous and widely used in trochlear dysplasia

TT–TG Distance Ask AI

TT–TG

- **Tibial tuberosity–trochlear groove distance**
- Two perpendicular lines relative to **posterior cortex** of femoral condyles
  - One → tibial tuberosity (tubercle); one → trochlear groove
- More than 20 mm → abnormal
- Measurement
  - Perpendicular → deepest trochlear groove point to posterior condylar line
  - Line transferred → projected to trochlear groove and tibial tuberosity
  - Two **CT cuts** → tibial tuberosity cut + trochlear groove depth cut
  - Images **superimposed** → distance between the two lines
  - More than 20 mm → sign of trochlear dysplasia
- Illustrative drawing
  - **Deepest point** of trochlear groove
  - Tangent to posterior femoral condyles
  - **Most prominent point** on tibial tuberosity
  - Perpendiculars → meet posterior condylar tangent
  - Distance between the two points = TT–TG; more than 20 mm → dysplasia
- Same steps repeated → images superimposed; more than 20 mm → abnormal

Limitations and TT–PCL Distance Ask AI

- TT–TG importance → declined greatly
  - Affected by **knee rotation**, knee flexion, trochlear dysplasia
  - → not a **reliable measurement**
- **TT–PCL distance** (tibial tuberosity–PCL) devised
  - **Constant** regardless of knee rotation, flexion, trochlear dysplasia
- No CT admission needed → measured on **MRI** (clearest modality for PCL)

	TT–TG	TT–PCL
Modality	CT (classically)	MRI
Abnormal	More than 20 mm	—
Knee rotation, flexion, dysplasia	Affected	Not affected

Next Lecture: Management Preview Ask AI

- Radiology section → finished
- Next lecture → **management** of patellofemoral instability
  - Syndromes vs **subjective patellar instability** → difference
  - Young patients aged 13 or 18 years vs 20 years and above
    - **Physis** → matters for MPFL fixation
    - Also for **trochlear sulcus deepening**, with its different parameters
    - And **anteromedialization** of tibial tuberosity → especially open physis
  - **Adolescents vs adults** → very important management considerations
- Large topic → divided over several sessions
- Management → to be completed next time

Patellofemoral Instability — Part 3

(Patellofemoral Pain and Management of Instability)

Prof Dr Amr Rashwan

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 281) “adductor longus tendon” → “adductor magnus tendon”

C2 (p. 281) “adductor longus attached at adductor tubercle” → “adductor magnus strip at adductor tubercle”

Queried - kept as spoken, but looks wrong

Q1 (p. 272) As spoken; normally the medial femoral condyle extends more distally, not the lateral.

Q2 (p. 272) As spoken; Dejour type C is convex lateral facet with hypoplastic medial facet, not hypoplastic LFC.

Q3 (p. 283) As spoken; AMZ shifts load medially, so medial facet arthritis is a contraindication, not indication.

Patellofemoral Pain Syndrome Ask AI

- Lecture: **Part 3** of patellar instability → rest of pathology + management
- Patellofemoral pain = "**black hole of orthopaedics**"
- Cause poorly understood → many theories
- Partly caused by **patellofemoral instability**
- Among the **most common musculoskeletal disorders** in routine practice
- Patient places hand over **anterior knee** → "this area hurts"

Synonyms Ask AI

PFPS

- **Anterior knee pain**
- **Chondromalacia**
- **Derangement of the knee** — cf. earlier lecture on internal derangement
- **Patellofemoral pain syndrome**

Theories Ask AI

PFPS

- **Chondromalacia** → softening of articular cartilage
- **Lateral patellar retinaculum** lesion
- **Peripatellar synovitis**
- Excessive lateral patellar tilt → **lateral patellar compression syndrome**
- **VMO dysplasia** → cannot counteract vastus lateralis
  - → patella shifts laterally → lateral patellar compression
- **Patellar malalignment**
  - One of the major causes of **patellar instability**
- Limited **contractile function** of muscle
- Exact theory → unknown

Source of Pain Ask AI

PFPS

- Candidate sources — nobody knows for sure:
  - Bone
  - **Articular cartilage**
  - Meniscus
  - Soft tissue
  - **Fat pad**
  - Capsule
  - Ligaments
- Candidate mechanisms:
  - **Malalignment**
  - Neurological
  - Patient **weight**
  - Altered knee **biomechanics**
    - Structural → **trochlear dysplasia**; cartilage damage with PF arthritis
    - Dynamic → **muscle imbalance**
- Most famous theory → **malalignment**

Contributing Factors Ask AI

PFPS

- **Malalignment** → all ↑ Q-angle → patellar instability
  - Lateral patellar tilt
  - ↑ **Femoral anteversion** with in-toeing
  - Genu valgum
  - Genu recurvatum
  - External tibial torsion
  - Pronated foot
- **Dysplasia**
  - Patella → **patella alta**
  - Trochlea → medial trochlear facet dysplasia / hypoplastic lateral trochlear facet
    - = **Trochlear dysplasia** (dysplastic trochlea)
- **Damage**
  - Patellofemoral arthritis
  - **Stress fractures**
  - Acute patellar dislocation → weakened **MPFL**
- **Dynamic** (muscular)
  - Excessive muscular **tightness**
  - Muscle **fatigue** — tires quickly
  - Imbalance → weak **VMO** vs vastus lateralis
  - **Overuse**

Synovial Plica Ask AI

Plica

- **Plica** → PF pain with normal MRI
  - Cartilage, menisci, cruciates, collaterals → all normal
- Exam → palpable **cord** over medial patella
- Genuinely painful at times → especially **women and young girls**
- Arthroscopic excision → most pain + **tenderness** resolve
- **Patellar tendinitis** (jumper's knee) → another cause of PFPS

Consequences of Recurrent Patellar Dislocation Ask AI

- Each dislocation →
  - **Articular cartilage** injury
  - **Osteochondral fracture** / separated osteochondral fragment
  - Reduced patient activity
- End result → **patellofemoral arthritis**

Causes of Patellar Instability Ask AI

- Management starts from the causes → final operative decision

Factors Increasing the Q-angle Ask AI

Q-angle

- **Femoral anteversion**
- **Genu valgum**
- **External tibial torsion**
- **Pronated foot**
- External tibial torsion + pronated foot → **lateralized tibial tuberosity** → ↑ Q-angle
- Genu valgum → **malalignment** → ↑ Q-angle
- ↑ **Femoral anteversion** →
  - Compensatory external tibial torsion / in-toeing
  - Femur rotated inwards → patella inwards
  - → ↑ Q-angle
- Clinical images
  - Genu valgum
  - **Recurvatum**
  - Pronated forefoot → **flatfoot**
  - **Ehlers–Danlos syndrome** → signs of hyperlaxity
  - ↑ Femoral anteversion
- External tibial torsion → **tibial tuberosity** rotated laterally
  - Seen from below sitting + in standing

Osseous Factors Ask AI

Anatomical

- Two groups → **general** vs **anatomical**
  - Anatomical → osseous (bone itself) / muscular
- **Trochlear dysplasia**
- **Patella alta**
  - High patella → no trochlear engagement before 40–50° flexion
  - → 0–50° range → **liable to instability**
  - Normal → engages within first 20° → PFJ **stabilization** begins
- Excessive lateral **patellar tilt**
  - Patella tight, displaced laterally → lateral patellar compression syndrome
- **Lateral femoral condyle hypoplasia**
  - Normal LFC → more proximal + more anterior → **wall / strut**
    - → Prevents lateral patellar dislocation
  - Hypoplasia → **dislocation** risk

Must-Know

- Normal patella engages trochlea within first 20° flexion
- **Patella alta** → no engagement before 40–50° → unstable 0–50°
- LFC → more proximal + anterior → strut against lateral dislocation

Summary of Causes Ask AI

Group	Causes
General	Ehlers–Danlos syndrome; miserable malalignment syndrome
Anatomical — osseous	Trochlear dysplasia; patella alta; excessive patellar tilt; LFC hypoplasia
Anatomical — muscular	Dysplastic VMO; lateral overpull by iliotibial band or vastus lateralis

- **Miserable triad** → femoral anteversion + genu valgum + external tibial torsion
  - → Tibial tuberosity moves laterally
- Keep in mind → examination + **pre-operative planning**

Imaging Examples Ask AI

- **Sulcus angle** ≈ 160° (upper-left image)
- **Skyline view** → patella displaced laterally, beyond trochlear sulcus
- **Hypoplastic LFC** → medial condyle extends lower → abnormal
  - Normal → lateral condyle extends more distally, standing *[Q1 · As spoken; normally the medial femoral condyle extends more distally, not the lateral.]*
  - Lower-right image → also hypoplastic LFC
- **Trochlear dysplasia** types on images:

Type	Pattern
A	Shallow
B	Convex or flat
C	Hypoplastic lateral femoral condyle *[Q2 · As spoken; Dejour type C is convex lateral facet with hypoplastic medial facet, not hypoplastic LFC.]*
D	Cliff pattern

- **Lateral patellar tilt** → big medial vs lateral distance difference
  - Cause → **tight structures**: retinaculum, iliotibial band, vastus lateralis
- Normal → prominent **LFC** → proper trochlear sulcus
  - → Patella can **engage**
- LFC disorders → **aplasia** / hypoplasia / dysplasia
- **Dysplasia** → normal trochlear groove replaced by:
  - Flat / convex surface
  - Anterior distal femur → one piece

Alternative Classification Ask AI

- Classification 1 → general vs anatomical (osseous / muscular)
- Classification 2 → table

Category	Causes
Soft tissue — global	Ehlers–Danlos syndrome
Soft tissue — medial	MPFL insufficiency (torn ligament); VMO dysplasia
Soft tissue — lateral	Vastus lateralis overdominance; iliotibial band / lateral retinaculum contracture → pulls patella laterally
Osseous	Patella alta; trochlear dysplasia; lower-limb malalignment
Gait	Genu valgum with valgus thrust; pelvic problems

- **Lower-limb malalignment** → all ↑ Q-angle
  - ↑ Femoral anteversion
  - Genu valgum
  - Genu recurvatum
  - External tibial torsion
  - Forefoot pronation
- Gait → genu valgum with **valgus thrust**
- Examine **pelvis** → pelvic core muscles affect patellar instability

Risk Factors and Types of Dislocation Ask AI

Risk Factors for Instability and Recurrence Ask AI

- Predisposition to instability / **recurrence**:
  - Age **< 18 years**
  - **Female** — most common
  - **Family history** → mother, older sister, aunt
  - Usually **bilateral**
  - Anatomical abnormality
    - Patella alta
    - Trochlear dysplasia
    - Abnormal **TT-TG distance**
    - ↑ Q-angle
  - **Quadriceps dysfunction**
  - **Hypermobility** → ligamentous laxity / Ehlers–Danlos
- Factors present → recurrence likely, even at **first episode**
  - Typically young girls

Types of Dislocation Ask AI

Type	Features
Acute	Acute trauma — direct blow, or femoral internal rotation on externally rotated fixed tibia (planted foot)
Recurrent	1–2 previous episodes; every 2–3 months to every 1–2 years
Habitual	Dislocates all the time; chronic dislocation

- Acute mechanism → body rotates, **tibia stays fixed**
- **Habitual** → patella may lie lateral to femoral condyle from the start

Clinical Assessment and Imaging Ask AI

- Instability confirmed → structured approach

Pre-operative Checklist Ask AI

- **Gait** → thrust while walking
- Lower limb
  - ↑ Femoral anteversion
  - Genu valgum
  - Genu recurvatum
  - External tibial torsion
- **Rotational profile** + lower-limb alignment
- Q-angle
- **Patellar tilt**
- Ligamentous laxity → **Beighton score**
- **Range of motion**
- Images → recurvatum, genu valgum, external tibial torsion, ↑ femoral anteversion
- Then → examination

Examination Tests Ask AI

Test	Technique → finding
Apprehension	Lateral push on patella → patient stops hand, guards, contracts quadriceps
Patellar tilt	Hold patella → elevate from lateral side
J sign	Flexion → extension → patella jumps laterally
Patellar glide	Push patella → quadrants moved medially
Q-angle	Measured

- **Apprehension** ≈ shoulder instability → abduction–external rotation → guarding
  - Previous dislocator knows the pain → senses imminent dislocation
- Positive apprehension + laxity → shoulder **sulcus sign**
  - Patellar instability below + shoulder instability above

Q-angle Ask AI

Q-angle

- **Q-angle** → between quadriceps line of pull and patellar tendon
- Larger → ↑ tendency to **lateral patellar subluxation**
- ↑ In:
  - Genu valgum
  - External tibial torsion
  - ↑ Femoral anteversion
  - Tight **lateral retinaculum**
- ↑ **Femoral anteversion** → femur rotated → ↑ Q-angle → lateral subluxation

Imaging Work-up Ask AI

- After gait, rotational profile, examination → imaging

Modality	Purpose
Long leg film	Limb alignment
AP view	—
True lateral	Trochlear dysplasia signs; patellar height
Skyline	Loose bodies / fragment; osteochondral fragment; trochlear dysplasia; patella alta
MRI	Osteochondral fragment; MPFL tear site; associated pathology
CT	TT-TG distance
Rotational profile	Internal torsion; external tibial torsion; femoral anteversion

- **True lateral** → essential
  - Without it → no **crossing sign**, double contour, supratrochlear spur
  - Trochlear dysplasia → visible only on true lateral
  - Not sent by radiographer → nothing can be judged
- MRI → MPFL torn from **patellar** vs femoral attachment
- MRI → associated injuries
  - Femoral internal rotation + external tibial torsion → **meniscal tear**
  - Same position → **ACL tear** with patellar dislocation
- **TT-PCL** distance → now preferred
  - TT-TG varies with leg rotation + flexion angle
- Long leg film → image
- True lateral → **Insall–Salvati** / Caton–Deschamps index → patella alta

Examination Under Anaesthesia Ask AI

EUA

- **Range of motion** → note when patella dislocates
- Ligamentous stability → **MCL**, posterolateral corner, ± ACL
- Compare findings with **other leg**

Management Principles Ask AI

Adult vs Paediatric Ask AI

- Management → **adult** vs paediatric
- Typical age → children 9–10 years; adolescents 15–16 years
- Surgery near open **physes**
  - Tibial tuberosity
  - Femoral condyle in MPFL reconstruction
- → **Physeal arrest** → growth problems

À La Carte Approach Ask AI

- Don't rush to surgery → confirm diagnosis; know each operation's **indications**
- **MPFL reconstruction** → most famous op in Egypt after conferences
  - → **Catastrophic results**
- Not every dislocation = **torn MPFL**
- MPFL reconstruction alone in most patients → foolish; address the underlying problem:
  - **Trochlear dysplasia**
  - **Patella alta**
  - Excessive tightness of **lateral structures**
  - Lateral patellar tilt
  - Hypoplastic LFC
- Most patients → MPFL **plus** something else
- Expectation "like a horse after six weeks" → catastrophic results
- **Failed MPFL** → re-tears → primary problem missed
- French **Lyon school** (France) → Henri Dejour → David Dejour
  - **À la carte approach** → tailor operation to the patient's problem
  - No single operation suits all
  - Options:
    - Trochleoplasty
    - **MPFL repair**
    - MPFL reconstruction
    - **Distal realignment**
    - Derotation osteotomy
- ↑ Femoral anteversion → MPFL alone insufficient
  - → **Derotation distal femoral osteotomy**
- Genu valgum → address it with **MPFL reconstruction**
- Each case different → **tailor** surgery

Non-operative Treatment Ask AI

Non-op

- **Indications**
  - **First-time** dislocator
    - No loose bodies, no osteochondral fragments
  - **Habitual dislocator** → conservative usually better
- Habitual ≈ shoulder **voluntary dislocator** → surgery fails
  - Especially Ehlers–Danlos
  - **Hyper-ligamentous laxity** → high failure rate
- Techniques
  - **Core and hip strengthening** → limb positioning + balance
    - Hip abductors, glutei, abdominal muscles
  - **Closed-chain** quadriceps strengthening
  - **J brace** → prevents lateral patellar jump
  - **Taping**
- Failure → surgery → **à la carte**
- Not everyone → **MPFL reconstruction**

Pitfall

- **MPFL reconstruction** alone → catastrophic failure if primary problem missed
- Trochlear dysplasia unaddressed → MPFL fails
- Genu valgum / recurvatum unaddressed → MPFL fails
- ↑ Femoral anteversion → needs derotation distal femoral osteotomy
- Habitual dislocator / Ehlers–Danlos → surgery fails

Surgical Indications and Operations Ask AI

Acute Dislocation — Same-day Surgery Ask AI

- e.g. dislocated at school → reduced by teacher / **school doctor**
- Operate same day if:
  - Large **haemarthrosis** → MPFL torn on MRI
  - **Loose fragment**
  - **Osteochondral fracture** → patella or femoral condyles
- Otherwise → **conservative**

High Yield

- First-time dislocation → mostly **conservative**
- Same-day surgery → loose body / fragment
- Same-day surgery → osteochondral fracture
- Same-day surgery → MPFL tear on MRI (large haemarthrosis)

Recurrent Dislocation — Proximal vs Distal Ask AI

- Recurrent → patella dislocated 2–3 times

Group	Procedures
Proximal — above patella	MPFL reconstruction; lateral retinacular release; medial plication / reefing; VMO advancement
Distal — below knee	Anteromedialization / medialization of tibial tuberosity → lateralized TT, ↑ Q-angle
Distal — below knee	Distalization of tibial tuberosity → patella alta

- **Trochleoplasty** → sometimes counted as proximal
  - Separate entity → designed for trochlear dysplasia

The Seven Operations Ask AI

- **Arthroscopic débridement** → remove loose body / repair ± fixation
  - Standard anterolateral + anteromedial portals → cannot assess
  - **Superolateral portal** → patellar tracking from above
  - Lateral patellar compression → nothing seen even in extension via standard portals
    - → Superolateral view of trochlear groove + tracking
- **MPFL repair** → suture MPFL
  - Bone fragment off medial patella
  - Avulsed from **medial epicondyle** of femur
- **MPFL reconstruction** → autograft / allograft
- **Fulkerson osteotomy** → anteromedialization / medialization of tibial tuberosity
- **Tibial tubercle distalization**
- **Lateral retinacular release**
- **Trochleoplasty**

> **Clinical Pearl**
> - Arthroscopy in instability → **superolateral portal**
> - Standard portals → cannot assess patellar tracking

Recap Ask AI

- Acute → operate if osteochondral fragment, loose body, or MPFL tear
  - MPFL tear → evident on **MRI**
- Recurrent → **proximal** or distal realignment

Arthroscopic Débridement Ask AI

Arthroscopy

- **Indications**
  - Displaced osteochondral fragment
  - Osteochondral fracture
  - Small unfixable **loose body**
- **First-time dislocator**
  - Dislocated today + loose body / osteochondral fragment
- Loose body too small to fix → remove
  - Pebble / marble in joint → **locking**
- Osteochondral fragment → **fixation**

Case: Volleyball Player Ask AI

- **Volleyball** player with patellar dislocation
- Poor-quality CT; uncooperative → pain + **haemarthrosis**
- Arthroscopy → whole posterior patella fractured
  - Anterior piece → held by quadriceps + **patellar tendon**
  - Cartilage behind → separated as one piece
- Converted → **open** procedure
- Sutures → four holes in patella + four in osteochondral piece → reattached
- **Fixation** options → by fragment size + position
  - Bioabsorbable screw
  - **Herbert screw**
  - Bioabsorbable pins

MPFL Repair and Older Soft-Tissue Procedures Ask AI

MPFL Repair Ask AI

MPFL

- First-time dislocator → MPFL off with **bone fragment**
- **Reattach** with:
  - Anchor
  - Herbert / **headless screw**
  - Pin
  - Bioabsorbable screw
- Three techniques
  - **Plication** → medial patellar retinaculum
  - **Anatomical repair** → MPFL itself
  - **VMO advancement** → onto medial patella
    - Strip from medial **quadriceps tendon**
    - Whole strip down → VMO turned over, advanced
    - → Sutured to **anteromedial half** of patella

Declining Procedures Ask AI

- **VMO advancement**, medial reefing, lateral release
  - → Much less common now
- Easy fashion → **lateral release** via 2 cm incision + two medial stitches
- Replaced → **MPFL repair** / reconstruction
- **Medial reefing** + lateral release → prevalent 8–9 years ago
  - Now → hardly done in **Europe / America**

MPFL Reconstruction Ask AI

- Torn MPFL shredded, no bony fragment → cannot **reattach**
- Recurrent instability without **malalignment** → isolated MPFL
- **Graft** → allograft / autograft
  - Egypt → **autograft** only
  - Gracilis, semitendinosus, quadriceps tendon, adductor tendon
- Femoral attachment → true lateral, intraoperative **C-arm** → Schöttle's point

Indications for Isolated MPFL Ask AI

MPFL-R

- **Skeletally mature** → immature no longer excluded
  - Immature → no **femoral fixation** → open physis
  - Graft looped around **adductor magnus tendon** *[C1 · transcript: "adductor longus tendon"]*
  - Or adductor magnus strip → attached at adductor tubercle *[C2 · transcript: "adductor longus attached at adductor tubercle"]*
    - → Turned **downwards** → sutured to medial patella
  - No nail / screw in **open physis**
    - → **Physeal arrest**
- Excessive lateral **laxity**
  - Not tight → no lateral patellar compression syndrome
- **Normal trochlea** / low-grade dysplasia
  - Type 1 / **Dejour type A**
- Normal **Q-angle**
- **TT-TG** < 20 mm

High Yield

- Isolated MPFL → **five indications**
- Skeletally mature + excessive lateral laxity
- Normal trochlea / low-grade dysplasia (Dejour A)
- Normal Q-angle
- TT-TG < 20 mm

- MPFL may also be combined → MPFL **plus** another procedure
- **Contraindication** → formerly skeletally immature (crossed out in red)
  - Now → many modifications
  - Graft through patella → looped around adductor tendon → **tied on itself**
  - Or adductor magnus strip → turned **distally** → sutured to medial patella

Graft Sources Ask AI

MPFL-R

- **Gracilis**
- **Semitendinosus**
- Medial part of **patellar tendon**
- Medial / central part of **quadriceps tendon**
- **Allografts**
- **Adductor tendon**
- Medial one-third **quadriceps tendon** → turned behind medial limb
  - → Fixed at MPFL attachment site
- **Adductor magnus** tendon → turned down
- Skeletally immature → no screw at all
  - **Anchor** in patella
  - Or anchor + sutured on itself
  - → Avoid **physeal arrest**

Femoral Attachment Ask AI

MPFL-R

Reference point	Graft position
Medial femoral epicondyle	1 cm proximal, 2 mm posterior
Adductor tubercle	4 mm distal, 2 mm anterior

- MPFL attachment → **debatable**
  - Somewhere between medial epicondyle and adductor tubercle
- Either reference → ends **midway** between the two points

Schöttle's Point Ask AI

MPFL-R

- **Schöttle's point** → radiological point under fluoroscopy
- Described as **best point** for femoral attachment near epicondyle
- Current studies → not perfect
- Still → **reference point** in most MPFL studies
- Three lines
  - Along **posterior femoral cortex** (distal femur)
  - Perpendicular → junction of posterior condyle + posterior cortex
  - Perpendicular → most posterior point of **Blumensaat line**
- Circle on image → C-arm → **drill** here

Exam Pearl

- **Schöttle's point** → line along posterior femoral cortex
- Perpendicular → posterior condyle–cortex junction
- Perpendicular → most posterior point of Blumensaat line

Patellar Fixation Ask AI

MPFL-R

- Two **tunnels** in patella
- Tunnels across whole length / width → risk **patellar fracture**
- Medial **trough** + two anchors → graft on screw
- Image → tunnel site; **semitendinosus** / gracilis harvest
- Skeletally immature
  - **Quadriceps turn-down flap**
  - Or adductor tendon strip

Tibial Tuberosity Osteotomies Ask AI

Fulkerson Osteotomy Ask AI

AMZ

- Fourth operation → **anterior + medial transfer** of tibial tuberosity
- Indication
  - **TT-TG** > 20 mm on CT
  - Alone or with **MPFL reconstruction**
- **Osteotomy**
  - TT with patellar tendon attachment → moved **medially**
  - Fixed → two screws
- Extend along anterior **tibial crest** → substantial fragment
  - → No fracture on drilling
- Image → before / after **anteriorization** + medialization
  - Fixed → two extra screws

Indications	Contraindications
Q-angle / TT-TG > 20 mm	Skeletally immature → physeal arrest
Patella alta	Probably: MPFL tear without concurrent MPFL reconstruction
Normal patellar glide — unrestricted	Patellofemoral arthritis
Medial facet arthritis → relieves PFJ reaction force	— *[Q3 · As spoken; AMZ shifts load medially, so medial facet arthritis is a contraindication, not indication.]*

- **PF arthritis** → medial TT shift ↓ joint reaction force
  - But → worsens arthritis
- **Fulkerson** → osteotomy extends down tibial crest
  - Room for two screws
- Moved **medially** + anteriorly → screws

Tibial Tuberosity Distalization Ask AI

Distal TT

- Fifth operation → **patella alta**
  - High patella → doesn't engage trochlear sulcus
  - → Dislocates early, **early degrees** of flexion
- Tibial tuberosity
  - → Moved 5–10 mm **distally**
  - Long patellar tendon in patella alta
  - Two screws

Lateral Retinacular Release Ask AI

- **Isolated procedure** → no longer indicated
- Only → **lateral patellar compression syndrome** / excessive lateral tilt
  - Residual lateral **tightness** after all other procedures
- Released structures
  - **Lateral patellar retinaculum**
  - **Lateral patellofemoral ligament**
  - **Lateral patellotibial ligament**

Trochleoplasty Ask AI

Principles Ask AI

Trochlea

- **Dejour** classification → explained last lecture
- Aim → new **trochlear sulcus**
  - Corrects flat / convex surface of dysplasia
  - → **Trough** → stabilizes patella → prevents lateral dislocation
- **Primary procedure** → trochlear dysplasia
- **Salvage procedure** → failed surgery / revision
  - Especially recurrent patellar instability

Types Ask AI

Trochlea

Type	Technique	Result
Lateral facet elevation (Albee)	Elevate lateral trochlear facet + bone graft beneath; groove untouched	Easy; ↑ joint reaction force → PF arthritis → not done
Deepening (Henri Dejour modification)	Remove osteochondral fragment beneath sulcus → split cartilage in two → deep groove	↓ Flatness + prominence → restores anatomy, creates trough
Recession wedge	Excise wedge with supratrochlear bump → reinsert remaining fragment	Corrects prominence + deepens floor; technically easiest

- **Albee** → trochlear groove still functions; only lateral facet raised
  - Patella strikes elevated facet → ↑ joint reaction force

Trochleoplasty in Egypt Ask AI

- Little done → **instruments** unavailable
- Own technique
  - **Shaver burr** → remove cancellous bone behind trochlear cartilage
  - Break trochlear cartilage
  - Fix → Herbert screws / headless screws / pins
- Image → trochlea appearance
- Behind cartilage → remove **osteochondral fragment**
  - Split → lateral + medial facets
  - Fix → most commonly **Herbert** / headless screws

Combined Procedures and Paediatric Instability Ask AI

Addressing Malalignment Ask AI

- Solution ≠ single op
  - MPFL
  - Trochleoplasty
  - Lateral retinacular release
  - TT distalization
  - TT anteromedialization
  - Arthroscopy
  - MPFL repair
- **Malalignment** → lateral high tibial osteotomy / opening wedge osteotomy
  - → Corrects genu valgum / recurvatum
  - → Concurrent **MPFL reconstruction** succeeds
- MPFL alone in **genu valgum** / recurvatum → fails

Paediatric Instability Ask AI

Paeds

- **Physes** → key concern
  - No **tibial tubercle osteotomies**
  - MPFL → soft-tissue procedures only
    - Turned quadriceps tendon strip
    - Turned **adductor tendon** part
    - Or anchor fixation
    - → No screw in physis → no physeal arrest

Trochleoplasty + MPFL Ask AI

- **Dysplastic trochlea** → trochleoplasty + MPFL reconstruction, all patients
- MPFL with unaddressed **trochlear dysplasia** → fails immediately

Lateral Release Complication Ask AI

- Post-op **dislocation** → iatrogenic
  - Probably medial, after isolated lateral release
- → Lateral release rarely done now

MPFL Reconstruction — Core Operation Ask AI

- **Arthroscopy journal** → MPFL reconstruction
  - → Core operation for all patients, within its indications
- Alone → only the **five indications**
  - Normal trochlea / low-grade dysplasia
  - Normal Q-angle
  - TT-TG < 20 mm
- Combined → **trochleoplasty**, TT distalization, anteromedialization
- Other pathology unaddressed → MPFL **fails**

Questions and Answers Ask AI

Patellofemoral Osteoarthritis Ask AI

PF OA

- Determine **pain source** → bone, muscle, other
- **Infrapatellar fat pad** problem → arthroscopic removal
  - → Relief
- Established PF arthritis → **patellofemoral arthroplasty**
  - Now available in Egypt — previously not
  - Replaces one of three **compartments**
  - Restores smooth trochlear surface + smooth patellar back

Working Near the Physis Ask AI

Physis

- **Anchor** → ≈ half a centimetre deep
- Full tunnel + 7 mm screw → smallest available in Egypt
- Anchor → no **physeal arrest**
  - **Fluoroscopic guidance** → direct anchor away from physis
- Screw tunnel → ≥ 4–5 cm in **distal femur**
  - → Hard to avoid physis → physeal arrest easy

Post-operative Rehabilitation Ask AI

Rehab

- **Hinged knee brace** → four weeks
- **Quad strengthening** from start → brace locked
- Brace opened → 45° → 60° → 75° → 90°
- Locked in extension → **partial weight bearing** / toe-touch

Acute vs Recurrent Protocol Ask AI

Rehab

Scenario	Management / risk
Acute — first dislocation	Osteochondral fragment / loose body excluded, MPFL only stretched → open knee support (patellar window) 4 weeks → reassess
Recurrent — 2–3 dislocations	Each episode → cartilage damage, osteochondral fracture, PF arthritis, pain, muscle weakness, muscle atrophy

- **First-time dislocation** → mostly conservative
- Except → three indications
  - **Loose body**
  - **Osteochondral fracture**
  - MRI → MPFL torn from patellar or femoral side

Trochleoplasty and Trochlear AVN Ask AI

AVN

- Trochleoplasty → **trochlear AVN** possible
- Very thin **burr**
- **Lyon school** → C-arm-type guide with thin burr
  - → Limited **subchondral bone** removal
- Current local practice → larger burr, wider opening → adequate **osteochondral flap**
- Leave reasonable **thickness** behind cartilage → avoids AVN

Habitual Dislocation at Ten Years Ask AI

- 10-year-old female → correction in **skeletally immature**?
- Assess well → at ten years, likely **anatomical abnormality**
- Check and address → prevents **habitual dislocation**
  - Ligamentous laxity
  - Genu valgum
  - Pronated forefoot
  - Recurvatum
- Habitual at 8–9 years → identify **anatomical abnormality**

Is MRI Mandatory? Ask AI

MRI

- **Acute dislocation** → MRI preferable in all
  - Loose / osteochondral fragment
  - MPFL integrity
  - **Haemarthrosis** extent

Why Partial Weight Bearing Ask AI

- Patella → not weight bearing; why **partial weight bearing**?
- After **distal realignment** (distalization / AMZ)
  - Avoid full quadriceps contraction
  - Probable risk → tuberosity fragment pulled off
- After **MPFL reconstruction** → weight bearing allowed
  - Provided → **brace** worn
- Certain **physiotherapy** school → no experience → no comment

Correction Beyond the Knee Ask AI

- **Flatfoot** → fibular piece → lateral subtalar region
  - → Corrects pronated foot
- Not all surgery at **patellar level**
  - **Subtrochanteric derotation osteotomy** → femoral anteversion
    - → Corrects patellar instability
  - **Flatfoot** correction → corrects patellar instability
  - **Tibial osteotomy** → genu recurvatum / valgum

Injections in PF Arthritis Ask AI

PF OA

- Useful → only without **extensive osteophytes**
  - Confirm on examination
- Osteophytes + rubbing → **hyaluronic acid** → no benefit
  - Unlike femorotibial osteoarthritis
- Relieves some **pain**